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Ubiquitous antigens for treatment of autoimmune or inflammatory diseases

Abstract

Described herein are methods for treating an autoimmune or inflammatory disease in a patient, comprising administering a composition comprising: a) a plurality of antigen-major histocompatibility class II complexes (antigen-MHCII), each antigen-MHCII of the plurality comprising a ubiquitous autoantigen associated with a binding groove of an MHC class II molecule, wherein the ubiquitous autoantigen is chosen from PDC-E2.sub.353-367, PDC-E2.sub.72-86, and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94, and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135, and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801; and b) a nanoparticle core possessing a diameter of between 1 and about 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core or a biocompatible layer surrounding the nanoparticle core; and wherein the autoimmune or inflammatory disease is chosen from multiple sclerosis and psoriasis.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4367110	12/1982	Yoshikawa	N/A	N/A
4414148	12/1982	Jansen	N/A	N/A
4452901	12/1983	Gordon	N/A	N/A
4478946	12/1983	Van der Merwe	N/A	N/A
4554101	12/1984	Hopp	N/A	N/A
4569789	12/1985	Blattler	N/A	N/A
4589071	12/1985	Yamamuro	N/A	N/A
4589330	12/1985	Teron	N/A	N/A
4659839	12/1986	Nicolotti	N/A	N/A
4671958	12/1986	Rodwell	N/A	N/A
4680338	12/1986	Sundoro	N/A	N/A
4699784	12/1986	Shih	N/A	N/A
4818542	12/1988	Deluca	N/A	N/A
4859839	12/1988	Tetelman	N/A	N/A
5258499	12/1992	Konigsberg	N/A	N/A
5543391	12/1995	Yatvin	N/A	N/A

5676926	12/1996	Platzek	N/A	N/A
5676928	12/1996	Klaveness	N/A	N/A
5731168	12/1997	Carter	N/A	N/A
5807706	12/1997	Carter	N/A	N/A
5821333	12/1997	Carter	N/A	N/A
5840839	12/1997	Wang	N/A	N/A
6103379	12/1999	Margel	N/A	N/A
6387498	12/2001	Coulter	N/A	N/A
6651655	12/2002	Licalsi	N/A	N/A
6688494	12/2003	Pozarnsky	N/A	N/A
6712997	12/2003	Won	N/A	N/A
6797514	12/2003	Berenson	N/A	N/A
6846474	12/2004	Nayfeh	N/A	N/A
6929675	12/2004	Bunge	N/A	N/A
7060121	12/2005	Lin	N/A	N/A
7090973	12/2005	Breton	N/A	N/A
7183065	12/2006	Braun	N/A	N/A
7285289	12/2006	Nagy	N/A	N/A
7326399	12/2007	Zhou	N/A	N/A
7332586	12/2007	Franzen	N/A	N/A
7361733	12/2007	Hershberg	N/A	N/A
7572631	12/2008	Berenson	N/A	N/A
7642228	12/2009	Carter	N/A	N/A
7695936	12/2009	Carter	N/A	N/A
7785801	12/2009	Tuereci	N/A	N/A
7795224	12/2009	Eisenbach	N/A	N/A
7812116	12/2009	Bae	N/A	N/A
7816814	12/2009	Hennessy	N/A	N/A
8216805	12/2011	Carter	N/A	N/A
8354110	12/2012	Santamaria	N/A	N/A
8679785	12/2013	Carter	N/A	N/A
8835144	12/2013	Jiang	N/A	N/A
9149440	12/2014	Turos	N/A	N/A
9511151	12/2015	Santamaria	N/A	N/A
9603948	12/2016	Santamaria	N/A	N/A
10004703	12/2017	Jacobson	N/A	N/A
10012045	12/2017	Lajesic	N/A	N/A
10080808	12/2017	Santamaria	N/A	N/A
10124045	12/2017	Santamaria	N/A	N/A
10172955	12/2018	Santamaria	N/A	N/A
10441556	12/2018	Jacobson	N/A	N/A
10485882	12/2018	Santamaria	N/A	N/A
10632193	12/2019	McCreedy	N/A	N/A
10905773	12/2020	Santamaria	N/A	N/A
10988516	12/2020	Santamaria	N/A	N/A
11000596	12/2020	Santamaria	N/A	N/A
11338024	12/2021	Santamaria	N/A	N/A
12011480	12/2023	Santamaria	N/A	N/A
2003/0068363	12/2002	Clark	N/A	N/A
2003/0124149	12/2002	Shalaby	N/A	N/A

2004/0115216	12/2003	Schneck	N/A	N/A
2004/0137642	12/2003	Erfle	N/A	N/A
2004/0197304	12/2003	Chen	N/A	N/A
2004/0224402	12/2003	Bonyhadi	N/A	N/A
2004/0265392	12/2003	Tovar	N/A	N/A
2005/0003431	12/2004	Wucherpennig	N/A	N/A
2005/0118102	12/2004	Xiang	N/A	N/A
2005/0129617	12/2004	Tan	N/A	N/A
2005/0202032	12/2004	Kaufman	N/A	N/A
2005/0208120	12/2004	Albani	N/A	N/A
2006/0216239	12/2005	Zhang	N/A	N/A
2006/0219239	12/2005	Plaschkas	N/A	N/A
2006/0246524	12/2005	Bauer	548/518	B82Y 5/00
2007/0054337	12/2006	Ferning	N/A	N/A
2007/0059775	12/2006	Hultman	N/A	N/A
2007/0129307	12/2006	Tan	N/A	N/A
2007/0154953	12/2006	Brunner	N/A	N/A
2009/0155292	12/2008	Santamaria	N/A	N/A
2009/0258355	12/2008	Maye	N/A	N/A
2010/0061984	12/2009	Greene	N/A	N/A
2010/0095544	12/2009	Haseloh	N/A	N/A
2010/0104503	12/2009	Mellman	N/A	N/A
2010/0303730	12/2009	Hegmann	N/A	N/A
2010/0303866	12/2009	Saint-Remy	N/A	N/A
2011/0029121	12/2010	Amit	N/A	N/A
2011/0059121	12/2010	Santamaria	N/A	N/A
2011/0250146	12/2010	Zhang	N/A	N/A
2011/0318380	12/2010	Brix	N/A	N/A
2012/0077686	12/2011	Weiner	N/A	N/A
2012/0093934	12/2011	Santamaria	N/A	N/A
2012/0121649	12/2011	Santamaria	N/A	N/A
2012/0252742	12/2011	Kranz	N/A	N/A
2013/0089553	12/2012	Carter	N/A	N/A
2013/0128138	12/2012	Kuo	N/A	N/A
2013/0171179	12/2012	Burrows	N/A	N/A
2013/0302421	12/2012	Santamaria	N/A	N/A
2013/0330414	12/2012	Santamaria	N/A	N/A
2014/0105980	12/2013	Santamaria	N/A	N/A
2014/0294982	12/2013	Freund	N/A	N/A
2014/0341938	12/2013	Rademacher	N/A	N/A
2014/0370099	12/2013	Green	N/A	N/A
2015/0068613	12/2014	Taskar	N/A	N/A
2015/0125536	12/2014	Santamaria	N/A	N/A
2015/0150996	12/2014	Miller	N/A	N/A
2015/0209446	12/2014	Santamaria	N/A	N/A
2015/0250871	12/2014	Santamaria	N/A	N/A
2015/0344586	12/2014	Georges	N/A	N/A
2015/0374815	12/2014	Kishimoto	N/A	N/A
2016/0068613	12/2015	Regula	N/A	N/A
2016/0271237	12/2015	Santamaria	N/A	N/A

2017/0095544	12/2016	Santamaria	N/A	N/A
2017/0274096	12/2016	Santamaria	N/A	N/A
2017/0312348	12/2016	Santamaria	N/A	N/A
2017/0333540	12/2016	Santamaria	N/A	N/A
2018/0127481	12/2017	Santamaria	N/A	N/A
2019/0060427	12/2018	Santamaria	N/A	N/A
2019/0060484	12/2018	Santamaria	N/A	N/A
2019/0076545	12/2018	Santamaria	N/A	N/A
2019/0134171	12/2018	Santamaria	N/A	N/A
2020/0009265	12/2019	Santamaria	N/A	N/A
2020/0057048	12/2019	Santamaria	N/A	N/A
2021/0145949	12/2020	Santamaria	N/A	N/A
2021/0205470	12/2020	Santamaria	N/A	N/A
2021/0230237	12/2020	Santamaria	N/A	N/A
2022/0401534	12/2021	Santamaria	N/A	N/A
2023/0355730	12/2022	Santamaria	N/A	N/A
2024/0201171	12/2023	Santamaria	N/A	N/A
2024/0299536	12/2023	Santamaria	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2517097	12/2003	CA	N/A
2717719	12/2008	CA	N/A
2868551	12/2012	CA	N/A
101678090	12/2009	CN	N/A
0188256	12/1985	EP	N/A
1088256	12/2000	EP	N/A
2131856	12/2008	EP	N/A
2614834	12/2012	EP	N/A
2621523	12/2012	EP	N/A
3065771	12/2015	EP	N/A
3067366	12/2015	EP	N/A
3269384	12/2017	EP	N/A
3291832	12/2017	EP	N/A
3539564	12/2018	EP	N/A
2003231698	12/2002	JP	N/A
2005538083	12/2004	JP	N/A
2006522319	12/2005	JP	N/A
2007508503	12/2006	JP	N/A
2008514686	12/2007	JP	N/A
2010522695	12/2009	JP	N/A
2012505249	12/2011	JP	N/A
2013538208	12/2012	JP	N/A
2015063616	12/2014	JP	N/A
20120100840	12/2011	KR	N/A
1990007339	12/1989	WO	N/A
1992018150	12/1991	WO	N/A
1993001716	12/1992	WO	N/A
1993016725	12/1992	WO	N/A
1994009823	12/1993	WO	N/A

1996018105	12/1995	WO	N/A
1998006749	12/1997	WO	N/A
1999014236	12/1998	WO	N/A
1999042597	12/1998	WO	N/A
1999064926	12/1998	WO	N/A
2000043662	12/1999	WO	N/A
2000067788	12/1999	WO	N/A
2001024764	12/2000	WO	N/A
2002080963	12/2001	WO	N/A
2004006951	12/2003	WO	N/A
2004076909	12/2003	WO	N/A
2004078909	12/2003	WO	N/A
2005033267	12/2004	WO	N/A
2005036035	12/2004	WO	N/A
2006016113	12/2005	WO	N/A
2006037979	12/2005	WO	N/A
2006054806	12/2005	WO	N/A
2006080951	12/2005	WO	N/A
2007024026	12/2006	WO	N/A
2008051245	12/2007	WO	N/A
2008109852	12/2007	WO	N/A
2008118861	12/2007	WO	N/A
2009003492	12/2008	WO	N/A
2009031258	12/2008	WO	N/A
2009040811	12/2008	WO	N/A
2009064273	12/2008	WO	N/A
2009078799	12/2008	WO	N/A
2009094273	12/2008	WO	N/A
2009111588	12/2008	WO	N/A
2009126835	12/2008	WO	N/A
2010025324	12/2009	WO	N/A
2010027827	12/2009	WO	N/A
2010037395	12/2009	WO	N/A
2010037397	12/2009	WO	N/A
2010042876	12/2009	WO	N/A
2010080032	12/2009	WO	N/A
2010085509	12/2009	WO	N/A
2011073685	12/2010	WO	N/A
2011104497	12/2010	WO	N/A
2012012874	12/2011	WO	N/A
2012031258	12/2011	WO	N/A
2012041968	12/2011	WO	N/A
2012062904	12/2011	WO	N/A
2012121528	12/2011	WO	N/A
2013043662	12/2012	WO	N/A
2013072051	12/2012	WO	N/A
2013103438	12/2012	WO	N/A
2013144811	12/2012	WO	N/A
2014080286	12/2013	WO	N/A
2014096015	12/2013	WO	N/A

2015063616	12/2014	WO	N/A
2016081854	12/2015	WO	N/A
2016097334	12/2015	WO	N/A
2016145605	12/2015	WO	N/A
2016146505	12/2015	WO	N/A
2016160721	12/2015	WO	N/A
2016193299	12/2015	WO	N/A
2016198932	12/2015	WO	N/A
2017044672	12/2016	WO	N/A
2018087597	12/2017	WO	N/A
2018185564	12/2017	WO	N/A
2019106435	12/2018	WO	N/A

OTHER PUBLICATIONS

Lieo, A., et al. Semin. Liver. Dis.; 34(3):273-284 (Year: 2014). cited by examiner

Schaefer, W. et al., “Heavy and light chain pairing of bivalent quadroma and knobs-into-holes antibodies analyzed by UHR-ESI-QTOF mass spectrometry”, MAbs., 8(1):49-55, (2016). cited by applicant

Umeshappa, C. et al., “Suppression of A Broad Spectrum of Liver Autoimmune Pathologies by Single Peptide-MHC-Based Nanomedicines”, Nat Commun., 10(1):2150, (2019). cited by applicant

GenBank Accession No. AAC14923.1, “T cell receptor alpha chain [*Homo sapiens*]”, retrieved from [https://www.ncbi.nlm.nih.gov/protein/AAC14923.1?report=genbank&log\\$=protalign&blast](https://www.ncbi.nlm.nih.gov/protein/AAC14923.1?report=genbank&log$=protalign&blast), (1998). cited by applicant

t Hart, B. et al., “Modelling of Multiple Sclerosis: Lessons Learned in a Non-Human Primate”, Lancet Neurol., 3(10):588-97, (2004). cited by applicant

Aichele, P. et al., “Peptide-induced T-cell Tolerance to Prevent Autoimmune Diabetes in a Transgenic Mouse Model”, Proc Natl Acad Sci USA, 91(2):444-8, (1994). cited by applicant

Altman, J. et al., “Phenotypic Analysis of Antigen-Specific T Lymphocytes”, Science, 274(5284):94-6, (1996). cited by applicant

Altschul, S. et al., “Basic Local Alignment Search Tool”, J Mol Biol., 215(3):403-10, (1990). cited by applicant

Amrani, A. et al., “Expansion of the Antigenic Repertoire of a Single T Cell Receptor Upon T Cell Activation”, J Immunol., 167(2):655-66, (2001). cited by applicant

Amrani, A. et al., “Progression of Autoimmune Diabetes Driven by Avidity Maturation of a T-cell Population,” Nature, 406(6797):739-42, (2000). cited by applicant

Anderson, B. et al., “Prevalent CD8(+) T Cell Response Against One peptide/MHC Complex in Autoimmune Diabetes”, Proc Natl Acad Sci USA, 96(16):9311-6, (1999). cited by applicant

Anderton, S. et al., “Hierarchy in the Ability of T Cell Epitopes to Induce Peripheral Tolerance to Antigens From Myelin”, Eur J Immunol., 28(4):1251-61, (1998). cited by applicant

Anonymous, Diabetes Prevention Trial—Type 1 Diabetes Study Group, “Effects of Insulin in Relatives of Patients With Type 1 Diabetes Mellitus”, N Engl J Med., 346(22):1685-91, (2002). cited by applicant

Anonymous, Website article from kidshealth.org/PageManager.jsp?dn=KidsHealth&lic=1&ps=107&cat_id=139&article; downloaded Nov. 9, 2010: 2 pages total. cited by applicant

Appay, V. et al., “HIV-specific CD8(+) T Cells Produce Antiviral Cytokines but Are Impaired in Cytolytic Function”, J Exp Med., 192(1):63-72, (2000). cited by applicant

Asai, H. et al., “Co-introduced Functional CCR2 Potentiates in Vivo Anti-Lung Cancer Functionality Mediated by T Cells Double Gene-Modified to Express WT1-specific T-cell”, PLoS

One, 8(2):e56820, (2013). cited by applicant

Azuma, M. et al., "T Cell Costimulation and Diseases," *Kokubyo Gakkai Zasshi*, 67(3):233-9, (2000). cited by applicant

Babbe, H. et al., "Clonal Expansions of CD8(+) T Cells Dominate the T Cell Infiltrate in Active Multiple Sclerosis Lesions as Shown by Micromanipulation and Single Cell Polymerase Chain Reaction", *J Exp Med.*, 192(3):393-404, (2000). cited by applicant

Bacchetta, R. et al., "High Levels of Interleukin 10 Production in Vivo Are Associated With Tolerance in SCID Patients Transplanted With HLA Mismatched Hematopoietic Stem Cells ", *J Exp Med.*, 179(2):493-502, (1994). cited by applicant

Bachmann, M. et al., "Developmental Regulation of Lck Targeting to the CD8 Coreceptor Controls Signaling in Naive and Memory T Cells", *J Exp Med.*, 189(10):1521-30, (1999). cited by applicant

Bahcetepe, N. et al., "The Role of HLA Antigens in the Aetiology of Psoriasis", *Med Glas (Zenica)*, 10(2):339-42, (2013). cited by applicant

Bailey-Bucktrout, S. et al., "Self-antigen-driven Activation Induces Instability of Regulatory T Cells During an Inflammatory Autoimmune Response", *Immunity*, 39(5):949-62, (2013). cited by applicant

Baker, D. et al., "Critical Appraisal of Animal Models of Multiple Sclerosis", *Mult Scler.*, 17(6):647-57, (2011). cited by applicant

Bakker, A. et al., "MHC Multimer Technology: Current Status and Future Prospects", *Curr Opin Immunol.*, 17(4):428-33, (2005). cited by applicant

Barber, D. et al., "Restoring Function in Exhausted CD8 T Cells During Chronic Viral Infection", *Nature*, 439(7077):682-7, (2006). cited by applicant

Becker, T. et al., "Interleukin 15 Is Required for Proliferative Renewal of Virus-Specific Memory CD8 T Cells", *J Exp Med.*, 195(12):1541-8, (2002). cited by applicant

Behan, P. et al., "The Sad Plight of Multiple Sclerosis Research (Low on Fact, High on Fiction): Critical Data to Support It Being a Neurocristopathy", *Inflammopharmacology*, 18(6):265-90, (2010). cited by applicant

Chen, Q. et al., "IL-2 Controls the Stability of Foxp3 Expression in TGF-beta-induced Foxp3+ T Cells in Vivo", *J Immunol.*, 186(11):6329-37, (2011). cited by applicant

Choudhuri, K. et al., "Signaling Microdomains in T Cells", *FEBS Lett.*, 584(24):4823-31, (2010). cited by applicant

Cirillo, C. et al., "S100B Protein in the Gut: The Evidence for Enteroglial-Sustained Intestinal Inflammation", *World J Gastroenterol.*, 17(10):1261-6, (2011). cited by applicant

Clemente Casares, J. "pMHC-Class II Nanovaccine to Treat Autoimmune Diseases", Doctor of Philosophy Thesis, Calgary University, Alberta, Canada, 391 pages, retrieved from: <http://theses.ucalgary.ca/handle/11023/1589>; (2014). cited by applicant

Clemente-Casares, X. et al., "Expanding Antigen-Specific Regulatory Networks to Treat Autoimmunity", *Nature*, 530(7591):434-40, (2016). cited by applicant

Clemente-Casares, X. et al., "Peptide-MHC-based Nanovaccines for the Treatment of Autoimmunity: A "One Size Fits All" Approach?", *J Mol Med.*, 89(8):733-42, (2011). cited by applicant

Cnop, M. et al., "Mechanisms of Pancreatic Beta-Cell Death in Type 1 and Type 2 Diabetes: Many Differences, Few Similarities", *Diabetes*, 54(Suppl 2):S97-S107, (2005). cited by applicant

Constantinescu, C. et al., "Experimental Autoimmune Encephalomyelitis (EAE) as a Model for Multiple Sclerosis (MS)", *Br J Pharmacol.*, 164(4):1079-106, (2011). cited by applicant

Corrigall, V. et al., "Autoantigens and Immune Pathways in Rheumatoid Arthritis", *Grit Rev Immunol.*, 22(4):281-93, (2002). cited by applicant

Croxford, A. et al., "Mouse Models for Multiple Sclerosis: Historical Facts and Future Implications", *Biochim Biophys Acta.*, 1812(2):177-83, (2011). cited by applicant

Cuív, P. et al., "Draft Genome Sequence of *Bacteroides Vulgatus* PC510, a Strain Isolated From

Human Feces”, *J Bacteriol.*, 193(15):4025-6, (2011). cited by applicant

Culina, S. et al., “Antigen-Based Immune Therapeutics for Type 1 Diabetes: Magic Bullets or Ordinary Blanks?”, *Clin Dev Immunol.*, 2011:286248, (2011). cited by applicant

Daperno, M. et al., “Results of the 2nd Part Scientific Workshop of the ECCO. II: Measures and Markers of Prediction to Achieve, Detect, and Monitor Intestinal Healing in Inflammatory Bowel Disease”, *J Crohns Colitis.*, 5(5):484-98, (2011). cited by applicant

Dave, M. et al., “Mucosal Healing in Inflammatory Bowel Disease-A True Paradigm of Success?”, *Gastroenterol hepatol.*, 8(1):29-38, (2012). cited by applicant

Davies, R. et al., “Engineered Particle Surfaces”, *Adv Mat.*, 10(15):1264-70, (1998). cited by applicant

De Plaen, E. et al., “Immunogenic (Tum-) Variants of Mouse Tumor P815: Cloning of the Gene of Tum- Antigen P91A and Identification of the Tum- Mutation”, *Proc Natl Acad Sci USA*, 85(7):2274-8, (1988). cited by applicant

Denic, A. et al., “The Relevance of Animal Models in Multiple Sclerosis Research”, *Pathophysiology*, 18(1):21-9, (2011). cited by applicant

Desreumaux, P. et al., “Safety and Efficacy of Antigen-Specific Regulatory T-cell Therapy for Patients With Refractory Crohn's Disease”, *Gastroenterology*, 143(5):1207-17, (2012). cited by applicant

Flad, T. et al., “Development of an MHC-class I Peptide Selection Assay Combining Nanoparticle Technology and Matrix-Assisted Laser Desorption/Ionisation Mass Spectrometry”, *J Immunol Methods*, 283(1-2):205-13, (2003). cited by applicant

Jarchum, I. et al., “Identification of Novel IGRP Epitopes Targeted in Type 1 Diabetes Patients”, *Clin Immunol.*, 127(3):359-65, (2008). cited by applicant

Jarchum, I. et al., “In Vivo Cytotoxicity of Insulin-Specific CD8+ T-cells in HLA-A*0201 Transgenic NOD Mice”, *Diabetes*, 56(10):2551-60, (2007). cited by applicant

Jarius, S. et al., “Mechanisms of Disease: Aquaporin-4 Antibodies in Neuromyelitis Optica”, *Nat Clin Pract Neurol.*, 4(4):202-14, (2008). cited by applicant

Jokerst, J. et al., “Nanoparticle PEGylation for Imaging and Therapy”, *Nanomedicine (Lond.)*, 6(4):715-28, (2011). cited by applicant

Judge, A. et al., “Interleukin 15 Controls Both Proliferation and Survival of a Subset of Memory-Phenotype CD8(+) T Cells”, *J Exp Med.*, 196(7):935-46, (2002). cited by applicant

Jun, H. et al., “A New Look at Viruses in Type 1 Diabetes”, *Diab Met Res.*, 19(1):8-31, (2003). cited by applicant

Jurewicz, A. et al., “MHC Class I-restricted Lysis of Human Oligodendrocytes by Myelin Basic Protein Peptide-Specific CD8 T Lymphocytes”, *J Immunol.*, 160(6):3056-9, (1998). cited by applicant

Kamanaka, M. et al., “Expression of interleukin-10 in Intestinal Lymphocytes Detected by an interleukin-10 Reporter Knockin Tiger Mouse”, *Immunity*, 25(6):941-52, (2006). cited by applicant

Kamikura, Y. et al., “Adhesion, Costimulatory Molecule, Trafficking, Homing: Cancer Immunotherapy and Costimulatory Molecule”, *Ann Rev., Immunity*, 162:2-13, (2004). cited by applicant

Kappos, L. et al., “Induction of a Non-Encephalitogenic Type 2 T Helper-Cell Autoimmune Response in Multiple Sclerosis After Administration of an Altered Peptide Ligand in a Placebo-Controlled, Randomized Phase II Trial. The Altered Peptide Ligand in Relapsing MS Study Group”, *Nat Med.*, 6(10):1176-82, (2000). cited by applicant

Karin, N. et al., “Reversal of Experimental Autoimmune Encephalomyelitis by a Soluble Peptide Variant of a Myelin Basic Protein Epitope: T Cell Receptor Antagonism and Reduction of Interferon Gamma and Tumor Necrosis Factor Alpha Production”, *J Exp Med.*, 180(6):2227-37, (1994). cited by applicant

Karlin, S. et al., “Applications and Statistics for Multiple High-Scoring Segments in Molecular

Sequences", *Proc Natl Acad Sci USA*, 90(12):5873-7, (1993). cited by applicant

Karounos, D. et al., "Metabolically Inactive Insulin Analog Prevents Type I Diabetes in Prediabetic NOD Mice", *J Clin Invest.*, 100(6):1344-8, (1997). cited by applicant

Kent, S. et al., "Expanded T Cells From Pancreatic Lymph Nodes of Type 1 Diabetic Subjects Recognize an Insulin Epitope", *Nature*, 435(7039):224-8, (2005). cited by applicant

Keymeulen, B. et al., "Insulin Needs After CD3-antibody Therapy in New-Onset Type 1 Diabetes", *N Engl J Med.*, 352(25):2598-608, (2005). cited by applicant

Kim, S. et al., "Induction and Visualization of Mucosal Memory CD8 T Cells Following Systemic Virus Infection", *J Immunol.*, 163(8):4125-32, (1999). cited by applicant

Kita, H. et al., "Quantitative and Functional Analysis of PDC-E2-specific Autoreactive Cytotoxic T Lymphocytes in Primary Biliary Cirrhosis", *J Clin Invest.*, 109(9):1231-40, (2002). cited by applicant

Komai-Koma, M. et al., "TLR2 Is Expressed on Activated T Cells as a Costimulatory Receptor", *Proc Natl Acad Sci USA*, 181(9):3829-34, (2004). cited by applicant

Longhi, M. et al., "Autoantigen-specific Regulatory T Cells, a Potential Tool for Immune-Tolerance Reconstitution in type-2 Autoimmune Hepatitis", *Hepatology*, 53(2):536-47, (2011). cited by applicant

Lowery, A. et al., "Immunonanoshells for Targeted Photothermal Ablation of Tumor Cells", *Int J Nanomedicine*, 1(2):149-54, (2006). cited by applicant

Lupas, A., "Coiled Coils: New Structures and New Functions", *TIB* 21, 375-82, (1996). cited by applicant

Ma, Z. et al., "TCR Triggering by pMHC Ligands Tethered on Surfaces via Poly(ethylene Glycol) Depends on Polymer Length", *PLoS One*, 9(11):e112292, (2014). cited by applicant

Mallone, R. et al., "CD8+ T-cell Responses Identify Beta-Cell Autoimmunity in Human Type 1 Diabetes", *Diabetes*, 56(3):613-21, (2007). cited by applicant

Mallone, R. et al., "T Cell Recognition of Autoantigens in Human Type 1 Diabetes: Clinical Perspectives", *Clin Dev Immunol.*, 2011:513210, (2011). cited by applicant

Marée, A. et al., "Modeling Competition Among Autoreactive CD8+ T Cells in Autoimmune Diabetes: Implications for Antigen-Specific Therapy", *Int Immunol.*, 18(7):1067-77, (2006). cited by applicant

Mars, L. et al., "CD8 T Cell Responses to Myelin Oligodendrocyte Glycoprotein-Derived Peptides in Humanized HLA-A*0201-transgenic Mice". *J Immunol.*, 179(8):5090-8, (2007). cited by applicant

Marsh, S. et al., "Nomenclature for Factors of the HLA System, Update Oct. 2010", *Tissue Antigens*, 77(4):349-54, (2011). cited by applicant

Martínez-Martín, N et al., "Cooperativity Between T Cell Receptor Complexes Revealed by Conformational Mutants of CD3epsilon", *Sci Signal.*, 2(83):ra43, (2009). cited by applicant

Marwaha, A. et al., "Cutting Edge: Increased IL-17-secreting T Cells in Children With New-Onset Type 1 Diabetes", *J Immunol.*, 185(7):3814-8, (2010). cited by applicant

Mazzarella, G., "Effector and Suppressor T Cells in Celiac Disease", *World J Gastroenterol.*, 21(24):7349-56, (2015). cited by applicant

McCarthy, D. et al., "Mouse Models of Multiple Sclerosis: Experimental Autoimmune Encephalomyelitis and Theiler's Virus-Induced Demyelinating Disease", *Methods Mol Biol.*, 900:281-401, (2012). cited by applicant

McClymont, S. et al., "Plasticity of Human Regulatory T Cells in Healthy Subjects and Patients With Type 1 Diabetes", *J Immunol.*, 186(7):3918-26, (2011). cited by applicant

McKeithan, T., "Kinetic Proofreading in T-cell Receptor Signal Transduction", *Proc Natl Acad Sci USA*, 92(11):5042-6, (1995). cited by applicant

McLarnon, A., "IBD: Regulatory T-cell Therapy Is a Safe and Well-Tolerated Potential Approach for Treating Refractory Crohn's Disease", *Nat Rev Gastroenterol Hepatol.*, 9(10):559, (2012). cited

by applicant

Mei, X. et al., Chemical Industry Press, Biotechnology Pharmaceutical Preparation: Foundation and Application: 199, (2004). cited by applicant

Merchant, A. et al., "An Efficient Route to Human Bispecific IgG", Nat Biotechnol., 16(7):677-81, (1998). cited by applicant

Mescher, M. et al., "Signals Required for Programming Effector and Memory Development by CD8+ T Cells", Immunol Rev., 211:81-92, (2006). cited by applicant

Mestas, J. et al., "Of Mice and Not Men: Differences Between Mouse and Human Immunology", J Immunol., 172(5):2731-8, (2004). cited by applicant

Metzler, B. et al., "Inhibition of Experimental Autoimmune Encephalomyelitis by Inhalation but Not Oral Administration of the Encephalitogenic Peptide: Influence of MHC Binding Affinity", Int Immunol., 5(9):1159-65, (1993). cited by applicant

Miguel-Sancho, N. et al., "Development of Stable, Water-Dispersible, and Biofunctionalizable Superparamagnetic Iron Oxide Nanoparticles", Chem Mat., 23(11):2795-802, (2011). cited by applicant

Miller, S. et al., "The Induction of Cell-Mediated Immunity and Tolerance With Protein Antigens Coupled to Syngeneic Lymphoid Cells", J Exp Med., 149(3):758-66, (1979). cited by applicant

Miyara, M. et al., "Functional Delineation and Differentiation Dynamics of Human CD4+ T Cells Expressing the FoxP3 Transcription Factor", Immunity, 30(6):899-911, (2009). cited by applicant

Mondini, S. et al., "One-Step Synthesis and Functionalization of Hydroxyl-Decorated Magnetite Nanoparticles", J Colloid Interface Sci., 322(1):173-9, (2008). cited by applicant

Moore, A. et al., "Tracking the Recruitment of Diabetogenic CD8+ T-cells to the Pancreas in Real Time", Diabetes, 53(6):1459-66, (2004). cited by applicant

Mukherjee, R. et al., "Identification of CD4+ T Cell-Specific Epitopes of Islet-Specific glucose-6-phosphatase Catalytic Subunit-Related Protein: A Novel Beta Cell Autoantigen in Type 1 Diabetes", J Immunol., 174(9):5306-15, (2005). cited by applicant

Musacchio, T. et al., "PEG-PE Micelles Loaded With Paclitaxel and Surface-Modified by a PBR-ligand: Synergistic Anticancer Effect", Mol Pharm., 6(2):468-79, (2009). cited by applicant

Nakayama, M. et al., "Prime Role for an Insulin Epitope in the Development of Type 1 Diabetes in NOD Mice", Nature, 435(7039):220-3, (2005). cited by applicant

Nelson, J. et al., "6 Types of Asthma and How They're Treated", Mother Nature Network, mnn.com, 4 pages, (2015). cited by applicant

Noren, C. et al., "A General Method for Site-Specific Incorporation of Unnatural Amino Acids Into Proteins", Science, 244(4901):182-8, (1989). cited by applicant

Oh, S. et al., "IL-15/IL-15R α -mediated Avidity Maturation of Memory CD8+ T Cells", Proc Natl Acad Sci USA, 101:15154-9, (2004). cited by applicant

Oleszak, E. et al., "Theiler's Virus Infection: A Model for Multiple Sclerosis," Clin Microbiol Rev., 17(1):174-207, (2004). cited by applicant

Onoda, T. et al., "Human CD4+ Central and Effector Memory T Cells Produce IL-21: Effect on Cytokine-Driven Proliferation of CD4+ T Cell Subsets", Int Immunol., 19(10):1191-9, (2007). cited by applicant

Ouyang, Q. et al., "Recognition of HLA Class I-restricted Beta-Cell Epitopes in Type 1 Diabetes", Diabetes, 55(11):3068-74, (2006). cited by applicant

Oyewumi, M. et al., Nano-microparticles as Immune Adjuvants: Correlating Particle Sizes and the Resultant Immune Responses, Expert Rev Vaccines, 9(9):1095-107, (2010). cited by applicant

Pachner, A., "Experimental Models of Multiple Sclerosis", Curr Opin Neurol., 24(3):291-9, (2011). cited by applicant

Packard, T. et al., "COPD Is Associated With Production of Autoantibodies to a Broad Spectrum of Self-Antigens, Correlative With Disease Phenotype", Immunol Res., 55(1-3):48-57, (2013). cited by applicant

Pakula, A. et al., "Genetic Analysis of Protein Stability and Function", *Annu Rev Genet.*, 23:289-310, (1989). cited by applicant

Palmer, J. et al., "Insulin Antibodies in Insulin-Dependent Diabetics Before Insulin Treatment", *Science*, 222(4630):1337-9, (1983). cited by applicant

Partch, R. et al., "Aerosol and Solution Modification of Particle-Polymer Interfaces", *J Adhesion*, 67(1-4):259-76, (1998). cited by applicant

Pascolo, S. et al., "HLA-A2.1-restricted Education and Cytolytic Activity of CD8(+) T Lymphocytes From beta2 Microglobulin (beta2m) HLA-A2.1 Monochain Transgenic H-2Db beta2m Double Knockout Mice", *J Exp Med.*, 185(12):2043-51, (1997). cited by applicant

Patel, J. et al., "Cationic Nanoparticles for Delivery of CpG Oligodeoxynucleotide and Ovalbumin: In Vitro and In Vivo Assessment", *J Biomed Nanotechnol.*, 3(1):97-106, (2007). cited by applicant

Pekarek, K. et al., "Double-walled Polymer Microspheres for Controlled Drug Release", *Nature*, 367(6460):258-60, (1994). cited by applicant

Peng, S. et al., "Synthesis and Characterization of Monodisperse Hollow Fe₃O₄ Nanoparticles", *Angew Chem Int Ed Engl.*, 46(22):4155-8, (2007). cited by applicant

Perrault, S. et al., "Mediating Tumor Targeting Efficiency of Nanoparticles Through Design", *Nano Lett*, 9(5):1909-15, (2009). cited by applicant

Petros, R. et al., Antibody Conjugation to PRINT Abstracts of Papers Am Chem Soc., 233(14):14, (2007). cited by applicant

Pinkse, G. et al., "Autoreactive CD8 T Cells Associated With Beta Cell Destruction in Type 1 Diabetes", *Proc Natl Acad Sci USA*, 102(51):18425-30, (2005). cited by applicant

Ponder, J. et al., "Tertiary Templates for Proteins. Use of Packing Criteria in the Enumeration of Allowed Sequences for Different Structural Classes", *J Mol Biol.*, 193(4):775-91, (1987). cited by applicant

Pot, C. et al., "Cutting Edge: IL-27 Induces the Transcription Factor c-Maf, Cytokine IL-21, and the Costimulatory Receptor ICOS That Coordinately Act Together to Promote Differentiation of IL-10- producing Tr1 Cells", *J Immunol.*, 183(2):797-801, (2009). cited by applicant

Purton, J. et al., "Antiviral CD4+ Memory T Cells Are IL-15 Dependent ," *J Exp Med.*, 204(4):951-61, (2007). cited by applicant

Quinn, M. et al., "How Do You Diagnose Rheumatoid Arthritis Early?", *Best Pract Res Clin Rheumatol.*, 15(1):49-66, (2001). cited by applicant

Ransohoff, R., "Animal Models of Multiple Sclerosis: The Good, the Bad and the Bottom Line", *Nat Neurosci.*, 15(8):1074-7, (2012). cited by applicant

Reijonen, H. et al., "Detection of GAD65-specific T-cells by Major Histocompatibility Complex Class II Tetramers in Type 1 Diabetic Patients and At-Risk Subjects", *Diabetes*, 51(5):1375-82, (2002). cited by applicant

Report to Congress, "Progress in Autoimmune Diseases Research", U.S. DHHS, NIH, 146 pages, (2005). cited by applicant

Riemekasten, G. et al., "Key Autoantigens in SLE", *Rheumatology (Oxford)*, 44(8):975-82, (2005). cited by applicant

Roncarolo, M. et al., "Clinical Tolerance in Allogeneic Hematopoietic Stem Cell Transplantation", *Immunol Rev.*, 241(1):145-63, (2011). cited by applicant

Rossi, M. et al., "Intravenous or Intranasal Administration of Gliadin Is Able to Down-Regulate the Specific Immune Response in Mice", *Scand J Immunol.*, 50(2):177-82, (1999). (1999). cited by applicant

Routsias, J. et al., "Autoimmune Response and Target Autoantigens in Sjogren's Syndrome", *Eur J Clin Invest.*, 40(11):1026-36, (2010). cited by applicant

Saengruengrit, C. et al., "The Combined Magnetic Field and Iron oxide-PLGA Composite Particles: Effective Protein Antigen Delivery and Immune Stimulation in Dendritic Cells", *J Colloid Interface Sci.*, 520:101-11, (2018). cited by applicant

Sahin, U. et al., "Human Neoplasms Elicit Multiple Specific Immune Responses in the Autologous Host", *Proc Natl Acad Sci USA*, 92(25):11810-3, (1995). cited by applicant

Sakaguchi, S. et al., "Foxp3+ CD25+ CD4+ Natural Regulatory T Cells in Dominant Self-Tolerance and Autoimmune Disease", *Immunol Rev.*, 212:8-27, (2006). cited by applicant

Santamaria, P. et al., "Beta-cell-cytotoxic CD8+ T Cells From Nonobese Diabetic Mice Use Highly Homologous T Cell Receptor Alpha-Chain CDR3 Sequences", *J Immunol.*, 154(5):2494-503, (1995). cited by applicant

Santamaria, P. et al., "Characterization of T Lymphocytes Infiltrating Human Pancreas Allograft Affected by Isletitis and Recurrent Diabetes", *Diabetes*, 41(1):53-61, (1992). cited by applicant

Santamaria, P. et al., "Skewed T-cell Receptor Usage and Junctional Heterogeneity Among Isletitis Alpha Beta and Gamma Delta T-cells in Human IDDM [Corrected]", *Diabetes*, 43(4):599-606, (1994). cited by applicant

Santamaria, P., "Effector Lymphocytes in Autoimmunity", *Curr Opin Immunol.*, 13(6):663-9, (2001). cited by applicant

Santamaria, P., "The Long and Winding Road to Understanding and Conquering Type 1 Diabetes", *Immunity*, 32(4):437-45, (2010). cited by applicant

Saragovi, H. et al., "Small Molecule and Protein-Based Neurotrophic Ligands: Agonists and Antagonists as Therapeutic Agents", *Exp Opin Ther Patents*, 9(6):737-51, (1999). cited by applicant

Saraiva, M. et al., "Interleukin-10 Production by Th1 Cells Requires interleukin-12-induced STAT4 Transcription Factor and ERK MAP Kinase Activation by High Antigen Dose", *Immunity*, 31(2):209-19, (2009). cited by applicant

Sato, "Diabetes and Cytokines. Roles of Cytokines in Diabetes Mellitus, Separate Volume", *J Clin Exp Med.*, cytokine-state of arts, 5 pages, (2004). cited by applicant

Sato, K. et al., "Marked Induction of c-Maf Protein During Th17 Cell Differentiation and Its Implication in Memory Th Cell Development", *J Biol Chem* . . . 286(17):14963-71, (2011). cited by applicant

Scaldaferri, F. et al., "Mucosal Biomarkers in Inflammatory Bowel Disease: Key Pathogenic Players or Disease Predictors?" *World J Gastroenterol.*, 16(21):2616-25, (2010). cited by applicant

Schamel, W. et al., "Organization of the Resting TCR in Nanoscale Oligomers ", *Immunol Rev.*, 251(1):13-20, (2013). cited by applicant

Schirle, M. et al., "Combining Computer Algorithms With Experimental Approaches Permits the Rapid and Accurate Identification of T Cell Epitopes From Defined Antigens", *J Immunol Meth.*, 257(1-2):1-16, (2001). cited by applicant

Schlesinger, S. et al., "Alphavirus Vectors for Gene Expression and Vaccines", *Curr Opin Biotechnol.*, 10(5):434-9, (Abstract only), (1999). cited by applicant

Schlichtholz, B. et al., The Immune Response to p53 in Breast Cancer Patients Is Directed Against Immunodominant Epitopes Unrelated to the Mutational Hot Spot, *Cancer Res.*, 52(22):6380-4, (1992). cited by applicant

Schneider, K. et al., "The End of the Era of Generosity? Global Health Amid Economic Crisis", *Philos Ethics Humanit Med.*, 4:1, (2009). cited by applicant

Schnütgen, F. et al., "A Directional Strategy for Monitoring Cre-mediated Recombination at the Cellular Level in the Mouse", *Nat Biotechnol.*, 21(5):562-6, (2003). cited by applicant

Schreiber, H. et al., "Using Carbon Magnetic Nanoparticles to Target, Track, and Manipulate Dendritic Cells", *J Immunol Methods*, 356(1-2):47-59, (2010). cited by applicant

Scott, R. et al., "Synthesis, Characterization, and Applications of Dendrimer-Encapsulated Nanoparticles", *J Phys Chem B.*, 109(2):692-704, (2005). cited by applicant

Serra, P. et al., "Nanoparticle-based Approaches to Immune Tolerance for the Treatment of Autoimmune Diseases", *Eur J Immunol.*, 48(5):751-6, (2018). cited by applicant

Serreze, D. et al., "Autoreactive Diabetogenic T-cells in NOD Mice Can Efficiently Expand From a

Greatly Reduced Precursor Pool”, *Diabetes*, 50(9):1992-2000, (2001). cited by applicant

Shanks, N. et al., “Are Animal Models Predictive for Humans?”, *Philos Ethics Humanit Med.*, 4(2):1-20, (2009). cited by applicant

Shao, K. et al., “Nanoparticle-Based Immunotherapy for Cancer”, *ACS Nano.*, 9(1):16-30, (2015). cited by applicant

Sherman, E. et al., “Functional Nanoscale Organization of Signaling Molecules Downstream of the T Cell Antigen Receptor”, *Immunity*, 35(5):705-20, (2011). cited by applicant

Shimoda, S. et al., “Identification and Precursor Frequency Analysis of a Common T Cell Epitope Motif in Mitochondrial Autoantigens in Primary Biliary Cirrhosis”, *J Clin Invest.*, 102(10):1831-40, (1998). cited by applicant

Shukla, S. et al., “Emerging Nanotechnologies for Cancer Immunotherapy”, *Exp Bioi Med* (Maywood)., 241(10):1116-26, (2016). cited by applicant

Sibley, R. et al., “Recurrent Diabetes Mellitus in the Pancreas Iso- And Allograft. A Light and Electron Microscopic and Immunohistochemical Analysis of Four Cases”, *Lab Invest.*, 53(2):132-44, (1985). cited by applicant

Singha, S. et al., “Peptide-MHC-based Nanomedicines for Autoimmunity Function as T-cell Receptor Microclustering Devices”, *Nat Nanotechnol.*, 12(7):701-10, (2017). cited by applicant

Sollid, L. et al., “Nomenclature and Listing of Celiac Disease Relevant Gluten T-cell Epitopes Restricted by HLA-DQ Molecules”, *Immunogenetics*, 64(6):455-60, (2012). cited by applicant

Somoza, N. et al., “Pancreas in Recent Onset Insulin-Dependent Diabetes Mellitus. Changes in HLA, Adhesion Molecules and Autoantigens, Restricted T Cell Receptor V Beta Usage, and Cytokine Profile”, *J Immunol.*, 153(3):1360-77, (1994). cited by applicant

Spada, F. et al., “Self-recognition of CD1 by Gamma/Delta T Cells: Implications for Innate Immunity”, *J Exp Med.*, 191(6):937-48, (2000). cited by applicant

Spensieri, F. et al., “Human Circulating influenza-CD4+ ICOS1+IL-21+ T Cells Expand After Vaccination, Exert Helper Function, and Predict Antibody Responses”, *Proc Natl Acad Sci USA*, 110(35):14330-5, (2013). cited by applicant

Sprent, J. et al., “T Cell Death and Memory”, *Science*, 293(5528):245-8, (2001). cited by applicant

Sprent, J. et al., “T Cell Memory”, *Annu Rev Immunol.*, 20:551-79, (2002). cited by applicant

Standifer, N. et al., “Identification of Novel HLA-A*0201-restricted Epitopes in Recent-Onset Type 1 Diabetic Subjects and Antibody-Positive Relatives”, *Diabetes*, 55(11):3061-7, (2006). cited by applicant

Steenblock, E. et al., “A Comprehensive Platform for Ex Vivo T-cell Expansion Based on Biodegradable Polymeric Artificial Antigen-Presenting Cells”, *Mol Ther.*, 16(4):765-72, (2008). (2008). cited by applicant

Stratmann, T. et al., “Susceptible MHC Alleles, Not Background Genes, Select an Autoimmune T Cell Reactivity”, *J Clin Invest.*, 112(6):902-14, (2003). cited by applicant

Stratmann, T. et al., “The I-Ag7 MHC Class II Molecule Linked to Murine Diabetes Is a Promiscuous Peptide Binder”, *J Immunol.*, 165(6):3214-25, (2000). cited by applicant

Sukhorukov, G. et al., Stepwise Polyelectrolyte Assembly on Particle Surfaces: A Novel Approach to Colloid Design, *Polymers Adv Tech.*, 9(10-11):759-67, (1998). cited by applicant

Sutton, I. et al., “Primary Biliary Cirrhosis: Seeking the Silent Partner of Autoimmunity”, *Gut*, 50(6):743-6, (2002). cited by applicant

Szczerkowska-Dobosz, A., “Human Leukocyte Antigens as Psoriasis Inheritance and Susceptibility Markers”, *Arch Immunol Ther Exp (Warsz.)*, 53(5):428-33, (2005). cited by applicant

Tait, B. et al., “HLA Antigens and Age at Diagnosis of Insulin-Dependent Diabetes Mellitus”, *Hum Immunol.*, 42(2):116-24, (1995). cited by applicant

Takahashi, F. et al., “Isolation and Characterization of a Colonic Autoantigen Specifically Recognized by Colon Tissue-Bound Immunoglobulin G From Idiopathic Ulcerative Colitis”, *J Clin Invest.*, 76(1):311-8, (1985). cited by applicant

Takaki, T. et al., "HLA-A*0201-restricted T Cells From Humanized NOD Mice Recognize Autoantigens of Potential Clinical Relevance to Type 1 Diabetes", *J Immunol.*, 176(5):3257-65, (2006). cited by applicant

Tan, J. et al., "Interleukin (IL)-15 and IL-7 Jointly Regulate Homeostatic Proliferation of Memory Phenotype CD8⁺ Cells but Are Not Required for Memory Phenotype CD4⁺ Cells", *J Exp Med.*, 195(12):1523-32, (2002). cited by applicant

Tanimura, K. et al., "B2-Glycoprotein I/HLA Class II Complexes Are Novel Autoantigens in Antiphospholipid Syndrome", *Blood*, 125(18):2835-44, (2015). cited by applicant

Tigges, M. et al., "Human Herpes Simplex Virus (HSV)-specific CD8⁺ CTL Clones Recognize HSV-2-infected Fibroblasts After Treatment With IFN-gamma or When Virion Host Shutoff Functions Are Disabled", *J Immunol.*, 156(10):3901-10, (1996). cited by applicant

Toes, R. et al., "Peptide Vaccination Can Lead to Enhanced Tumor Growth Through Specific T-cell Tolerance Induction", *Proc Natl Acad Sci USA*, 93(15):7855-60, (1996). cited by applicant

Toma, A. et al., "Recognition of a Subregion of Human Proinsulin by Class I-restricted T Cells in Type 1 Diabetic Patients", *Proc Natl Acad Sci USA*, 102(30):10581-6, (2005). cited by applicant

Trenttham, D. et al., "Effects of Oral Administration of Type II Collagen on Rheumatoid Arthritis", *Science*, 261(5129):1727-30, (1993). cited by applicant

Trudeau, J. et al., "Prediction of Spontaneous Autoimmune Diabetes in NOD Mice by Quantification of Autoreactive T Cells in Peripheral Blood", *J Clin Invest.*, 111(2):217-23, (2003). cited by applicant

Tsai, S. et al., "CD8⁺ T Cells in Type 1 Diabetes", *Adv Immunol.*, 100:79-124, (2008). cited by applicant

Tsai, S. et al., "Reversal of Autoimmunity by Boosting Memory-Like Autoregulatory T Cells", *Immunity*, 32(4):568-80, (2010). cited by applicant

Tsuchida, T. et al., "Autoreactive CD8⁺ T-cell Responses to Human Myelin Protein-Derived Peptides", *Proc Natl Acad Sci USA*, 91(23):10859-63, (1994). cited by applicant

Tufveson, G. et al., "New Immunosuppressants: Testing and Development in Animal Models and the Clinic: With Special Reference to DSG", *Immunological Rev.*, 136:99-109, (1993). cited by applicant

Ugel, S. et al., "In Vivo Administration of Artificial Antigen-Presenting Cells Activates Low-Avidity T Cells for Treatment of Cancer", *Cancer Res.*, 69(24):9376-84, (2009). cited by applicant

Unger, W. et al., "Human Clonal CD8 Autoreactivity to an IGRP Islet Epitope Shared Between Mice and Men", *Ann NY Acad Sci.*, 1103:192-5, (2007). cited by applicant

Vadasz, Z. et al., "B-regulatory Cells in Autoimmunity and Immune Mediated Inflammation", *FEBS Lett.*, 587(13):2074-8, (2013). cited by applicant

Vakil, R. et al., "Effect of Cholesterol on the Release of Amphotericin B From PEG-phospholipid Micelles", *Mol Pharm.*, 5(1):98-104, (2008). cited by applicant

Van Belle, T. et al., "Type 1 Diabetes: Etiology, Immunology, and Therapeutic Strategies", *Physiol Rev.*, 91(1):79-118, (2011). cited by applicant

Van Boekel, M. et al., "Autoantibody Systems in Rheumatoid Arthritis: Specificity, Sensitivity and Diagnostic Value", *Arthritis Res.*, 4(2):87-93, (2002). cited by applicant

Van Driel, I. et al., "Role of Regulatory T Cells in Gastrointestinal Inflammatory Disease", *J Gastroenterol Hepatol.*, 23(2):171-7, (2008). cited by applicant

Vandenbark, A. et al., "Recombinant TCR Ligand Induces Tolerance to Myelin Oligodendrocyte Glycoprotein 35-55 Peptide and Reverses Clinical and Histological Signs of Chronic Experimental Autoimmune Encephalomyelitis in HLA-DR2 Transgenic Mice", *J Immunol.*, 171(1):127-33, (2003). cited by applicant

Verdaguer, J. et al., "Acceleration of Spontaneous Diabetes in TCR-beta-transgenic Nonobese Diabetic Mice by Beta-Cell Cytotoxic CD8⁺ T Cells Expressing Identical Endogenous TCR-alpha Chains", *J Immunol.*, 157(10):4726-35, (1996). cited by applicant

Verdaguer, J. et al., "Spontaneous Autoimmune Diabetes in Monoclonal T Cell Nonobese Diabetic Mice", *J Exp Med.*, 186(10):1663-76, (1997). cited by applicant

Verdù, E. et al., "Oral Administration of Antigens From Intestinal Flora Anaerobic Bacteria Reduces the Severity of Experimental Acute Colitis in BALB/c Mice", *Clin Exp Immunol.*, 120(1):46-50, (2000). cited by applicant

Vincent, M. et al., "Understanding the Function of CD1-restricted T Cells", *Nat Immunol.*, 4(6):517-23, (2003). cited by applicant

Wainwright, S. et al., "HLA-F Is a Predominantly Empty, Intracellular, TAP-associated MHC Class Ib Protein With a Restricted Expression Pattern", *J Immunol.*, 164(1):319-28, (2000). cited by applicant

Walter, U. et al., "CD8+ T Cells in Autoimmunity", *Curr Opin Immunol.*, 17(6):624-31, (2005). cited by applicant

Wang, J. et al., "In Situ Recognition of Autoantigen as an Essential Gatekeeper in Autoimmune CD8+ T Cell Inflammation ", *Proc Natl Acad Sci USA*, 107(20):9317-22, (2010). cited by applicant

Wang, X. et al., "Induction of Potent CD8 T-Cell Responses by Novel Biodegradable Nanoparticles Carrying Human Immunodeficiency Virus Type 1 gp120," *J Virol.*, 81(18):10009-16, (2007). cited by applicant

Wang, Y. et al., "One-Pot Reaction to Synthesize Superparamagnetic Iron Oxide Nanoparticles by Adding Phenol as Reducing Agent and Stabilizer", *J Nanoparticle Res.*, 14:755, (2012). cited by applicant

Warnock, G. et al., "Normoglycaemia After Transplantation of Freshly Isolated and Cryopreserved Pancreatic Islets in Type 1 (Insulin-Dependent) Diabetes Mellitus ", *Diabetologia*, 34(1): 55-8, (1991). cited by applicant

Wei, C. et al., "Facile Synthesis of Superparamagnetic Magnetite Nanoparticles in Liquid Polyols", *J Colloid Interface Sci.*, 305(2):366-70, (2007). cited by applicant

Weiner, H. et al., "Double-blind Pilot Trial of Oral Tolerization With Myelin Antigens in Multiple Sclerosis", *Science*, 259(5099):1321-4, (1993). cited by applicant

Weiss, G. et al., "Covalent HLA-B27/peptide Complex Induced by Specific Recognition of an Aziridine Mimic of Arginine", *Proc Natl Acad Sci USA*, 93(20):10945-8, (1996). cited by applicant

Wekerle, H. et al., "Animal Models of Multiple Sclerosis", *Drug Discov Today: Disease Models*, 3(4):359-67, (2006). cited by applicant

Wen, Z., "Surface Effect of the Nanoparticles", *Intro Nat Sci.*, Nanjing University Press, pp. 373-374, (2007). cited by applicant

Williams, M. et al., "Developing and Maintaining Protective CD8+ Memory T Cells", *Immunol Rev.*, 211:146-53, (2006). cited by applicant

Wilson, J. et al., "pH-Responsive Nanoparticle Vaccines for Dual-Delivery of Antigens and Immunostimulatory Oligonucleotides", *ASC Nano.*, 7(5):3912-25, (2013). cited by applicant

Winer, S. et al., "Autoimmune Islet Destruction in Spontaneous Type 1 Diabetes Is Not Beta-Cell Exclusive", *Nat Med.*, 9(2):198-205, (2003). cited by applicant

Wong, F. et al., "Identification of an MHC Class I-restricted Autoantigen in Type 1 Diabetes by Screening an Organ-Specific cDNA Library", *Nat Med.*, 5(9):1026-31, (1999). cited by applicant

Wraith, D. et al., "Antigen Recognition in Autoimmune Encephalomyelitis and the Potential for Peptide-Mediated Immunotherapy", *Cell*, 59(2):247-55, (1989). cited by applicant

Wu, W. et al., "Magnetic Iron Oxide Nanoparticles: Synthesis and Surface Functionalization Strategies", *Nanoscale Res Lett.*, 3(11):397-415, (2008). cited by applicant

Wucherpfennig, K. et al., "Structural Basis for Major Histocompatibility Complex (MHC)-linked Susceptibility to Autoimmunity: Charged Residues of a Single MHC Binding Pocket Confer Selective Presentation of Self-Peptides in Pemphigus Vulgaris", *Proc Natl Acad Sci USA*, 92(25):11935-9, (1995). cited by applicant

Xie, J. et al., "Controlled PEGylation of Monodisperse Fe₃O₄ Nanoparticles for Reduced Non-Specific Uptake by Macrophage Cells", *Adv Mat.*, 19(20):3163-6, (2007). cited by applicant

Xie, J. et al., "One-Pot Synthesis of Monodisperse Iron Oxide Nanoparticles for Potential Biomedical Applications", *Pure Appl Chem.*, 78(5):1003-14, (2006). cited by applicant

Xu, C. et al., "Monodisperse Magnetic Nanoparticles for Biomedical Applications", *Polymer Int.*, 56(7):821-6, (2007). cited by applicant

Xu, H., "13.3.3 Relationship Between Gene Transduction and Nanoparticle Size", *Nano Med.*, 35S, 4 pages, (2004). cited by applicant

Xu, Z. et al., "Oleylamine as Both Reducing Agent and Stabilizer in a Facile Synthesis of Magnetite Nano Particles", *Chem Mat.*, 21:1778-80, (2009). cited by applicant

Yadav, V. et al., "Recombinant T-Cell Receptor Ligand (RTL) for Treatment of Multiple Sclerosis: A Double-Blind, Placebo-Controlled, Phase 1, Dose-Escalation Study", *Autoimmune Dis.*, 2012:954739, (2012). cited by applicant

Yamanouchi, J. et al., "Interleukin-2 Gene Variation Impairs Regulatory T Cell Function and Causes Autoimmunity", *Nat Genet.*, 39(3):329-37, (2007). cited by applicant

Yanaba, K. et al., "The Development and Function of Regulatory B Cells Expressing IL-10 (B10 Cells) Requires Antigen Receptor Diversity and TLR Signals", *J Immunol.*, 182(12):7459-72, (2009). cited by applicant

Yang, J. et al., "CD4⁺ T Cells From Type 1 Diabetic and Healthy Subjects Exhibit Different Thresholds of Activation to a Naturally Processed Proinsulin Epitope", *J Autoimmun.*, 31(1):30-41, (2008). cited by applicant

Yang, J. et al., "Islet-specific glucose-6-phosphatase Catalytic Subunit-Related Protein-Reactive CD4⁺ T Cells in Human Subjects", *J Immunol.*, 176(5):2781-9, (2006). cited by applicant

Yang, X. et al., "Opposing Regulation of the Locus Encoding IL-17 Through Direct, Reciprocal Actions of STAT3 and STAT5", *Nat Immunol.*, 12(3):247-54, (2011). cited by applicant

Yeste, A. et al., "Nanoparticle-mediated Codelivery of Myelin Antigen and a Tolerogenic Small Molecule Suppresses Experimental Autoimmune Encephalomyelitis", *Proc Natl Acad Sci USA*, 109(28):11270-5, (2012). cited by applicant

Ying, H. et al., "Cancer Therapy Using a Self-Replicating RNA Vaccine", *Nat Med.*, 5(7):823-7, (1999). cited by applicant

Yokosuka, T. et al., "Newly Generated T Cell Receptor Microclusters Initiate and Sustain T Cell Activation by Recruitment of Zap70 and SLP-76", *Nat Immunol.*, 6(12):1253-62, (2005). cited by applicant

Yoshida, K. et al., "Evidence for Shared Recognition of a Peptide Ligand by a Diverse Panel of Non-Obese Diabetic Mice-Derived, Islet-Specific, Diabetogenic T Cell Clones", *Int Immunol.*, 14(12):1439-47, (2002). cited by applicant

Yoshizaki, A. et al., "Regulatory B Cells Control T-cell Autoimmunity Through IL-21-dependent Cognate Interactions", *Nature*, 491(7423):264-8, (2012). cited by applicant

Yu, Y. et al., "Cutting Edge: Single-Chain Trimers of MHC Class I Molecules Form Stable Structures That Potently Stimulate Antigen-Specific T Cells and B Cells", *J Immunol.*, 168(7):3145-9, (2002). cited by applicant

Zajac, A. et al., "Viral Immune Evasion Due to Persistence of Activated T Cells Without Effector Function", *J Exp Med.*, 188(12):2205-13, (1998). cited by applicant

Zang, Y. et al., "Increased CD8⁺ Cytotoxic T Cell Responses to Myelin Basic Protein in Multiple Sclerosis", *J Immunol.*, 172(8):5120-7, (2004). cited by applicant

Zhang, S. et al., "HMGB1, an Innate Alarmin, in the Pathogenesis of Type 1 Diabetes", *Int J Clin Exp Pathol.*, 3(1):24-38, (2010). cited by applicant

Zhong, L. et al., "NSOM/QD-based Direct Visualization of CD3-induced and CD28-enhanced Nanospatial Coclustering of TCR and Coreceptor in Nanodomains in T Cell Activation", *PLoS One*, 4(6): e5945, (2009). cited by applicant

Zhou, L. et al., "Plasticity of CD4+ T Cell Lineage Differentiation", *Immunity*, 30(5):646-55, (2009). cited by applicant

Zhou, X. et al., "Instability of the Transcription Factor Foxp3 Leads to the Generation of Pathogenic Memory T Cells in Vivo", *Nat Immunol.*, 10(9):1000-7, (2009). cited by applicant

Zufferey, R. et al., "Woodchuck Hepatitis Virus Posttranscriptional Regulatory Element Enhances Expression of Transgenes Delivered by Retroviral Vectors", *J Virol.*, 73(4):2886-92, (1999). cited by applicant

AU Patent Application No. 2016203231; Examination Report No. 2, dated Nov. 30, 2017; 3 pages. cited by applicant

AU Patent Application No. 2016225913; Examination Report No. 1, dated Sep. 22, 2017; 4 pages. cited by applicant

Bibliographic data page from EPO website at espace.net.com/publication_Details/biblio?CC=WO&NR=2004078909A2&KC=..., downloaded Nov. 15, 2010, showing that W02004078909 was also published as US2007154953: 1 page total. cited by applicant

CA Patent Application No. 2,817,710; Office Action, dated Oct. 19, 2017; 4 pages. cited by applicant

CN Patent Application No. 201380022126.2; Office Action No. 4, dated Jul. 24, 2017; 2 pages. cited by applicant

Japan Intractable Diseases Information Center, "Crohn's Disease," retrieved from nanbyou.or.jp/entry/111, (2015). cited by applicant

JP Patent Application No. 2015-536240; Office Action, dated Aug. 16, 2017; 6 pages. cited by applicant

JP Patent Application No. 2016-159414; Office Action, dated Oct. 30, 2017; 4 pages. cited by applicant

JP Patent Application No. 2017-014194; Office Action, dated Sep. 13, 2017; 6 pages. cited by applicant

MX Patent Application No. MX/a/2013/003559; Office Action No. 3, dated Jul. 3, 2017; 5 pages. cited by applicant

MX Patent Application No. MX/a/2014/011623; Office Action, dated Jul. 28, 2017; 10 pages. cited by applicant

NZ Patent Application No. 706970; First Examination Report, dated Nov. 8, 2017; 4 pages. cited by applicant

RU Patent Application No. 2014141984; Second Office Action, dated Nov. 23, 2017; 6 pages. cited by applicant

RU Patent Application No. 2015116509; Office Action, dated Nov. 29, 2017; 9 pages. cited by applicant

Shaefer, W. et al., "Heavy and light chain pairing of bivalent quadroma and knobs-into-holes antibodies analyzed by UHR-ESI-QTOF mass spectrometry", *MAbs.*, 8(1):49-55, (2015). cited by applicant

U.S. Appl. No. 12/044,435; Applicant-Initiated Interview Summary, dated Dec. 13, 2012; 4 pages. cited by applicant

U.S. Appl. No. 12/044,435; Applicant-Initiated Interview Summary, dated Sep. 12, 2012; 6 pages. cited by applicant

U.S. Appl. No. 12/044,435; Final Office Action, dated Jun. 8, 2011; 24 pages. cited by applicant

U.S. Appl. No. 12/044,435; Non-Final Office Action, dated May 2, 2012; 12 pages. cited by applicant

U.S. Appl. No. 12/044,435; Non-Final Office Action, dated Nov. 24, 2010; 39 pages. cited by applicant

U.S. Appl. No. 12/044,435; Notice of Allowance, dated Sep. 12, 2012; 8 pages. cited by applicant

U.S. Appl. No. 12/848,055; Final Office Action, dated Aug. 23, 2012; 13 pages. cited by applicant

U.S. Appl. No. 12/848,055; Final Office Action, dated Dec. 24, 2014; 19 pages. cited by applicant

U.S. Appl. No. 12/848,055; Final Office Action, dated Jul. 12, 2013; 17 pages. cited by applicant

U.S. Appl. No. 12/848,055; Non-Final Office Action, dated Apr. 4, 2012; 22 pages. cited by applicant

U.S. Appl. No. 12/848,055; Non-Final Office Action, dated Dec. 19, 2012; 14 pages. cited by applicant

U.S. Appl. No. 12/848,055; Non-Final Office Action, dated Jun. 6, 2014; 24 pages. cited by applicant

U.S. Appl. No. 12/848,055; Non-Final Office Action, dated May 13, 2016; 27 pages. cited by applicant

U.S. Appl. No. 13/249,105; 1.132 Declaration, dated Jun. 2, 2017; 26 pages. cited by applicant

U.S. Appl. No. 13/249,105; 1.132 Declaration, dated Sep. 4, 2015; 13 pages. cited by applicant

U.S. Appl. No. 13/249,105; Final Office Action, dated Apr. 11, 2018; 32 pages. cited by applicant

U.S. Appl. No. 13/249,105; Final Office Action, dated Nov. 30, 2015; 20 pages. cited by applicant

U.S. Appl. No. 13/249,105; Non-Final Office Action, dated Apr. 3, 2015; 31 pages. cited by applicant

U.S. Appl. No. 13/249,105; Non-Final Office Action, dated Sep. 8, 2017; 16 pages. cited by applicant

U.S. Appl. No. 13/294,109; 1.132 Declaration, dated Nov. 5, 2015; 21 pages. cited by applicant

U.S. Appl. No. 13/294,109; Examiner-Initiated Interview Summary, dated Apr. 4, 2016; 3 pages. cited by applicant

U.S. Appl. No. 13/294,109; Final Office Action, dated Jan. 12, 2015; 13 pages. cited by applicant

U.S. Appl. No. 13/294,109; Final Office Action, dated Nov. 13, 2013; 14 pages. cited by applicant

U.S. Appl. No. 13/294,109; Non-Final Office Action, dated Jun. 4, 2013; 23 pages. cited by applicant

U.S. Appl. No. 13/294,109; Notice of Allowance, dated Apr. 4, 2016; 15 pages. cited by applicant

U.S. Appl. No. 13/294,109; Notice of Allowance, dated Sep. 29, 2016; 31 pages. cited by applicant

U.S. Appl. No. 13/294,109; Notice of Appeal, dated Jul. 10, 2015; 4 pages. cited by applicant

U.S. Appl. No. 13/712,832; Non-Final Office Action, dated Feb. 27, 2015; 34 pages. cited by applicant

U.S. Appl. No. 13/830,521; 1.132 Declaration, dated Dec. 24, 2014; 8 pages. cited by applicant

U.S. Appl. No. 13/830,521; Final Office Action, dated Mar. 5, 2015; 15 pages. cited by applicant

U.S. Appl. No. 13/830,521; Non-Final Office Action, dated Jul. 25, 2014; 36 pages. cited by applicant

U.S. Appl. No. 13/830,521; Non-Final Office Action, dated Jun. 28, 2016; 13 pages. cited by applicant

U.S. Appl. No. 13/842,302; 1.132 Declaration, dated Feb. 17, 2016; 46 pages. cited by applicant

U.S. Appl. No. 13/842,302; 1.132 Declaration, dated Feb. 23, 2016; 2 pages. cited by applicant

U.S. Appl. No. 13/842,302; Examiner-Initiated Interview Summary, dated Oct. 22, 2020; 1 page. cited by applicant

U.S. Appl. No. 13/842,302; Final Office Action, dated Dec. 28, 2018; 21 pages. cited by applicant

U.S. Appl. No. 13/842,302; Final Office Action, dated Feb. 18, 2015; 25 pages. cited by applicant

U.S. Appl. No. 13/842,302; Final Office Action, dated May 3, 2017; 31 pages. cited by applicant

U.S. Appl. No. 13/842,302; Non-Final Office Action, dated Apr. 29, 2020; 22 pages. cited by applicant

U.S. Appl. No. 13/842,302; Non-Final Office Action, dated Apr. 30, 2014; 29 pages. cited by applicant

U.S. Appl. No. 13/842,302; Non-Final Office Action, dated Apr. 30, 2018; 46 pages. cited by applicant

U.S. Appl. No. 13/842,302; Non-Final Office Action, dated Jul. 6, 2016; 28 pages. cited by applicant

applicant

U.S. Appl. No. 13/842,302; Notice of Allowance, dated Oct. 22, 2020; 31 pages. cited by applicant

U.S. Appl. No. 14/531,707; Examiner-Initiated Interview Summary, dated Jun. 18, 2018; 1 page.

cited by applicant

U.S. Appl. No. 14/531,707; Non-Final Office Action, dated Oct. 3, 2017; 14 pages. cited by applicant

U.S. Appl. No. 14/531,707; Notice of Allowance, dated Jul. 20, 2018; 4 pages. cited by applicant

U.S. Appl. No. 14/531,707; Notice of Allowance, dated Jun. 18, 2018; 58 pages. cited by applicant

U.S. Appl. No. 14/684,153; Non-Final Office Action, dated Jun. 30, 2016; 32 pages. cited by applicant

U.S. Appl. No. 14/684,153; Notice of Allowance, dated Jan. 26, 2017; 7 pages. cited by applicant

U.S. Appl. No. 15/348,959; Applicant-Initiated Interview Summary, dated Aug. 30, 2017; 2 pages. cited by applicant

U.S. Appl. No. 15/348,959; Applicant-Initiated Interview Summary, dated Jul. 3, 2019; 2 pages. cited by applicant

U.S. Appl. No. 15/348,959; Examiner-Initiated Interview Summary, dated Jul. 3, 2019; 2 pages. cited by applicant

U.S. Appl. No. 15/348,959; Final Office Action, dated Jan. 12, 2018; 30 pages. cited by applicant

U.S. Appl. No. 15/348,959; First-Action Interview Office Action, dated Aug. 30, 2017; 25 pages. cited by applicant

U.S. Appl. No. 15/348,959; Non-Final Office Action, dated Feb. 21, 2019; 7 pages. cited by applicant

U.S. Appl. No. 15/348,959; Notice of Allowance, dated Jul. 3, 2019; 12 pages. cited by applicant

U.S. Appl. No. 15/348,959; Pre-Interview First Office Action, dated Apr. 13, 2017; 20 pages. cited by applicant

U.S. Appl. No. 15/433,898; Non-Final Office Action, dated Sep. 28, 2017; 31 pages. cited by applicant

U.S. Appl. No. 15/433,898; Notice of Allowance, dated May 17, 2018; 34 pages. cited by applicant

U.S. Appl. No. 15/572,137; Final Office Action, dated Feb. 8, 2022; 17 pages. cited by applicant

U.S. Appl. No. 15/572,137; Final Office Action, dated Feb. 25, 2021; 23 pages. cited by applicant

U.S. Appl. No. 15/572,137; Non-Final Office Action, dated Jul. 23, 2021; 16 pages. cited by applicant

U.S. Appl. No. 15/572,137; Non-Final Office Action, dated Sep. 28, 2022; 26 pages. cited by applicant

U.S. Appl. No. 15/610,550; Applicant-Initiated Interview Summary, date of interview Mar. 25, 2020; 3 pages. cited by applicant

U.S. Appl. No. 15/610,550; Final Office Action, dated Jun. 19, 2020; 16 pages. cited by applicant

U.S. Appl. No. 15/610,550; Non-Final Office Action, dated Nov. 4, 2019; 50 pages. cited by applicant

U.S. Appl. No. 15/807,415; Final Office Action, dated Aug. 30, 2022; 26 pages. cited by applicant

U.S. Appl. No. 15/807,415; Final Office Action, dated Dec. 30, 2020; 33 pages. cited by applicant

U.S. Appl. No. 15/807,415; Non-Final Office Action, dated Aug. 19, 2021; 17 pages. cited by applicant

U.S. Appl. No. 15/807,415; Non-Final Office Action, dated Jul. 29, 2020; 74 pages. cited by applicant

U.S. Appl. No. 15/999,192; Non-Final Office Action, dated Apr. 15, 2020; 71 pages. cited by applicant

U.S. Appl. No. 15/999,192; Notice of Allowance, dated Oct. 13, 2020; 15 pages. cited by applicant

U.S. Appl. No. 16/132,000; Examiner-Initiated Interview Summary, dated Jan. 7, 2022; 1 page. cited by applicant

U.S. Appl. No. 16/132,000; Notice of Allowance, dated Jan. 21, 2022; 15 pages. cited by applicant

U.S. Appl. No. 16/156,326; Final Office Action, dated Apr. 20, 2022; 40 pages. cited by applicant

U.S. Appl. No. 16/200,199; Non-Final Office Action, dated Aug. 27, 2020; 54 pages. cited by applicant

U.S. Appl. No. 16/200,199; Notice of Allowance, dated Jan. 13, 2021; 11 pages. cited by applicant

U.S. Appl. No. 16/603,180; Final Office Action, dated Mar. 1, 2023; 29 pages. cited by applicant

U.S. Appl. No. 16/603,180; Non-Final Office Action, dated Jul. 19, 2022; 54 pages. cited by applicant

U.S. Appl. No. 17/153,212; Applicant-Initiated Interview Summary, dated Aug. 6, 2024; 2 pages. cited by applicant

U.S. Appl. No. 18/048,339; Application as filed, dated Oct. 20, 2022; 83 pages. cited by applicant

Betts, R. et al., "CDS(+) T Cells in Asthma: Friend or Foe?", *Pharmacol Ther.*, 121(2):123-31, (2009). cited by applicant

Bianchi, A. et al., "Parenteral Vaccination of Mice and Piglets With F4+ *Escherichia coli* Suppresses the Enteric anti-F4 Response Upon Oral Infection", *Vaccine*, 14(3):199-206, (1996). cited by applicant

Bibliographic data page from EPO website at https://www.espace.net.com/publication_Details/biblio?CC=WO&NR=2004078909A2&KC=..., downloaded Nov. 15, 2010, showing that W02004078909 was also published as U.S. Pat. No. 2007154953: 1 page total. cited by applicant

Bielekova, B. et al., "Encephalitogenic Potential of the Myelin Basic Protein Peptide (Amino Acids 83-99) in Multiple Sclerosis: Results of a Phase II Clinical Trial With an Altered Peptide Ligand", *Nat Med.*, 6(10):1167-75, (2000). cited by applicant

Blancou, P. et al., "Immunization of HLA Class I Transgenic Mice Identifies Autoantigenic Epitopes Eliciting Dominant Responses in Type 1 Diabetes Patients", *J Immunol.*, 178(11):7458-66, (2007). cited by applicant

Bossuyt, X. et al., "Serologic Markers in Inflammatory Bowel Disease", *Clin Chem.*, 52(2):171-81, (2006). cited by applicant

Bottazzo, G. et al., "In Situ Characterization of Autoimmune Phenomena and Expression of HLA Molecules in the Pancreas in Diabetic Insulitis", *N Engl J Med.*, 313(6):353-60, (1985). cited by applicant

Bottini, M. et al., "Luminescent Silica Nanobeads: Characterization and Evaluation as Efficient Cytoplasmatic Transporters for T-lymphocytes", *J Am Chem Soc.*, 129(25):7814-23, (2007). cited by applicant

Bour-Jordan, H. et al., "B Cell Depletion: A Novel Therapy for Autoimmune Diabetes?", *J Clin Invest.*, 117(12):3642-5, (2007). cited by applicant

Braud, V. et al., "Functions of Nonclassical MHC and non-MHC-encoded Class I Molecules", *Curr Opin Immunol.*, 11(1):100-8, (1999). cited by applicant

Buenafe, A. et al., "Regulatory T Cells Play a Role in T-cell Receptor CDR2 Peptide Regulation of Experimental Autoimmune Encephalomyelitis", *Immunology*, 135(2):168-79, (2012). cited by applicant

Bunnell, S. et al., "T Cell Receptor Ligation Induces the Formation of Dynamically Regulated Signaling Assemblies", *J Cell Biol.*, 158(7):1263-75, (2002). cited by applicant

Burke, R. et al., "The Influence of Adjuvant on the Therapeutic Efficacy of a Recombinant Genital Herpes Vaccine", *J Inf Dis.*, 170(5):1110-9, (1994). cited by applicant

Burton, B. et al., "Sequential Transcriptional Changes Dictate Safe and Effective Antigen-Specific Immunotherapy", *Nat Commun.*, 5:4741, (2014). cited by applicant

Cao, K. et al., "Analysis of the Frequencies of HLA-A, B, and C Alleles and Haplotypes in the Five Major Ethnic Groups of the United States Reveals High Levels of Diversity in These Loci and Contrasting Distribution Patterns in These Populations", *Hum Immunol.*, 62(9):1009-30, (2001).

cited by applicant

Caruso, F. et al., "Investigation of Electrostatic Interactions in Polyelectrolyte Multilayer Films: Binding of Anionic Fluorescent Probes to Layers Assembled onto Colloids", *Macromol.*, 32(7):2317-28, (1999). cited by applicant

Caruso, F. et al., "Protein Multilayer Formation on Colloids Through a Stepwise Self-Assembly Technique", *J Am Chem Soc.*, 121(25):6039-46, (1999). cited by applicant

Chang, J. et al., "Design, Engineering, and Production of Human Recombinant T Cell Receptor Ligands Derived From Human Leukocyte Antigen DR2", *J Biol Chem.*, 276(26):24170-6, (2001). cited by applicant

Chatenoud, L., "Do NKT Cells Control Autoimmunity?", *J Clin Invest.*, 110(6):747-8, (2002). cited by applicant

Dieterich, W. et al., "Identification of Tissue Transglutaminase as the Autoantigen of Celiac Disease", *Nat Med.*, 3(7):797-801, (1997). cited by applicant

Dilorenzo, T. et al., "Major Histocompatibility Complex Class I-restricted T Cells Are Required for All but the End Stages of Diabetes Development in Nonobese Diabetic Mice and Use a Prevalent T Cell Receptor Alpha Chain Gene Rearrangement", *Proc Natl Acad Sci USA*, 95(21):12538-43, (1998). cited by applicant

Diwan, M. et al., "Biodegradable Nanoparticle Mediated Antigen Delivery to Human Cord Blood Derived Dendritic Cells for Induction of Primary T Cell Responses", *J Drug Target*, 11(8-10):495-507, (2003). cited by applicant

Dominguez, A. et al., "Targeting the Tumor Microenvironment With anti-neu/anti-CD40 Conjugated Nanoparticles for the Induction of Antitumor Immune Responses", *Vaccine*, 28(5):1383-90, (2010). cited by applicant

Dranoff, G., "Targets of Protective Tumor Immunity", *Ann NY Acad Sci.*, 1174:74-80, (2009). cited by applicant

Dressel, A. et al., "Autoantigen Recognition by Human CD8 T Cell Clones: Enhanced Agonist Response Induced by Altered Peptide Ligands", *J Immunol.*, 159(10):4943-51, (1997). cited by applicant

Edelman, G. et al., "The Covalent Structure of an Entire gammaG Immunoglobulin Molecule", *Proc Natl Acad Sci USA*, 63(1):78-85, (1969). cited by applicant

Eggna, M. et al., "Identification of Histone H1 as a Cognate Antigen of the Ulcerative Colitis-Associated Marker Antibody pANCA", *J Autoimmun.*, 14(1):83-97, (2000). cited by applicant

Ellman, J. et al., "Biosynthetic Method for Introducing Unnatural Amino Acids Site-Specifically Into Proteins", *Methods Enzymol.*, 202:301-36, (1991). cited by applicant

Fennessy, M. et al., "A Gene in the HLA Class I Region Contributes to Susceptibility to IDDM in the Finnish Population. Childhood Diabetes in Finland (DiMe) Study Group", *Diabetologia.*, 37(9):937-45, (1994). cited by applicant

Fifis, T. et al., "Short Peptide Sequences Containing MHC Class I and/or Class II Epitopes Linked to Nano-Beads Induce Strong Immunity and Inhibition of Growth of Antigen-Specific Tumour Challenge in Mice", *Vaccine*, 23(2):258-66, (2004). cited by applicant

Firestein, G., "Evolving Concepts of Rheumatoid Arthritis", *Nature*, 423(6937):356-61, (2003). cited by applicant

Flad, T. et al., "Development of an MHC-class I Peptide Selection Assay Combining Nanoparticle Technology and Matrix-Assisted Laser Desorption/Ionisation Mass Spectrometry", *J Immunol Methods*, 283(1-2):205-13, (2003). cited by applicant

Frankel, A. et al., "Characterization of Diphtheria Fusion Proteins Targeted to the Human interleukin-3 Receptor", *Protein Eng.*, 13(8):575-81, (2000). cited by applicant

Friedman, A. et al., "The Smart Targeting of Nanoparticles", *Curr Pharm Des.*, 19(35):6315-29, (2013). cited by applicant

Gagliani, N. et al., "Coexpression of CD49b and LAG-3 Identifies Human and Mouse T

Regulatory Type 1 Cells”, *Nat Med.*, 19(6):739-46, (2013). cited by applicant

Garboczi, D. et al., “HLA-A2-peptide Complexes: Refolding and Crystallization of Molecules Expressed in *Escherichia coli* and Complexed With Single Antigenic Peptides”, *Proc Natl Acad Sci USA*, 89(8):3429-33, (1992). cited by applicant

GenBank accession No. NM_001008228.2, accessed at <https://www.ncbi.nlm.nih.gov/nuccore/281371473/?report=genbank> on Jan. 16, 2018, 1 page. cited by applicant

Getts, D. et al., “Microparticles Bearing Encephalitogenic Peptides Induce T-cell Tolerance and Ameliorate Experimental Autoimmune Encephalomyelitis”, *Nat Biotechnol.*, 30(12):1217-24, (2012). cited by applicant

Gil, D. et al., “Recruitment of Nck by CD3 Epsilon Reveals a Ligand-Induced Conformational Change Essential for T Cell Receptor Signaling and Synapse Formation”, *Cell*, 109(7):901-12, (2002). cited by applicant

Gill, R. et al., “Characterization of Primary T Cell Subsets Mediating Rejection of Pancreatic Islet Grafts”, *J Immunol.*, 143(7):2176-8, (1989). cited by applicant

Gimmi, C. et al., “Human T-cell Clonal Anergy Is Induced by Antigen Presentation in the Absence of B7 Costimulation”, *Proc Natl Acad Sci USA*, 90(14):6586-90, (1993). cited by applicant

Giuliani, F. et al., “Additive Effect of the Combination of Glatiramer Acetate and Minocycline in a Model of MS”, *J Neuroimmunol.*, 158(1-2):213-21, (2005). cited by applicant

Gold, R. et al., “Understanding Pathogenesis and Therapy of Multiple Sclerosis via Animal Models: 70 Years of Merits and Culprits in Experimental Autoimmune Encephalomyelitis Research”, *Brain*, 129(Pt 8):1953-71, (2006). cited by applicant

Golman, B. et al., “Fine Particle Coating by Chemical Vapor Deposition for Functional Materials”, *Trends Chem Engin.*, 6:1-16, (2000). cited by applicant

Gong, W. et al., “Immobilized MHC Class I Chain-Related Protein A Synergizes With IL-15 and Soluble 4-1BB Ligand to Expand NK Cells With High Cytotoxicity Ex Vivo”, *Cell Mol Immunol.*, 7(6):477-84, (2010). cited by applicant

Gregori, S. et al., “Re-establishing Immune Tolerance in Type 1 Diabetes via Regulatory T Cells”, *Novartis Found Symp.*, 292:174-83, (2008), (abstract), Abstract only. cited by applicant

Guarda, G. et al., “L-selectin-negative CCR7- Effector and Memory CD8+ T Cells Enter Reactive Lymph Nodes and Kill Dendritic Cells”, *Nat Immunol.*, 8(7):743-52, (2007). cited by applicant

Guidance for Industry, “Estimating the Maximum Safe Starting Dose in Initial Clinical Trials for Therapeutics in Adult Healthy Volunteers”, U.S. DHHS, FDA, CDER, 27 pages, (2005). cited by applicant

Gunn, J. et al., “A Multimodal Targeting Nanoparticle for Selectively Labeling T Cells”, *Small*, 4(6):712-5, (2008). cited by applicant

Guo, H. et al., “Protein Tolerance to Random Amino Acid Change,” *Proc Natl Acad Sci USA*, 101(25):9205-10, (2004). cited by applicant

Gupta, A. et al., “Synthesis and Surface Engineering of Iron Oxide Nanoparticles for Biomedical Applications”, *Biomaterials*, 26(18):3995-4021, (2005). cited by applicant

Ha, J. et al., “Immunoglobulin Fc Heterodimer Platform Technology: From Design to Applications in Therapeutic Antibodies and Proteins”, *Front Immunol.*, 7(394):1-16, (2016). cited by applicant

Hale, J. et al., “Distinct Memory CD4+ T Cells With Commitment to T Follicular Helper- And T Helper 1-cell Lineages Are Generated After Acute Viral Infection”, *Immunity*, 38(4):805-17, (2013). cited by applicant

Hall, C. et al., “Mapping Labeled Sites in *Escherichia coli* Ribosomal RNA: Distribution of Methyl Groups and Identification of a Photoaffinity-Labeled RNA Region Putatively at the Peptidyltransferase Center”, *Biochemistry*, 24(21):5702-11, (1985). cited by applicant

Hamilton-Williams, E. et al., “Transgenic Rescue Implicates beta2-microglobulin as a Diabetes Susceptibility Gene in Nonobese Diabetic (NOD) Mice”, *Proc Natl Acad Sci USA*, 98(20):11533-

8, (2001). cited by applicant

Han, B. et al., “Developmental Control of CD8 T Cell-Avidity Maturation in Autoimmune Diabetes”, *J Clin Invest.*, 115(7):1879-87, (2005). cited by applicant

Han, B. et al., “Prevention of Diabetes by Manipulation of anti-IGRP Autoimmunity: High Efficiency of a Low-Affinity Peptide”, *Nat Med.*, 11(6):645-52, (2005). cited by applicant

Han, G. et al., “Interleukin-17-Producing Gammadelta+ T Cells Protect NOD Mice from Type 1 Diabetes Through a Mechanism Involving Transforming Growth Factor-Beta”, *Immunology*, 129(2):197-206, (2010). cited by applicant

Hanprasopwattana, A. et al., “Titania Coatings on Monodisperse Silica Spheres (Characterization Using 2-Propanol Dehydration and TEM)”, *Langmuir*, 12(13):3173-9, (1996). cited by applicant

Harris, S. et al., “Prediction of Murine MHC Class I Epitopes in a Major House Dust Mite Allergen and Induction of T1-type CD8+ T Cell Responses”, *Int Immunol.*, 9(2):273-80, (1997). cited by applicant

Hassainya, Y. et al., “Identification of Naturally Processed HLA-A2—restricted Proinsulin Epitopes by Reverse Immunology”, *Diabetes*, 54(7):2053-9, (2006). cited by applicant

Herold, K. et al., “Anti-CD3 Monoclonal Antibody in New-Onset Type 1 Diabetes Mellitus”, *N Eng J Med.*, 346(22):1692-8, (2002). cited by applicant

Hirsch, D. et al., “Antigen-based Immunotherapy for Autoimmune Disease: Current Status”, *Immunotargets Ther.*, 4:1-11, (2014). cited by applicant

Hirschfield, G. et al., “The Immunobiology and Pathophysiology of Primary Biliary Cirrhosis”, *Annu Rev Pathol.*, 8:303-30, (2013). cited by applicant

HLA Nomenclature (2015) (Year: 2015), 2 pages. cited by applicant

Ho, K. et al., “The Clinical Relevance of Autoantibodies in Scleroderma”, *Arthritis Res Ther.*, 5(2):80-93, (2003). cited by applicant

Holgate, S. et al., “Treatment Strategies for Allergy and Asthma”, *Nat Rev Immunol.*, 8(3):218-30, (2008). cited by applicant

Holst, J. et al., “Generation of T-cell Receptor Retrogenic Mice”, *Nat Protoc.*, 1(1):406-17, (2006). cited by applicant

Honeyman, M. et al., “Analysis of Families at Risk for Insulin-Dependent Diabetes Mellitus Reveals That HLA Antigens Influence Progression to Clinical Disease”, *Mol Med.*, 1(5):576-82, (1995). cited by applicant

Hugues, S. et al., “Generation and use of alternative multimers of peptide/MHC complexes”, *J Immunol Methods*, 268(1):83-92, (2002). cited by applicant

Huppa, J. et al., “TCR-peptide-MHC Interactions in Situ Show Accelerated Kinetics and Increased Affinity”, *Nature*, 463(7283):963-7, (2010). cited by applicant

International Application No. PCT/EP2011/066994; International Preliminary Report on Patentability, date of issuance Apr. 2, 2013; 5 pages. cited by applicant

International Application No. PCT/EP2011/066994; International Search Report and Written Opinion of the International Searching Authority, date of mailing Nov. 21, 2011; 7 pages. cited by applicant

International Application No. PCT/1B2013/052352; International Preliminary Report on Patentability, date of issuance Oct. 1, 2014; 6 pages. cited by applicant

International Application No. PCT/1B2013/052352; International Search Report and Written Opinion of the International Searching Authority, date of mailing Oct. 2, 2013; 9 pages. cited by applicant

International Application No. PCT/EP2011 /069931; International Preliminary Report on Patentability (Ch. 2), date of issuance May 16, 2013; 21 pages. cited by applicant

International Application No. PCT/EP2011 /069931; International Search Report and Written Opinion of the International Searching Authority, date of mailing Jul. 10, 2012; 24 pages. cited by applicant

International Application No. PCT/IB2013/003033; International Preliminary Report on Patentability, date of issuance Apr. 14, 2015; 6 pages. cited by applicant

International Application No. PCT/IB2013/003033; International Search Report and Written Opinion of the International Searching Authority, date of mailing Jul. 14, 2014; 9 pages. cited by applicant

International Application No. PCT/IB2014/003014; International Preliminary Report on Patentability, date of issuance May 10, 2016; 9 pages. cited by applicant

International Application No. PCT/IB2014/003014; International Search Report and Written Opinion of the International Searching Authority, date of mailing May 12, 2015; 14 pages. cited by applicant

International Application No. PCT/IB2016/000691; International Preliminary Report on Patentability, date of issuance Nov. 7, 2017; 31 pages. cited by applicant

International Application No. PCT/IB2016/000691; International Search Report and Written Opinion of the International Searching Authority, date of mailing Mar. 7, 2017; 44 pages. cited by applicant

International Application No. PCT/IB2017/001508; International Search Report and Written Opinion of the International Searching Authority, date of mailing Mar. 26, 2018; 13 pages. cited by applicant

International Application No. PCT/IB2017/001508; International Preliminary Report on Patentability, date of issuance May 14, 2019; 9 pages. cited by applicant

International Application No. PCT/IB2018/000510; International Preliminary Report on Patentability, date of issuance Oct. 8, 2019; 4 pages. cited by applicant

International Application No. PCT/IB2018/000510; International Search Report and Written Opinion of the International Searching Authority, date of mailing Nov. 8, 2018; 8 pages. cited by applicant

International Application No. PCT/IB2018/001520; International Preliminary Report on Patentability, date of issuance Jun. 2, 2020; 6 pages. cited by applicant

International Application No. PCT/IB2018/001520; International Search Report and Written Opinion of the International Searching Authority, date of mailing, Apr. 15, 2019; 10 pages. cited by applicant

International Application No. PCT/US2008/056279; International Preliminary Report on Patentability, date of issuance Sep. 8, 2009; 11 pages. cited by applicant

Invernizzi, P. et al., "Classical HLA-DRB1 and DPB1 Alleles Account for HLA Associations With Primary Biliary Cirrhosis", *Genes Immun.*, 13(6):461-8, (2012). cited by applicant

Itoh, N. et al., "Mononuclear Cell Infiltration and Its Relation to the Expression of Major Histocompatibility Complex Antigens and Adhesion Molecules in Pancreas Biopsy Specimens From Newly Diagnosed Insulin-Dependent Diabetes Mellitus Patients", *J Clin Invest.*, 92(5):2313-22, (1993). cited by applicant

Komatsu, N. et al., "Heterogeneity of Natural Foxp3+ T Cells: A Committed Regulatory T-cell Lineage and an Uncommitted Minor Population Retaining Plasticity", *Proc Natl Acad Sci USA*, 106(6):1903-8, (2009). cited by applicant

Komatsu, N. et al., "Pathogenic Conversion of Foxp3+ T Cells Into TH17 Cells in Autoimmune Arthritis", *Nat Med.*, 20(1):62-8, (2014). cited by applicant

Krishnamoorthy, G. et al., "Myelin-specific T Cells Also Recognize Neuronal Autoantigen in a Transgenic Mouse Model of Multiple Sclerosis", *Nat Med.*, 15(6):626-33, (2009). cited by applicant

Kukreja, A. et al., "NKT Cells and Type-1 Diabetes and the "hygiene Hypothesis" to Explain the Rising Incidence Rate", *Diabet Tech Ther.*, 4(3):323-33, (2002). cited by applicant

Kulmala, P., "Prediabetes in Children: Natural History, Diagnosis, and Preventive Strategies", *Pediatr Drugs*, 5(4):211-21, (2003). cited by applicant

Kwong, B. et al., "Synthesis and Characterization of Antibody-Nanoparticle Conjugates for Locally Sequestered Tumor Immunotherapy", Abstracts of Papers Am Chem Soc., 240:61, (2010). cited by applicant

Kyger, M. et al., "Effective Arrestin-Specific Immunotherapy of Experimental Autoimmune Uveitis with RTL: A Prospect for Treatment of Human Uveitis", Transl Vis Sci Technol., 2(2):1-15, (2013). cited by applicant

Kyung-Yu, M. et al., "Targeting Strategies for Multifunctional Nanoparticles in Cancer Imaging and Therapy", Theranostics, 2(1):3-44, (2012). cited by applicant

Laurence, A. et al., "T(H)-17 Differentiation: Of Mice and Men", Nat Immunol., 8(9):903-5, (2007). cited by applicant

Leavenworth, J. et al., "Amelioration of Arthritis Through Mobilization of Peptide-Specific CD8+ Regulatory T Cells", J Clin Invest., 123(3):1382-9, (2013). cited by applicant

Lechner, F. et al., "Analysis of Successful Immune Responses in Persons Infected With Hepatitis C Virus", J Exp Med., 191(9):1499-510, (2000). cited by applicant

Lee, Y. et al., "Biodegradable Nanoparticles Containing TLR3 or TLR9 Agonists Together With Antigen Enhance MHC-restricted Presentation of the Antigen", Arch Pharm Res., 33(11):1859-66, (2010). cited by applicant

Levings, M. et al., "T-regulatory 1 Cells: A Novel Subset of CD4 T Cells With Immunoregulatory Properties", J Allergy Clin Immunol., 106(1 Pt 2):S109-12, (2000). cited by applicant

Liblau, R. et al., "Autoreactive CD8 T Cells in Organ-Specific Autoimmunity: Emerging Targets for Therapeutic Intervention", Immunity, 17(1):1-6, (2002). cited by applicant

Lieberman, S. et al., "A Comprehensive Guide to Antibody and T-cell Responses in Type 1 Diabetes", Tissue Antigens, 62(5):359-77, (2003). cited by applicant

Lieberman, S. et al., "Identification of the Beta Cell Antigen Targeted by a Prevalent Population of Pathogenic CD8+ T Cells in Autoimmune Diabetes", Proc Natl Acad Sci USA, 100(14):8384-8, (2003). cited by applicant

Lieberman, S. et al., "Individual Nonobese Diabetic Mice Exhibit Unique Patterns of CD8+ T Cell Reactivity to Three Islet Antigens, Including the Newly Identified Widely Expressed *Dystrophin* Kinase", J Immunol., 173(11):6727-34, (2004). cited by applicant

Lillemeier, B. et al., "TCR and Lat Are Expressed on Separate Protein Islands on T Cell Membranes and Concatenate During Activation", Nat Immunol., 11(1):90-6 (2010). cited by applicant

Lleo, A. et al., "Etiopathogenesis of Primary Biliary Cirrhosis", World J Gastroenterol., 14(21):3328-37, (2008). cited by applicant

Singh, N. et al., "Emerging Concepts in TCR Specificity: Rationalizing and (Maybe) Predicting Outcomes", J Immunol., 199(7):2203-13, (2017). cited by applicant

IL Patent Application No. 249165; Office Action, dated Dec. 17, 2017; 5 pages. cited by applicant

Kozono, H. et al., "Production of Soluble MHC Class II Proteins With Covalently Bound Single Peptides", 369 (6476): 151-4, (1994). cited by applicant

"Homology", Definition of homology by Merriam-Webster dictionary, 13 pages, (2019). cited by applicant

Celik, A. et al., "The diversity of the HLA-E-restricted peptide repertoire explains the immunological impact of the Arg107Gly mismatch", Immunogenetics, 68(1):29-41, (2016). cited by applicant

Cochran, J. et al., "The relationship of MHC-peptide binding and T cell activation probed using chemically defined MHC class II oligomers", Immunity, 12(3):241-50, (2000). cited by applicant

GenBank Accession No. AAC14923.1, "T cell receptor alpha chain [*Homo sapiens*]", retrieved from ncbi.nlm.nih.gov/protein/AAC14923.1?report=genbank&log\$=protalign&blast, (1998). cited by applicant

Hemmer, B. et al., "Minimal Peptide Length Requirements for CD4(+) T Cell Clones—

Implications for Molecular Mimicry and T Cell Survival”, *Int Immunol.*, 12(3):375-83, (2000). cited by applicant

Kalandadze, A. et al., “Replacement of the Hydrophobic Transmembrane Region by a Leucine Zipper Dimerization Motif Allows the Assembly and Secretion of Soluble DR $\alpha\beta$ Heterodimers”, *J Biol Chem.*, 271(33):20156-20162, (1996). cited by applicant

Kalergis, A. et al., “A simplified procedure for the preparation of MHC/peptide tetramers: chemical biotinylation of an unpaired cysteine engineered at the C-terminus of MHC-I”, *J Immunol Meth.*, 234(1-2):61-70, (2000). cited by applicant

Koonin, E. et al., “Chapter 2 Evolutionary Concept in Genetic and Genomics”, NCBI Bookshelf, Bookshelf ID: NBK20255, 23 pages, (2003). cited by applicant

Nishioka, Y. et al., “CD1d-Restricted Type II NKT Cells Reactive With Endogenous Hydrophobic Peptides”, *Front Immunol.*, 9:548, 6 pages, (2018). cited by applicant

Pfister, D. et al., “Process for protein PEGylation”, *J Control Release*, 180:134-49, (2014). cited by applicant

Schumacher, T. et al., “Neoantigens in cancer immunotherapy”, *Science*, 348(6230):69-74, (2015). cited by applicant

Serra, P. et al., “Increased Yields and Biological Potency of Knob-Into-Hole-Based Soluble MHC Class II Molecules”, *Nat Commun.*, 10(1):4917, (2019). cited by applicant

Serra, P. et al., “Peptide-MHC-Based Nanomedicines for the Treatment of Autoimmunity: Engineering, Mechanisms, and Diseases”, *Front Immunol.*, 11:621774, (2021). cited by applicant

Shimoda, S. et al., “HLA DRB4 0101-restricted immunodominant T cell autoepitope of pyruvate dehydrogenase complex in primary biliary cirrhosis: evidence of molecular mimicry in human autoimmune diseases”, *J Exp Med.*, 181 (5):1835-45, (1995). cited by applicant

Wieczorek, M. et al., “Major Histocompatibility Complex (MHC) Class I and MHC Class II Proteins: Conformational Plasticity in Antigen Presentation”, *Front Immunol.*, 8(292):1-16, (2017). cited by applicant

Yang, Y. et al., “Antigen-specific nanomedicines for the treatment of autoimmune disease: target cell types, mechanisms and outcomes”, *Curr Opin Biotechnol.*, 74: 285-92, (2022). cited by applicant

CO Patent Application No. NC2017/0011437; Office Action, dated Nov. 19, 2017; 3 pages. cited by applicant

EP Patent Application No. 13856460.4; Communication, dated Nov. 15, 2017; 6 pages. cited by applicant

EP Patent Application No. 13856460.4; Extended European Search Report, mailed Feb. 26, 2016; 9 pages. cited by applicant

EP Patent Application No. 14184505.7; Communication, mailed Oct. 22, 2015; 4 pages. cited by applicant

EP Patent Application No. 17173410.6; Extended European Search Report, dated Dec. 15, 2017; 8 pages. cited by applicant

GenBank accession No. NM_001008228.2, accessed at ncbi.nlm.nih.gov/nuccore/281371473/?report=genbank on Jan. 16, 2018, 1 page. cited by applicant

GenBank accession No. NP_001008229.1, accessed at ncbi.nlm.nih.gov/protein/56788389/?report=genpept on Jan. 16, 2018, 3 pages. cited by applicant

Application No. PCT/EP2011 /069931; International Preliminary Report on Patentability (Ch. 2), date of issuance May 16, 2013; 21 pages. cited by applicant

Application No. PCT/EP2011 /069931; International Search Report and Written Opinion of the International Searching Authority, date of mailing Jul. 10, 2012; 24 pages. cited by applicant

Application No. PCT/EP2011/066994; International Preliminary Report on Patentability, date of issuance Apr. 2, 2013; 5 pages. cited by applicant

Application No. PCT/EP2011/066994; International Search Report and Written Opinion of the

International Searching Authority, date of mailing Nov. 21, 2011; 7 pages. cited by applicant
Application No. PCT/IB2013/003033; International Preliminary Report on Patentability, date of issuance Apr. 14, 2015; 6 pages. cited by applicant
Application No. PCT/IB2013/003033; International Search Report and Written Opinion of the International Searching Authority, date of mailing Jul. 14, 2014; 9 pages. cited by applicant
Application No. PCT/IB2014/003014; International Preliminary Report on Patentability, date of issuance May 10, 2016; 9 pages. cited by applicant
Application No. PCT/IB2014/003014; International Search Report and Written Opinion of the International Searching Authority, date of mailing May 12, 2015; 14 pages. cited by applicant
Nanjundappa, R. et al., "A Gut Microbial Mimic that Hijacks Diabetogenic Autoreactivity to Suppress Colitis", *Cell*, 171(3):655-67, (2017). cited by applicant
Chen, C. et al., "Induction of autoantigen-specific Th2 and Tr1 regulatory T cells and modulation of autoimmune diabetes", *J Immunol.*, 171(2):733-44, (2003). cited by applicant
Mallone, R. et al., "Functional avidity directs T-cell fate in autoreactive CD4+ T cells", *Blood*, 106(8):2798-805, (2005). cited by applicant
Matern, B. et al., "Insights into the polymorphism in HLA-DRA and its evolutionary relationship with HLA haplotypes", *HLA*, 95(2):117-27, (2020). cited by applicant
Rakoff-Nahoum, S. et al., "Role of toll-like receptors in spontaneous commensal-dependent colitis", *Immunity*, 25 (2):319-29, (2006). cited by applicant
U.S. Appl. No. 15/572,137; Notice of Allowance, dated Nov. 22, 2023; 66 pages. cited by applicant
U.S. Appl. No. 15/807,415; Applicant-Initiated Interview Summary, date of interview Aug. 23, 2023; 2 pages. cited by applicant
U.S. Appl. No. 15/807,415; Non-Final Office Action, dated May 25, 2023; 64 pages. cited by applicant
U.S. Appl. No. 17/097,682; Non-Final Office Action, dated Oct. 18, 2023; 60 pages. cited by applicant
U.S. Appl. No. 17/128,776; Non-Final Office Action, dated Dec. 8, 2023; 42 pages. cited by applicant
Clemente, J. et al., "The impact of the gut microbiota on human health: an integrative view", *Cell*, 148(6):1258-70, (2012). cited by applicant
Database Accession No. ADKO01000110, "Bacteroides vulgatus PC510 contig00041, whole genome shotgun sequence", Jul. 31, 2011; 6 pages. cited by applicant
Database Accession No. DVD94, "SubName: Full=Conserved domain protein", Database UniProtKB [online database] XP002712564; Jun. 15, 2010; retrieved Nov. 19, 2023; retrieved from www.uniprot.org/jobs/2013111951RNQ6FG0A.txt; 2 pages. cited by applicant
Diagnosis and Treatment Guidelines (for medical professionals): Japan Intractable Diseases Information Center, "Sjogren's Syndrome," Shengrensho University; retrieved Nov. 11, 2015; retrieved from <https://www.nanbyou.or.jp/entry/111> ; 6 pages. cited by applicant
GenBank Accession No. AAC08954, T cell receptor beta chain, [*Homo sapiens*], [online database], originally retrieved 2001, retrieved Nov. 8, 2024; retrieved from ncbi.nlm.nih.gov/protein/AAC08954; 3 pages. cited by applicant
GenBank Accession No. NP_001008229.1, Myelin-oligodendrocyte glycoprotein isoform alpha3 precursor [*Homo sapiens*], [online database], originally retrieved on Jan. 16, 2018, accessed at ncbi.nlm.nih.gov/protein/56788389/?report=genpept, (document dated Dec. 11, 2017); 3 pages. cited by applicant
International Application No. PCT/US2008/056279; International Search Report and Written Opinion of the International Searching Authority, date of mailing Oct. 22, 2008; 14 pages. cited by applicant
Japan Intractable Diseases Information Center, "Crohn's Disease," retrieved from nanbyou.or.jp/entry/81, 6 pages (2015). cited by applicant

McKown, K. et al., "Lack of Efficacy of Oral Bovine Type II Collagen Added to Existing Therapy in Rheumatoid Arthritis," *Arthritis Rheum.*, 42(6):1204-8, (1999). cited by applicant

Roncarolo, M. et al., "Interleukin-10-secreting Type 1 Regulatory T Cells in Rodents and Humans", *Immunol Rev.*, 212:28-50, (2006). cited by applicant

U.S. Appl. No. 12/044,435; Advisory Action, dated Aug. 23, 2011; 4 pages. cited by applicant

U.S. Appl. No. 12/044,435; Applicant-Initiated Interview Summary, date of interview Dec. 11, 2012; 4 pages. cited by applicant

U.S. Appl. No. 12/044,435; Applicant-Initiated Interview Summary, date of interview Sep. 7, 2012; 6 pages. cited by applicant

U.S. Appl. No. 12/044,435; Applicant-Initiated Interview Summary, dated Jul. 16, 2012; 3 pages. cited by applicant

U.S. Appl. No. 12/044,435; Applicant-Initiated Interview Summary, dated Nov. 7, 2011; 3 pages. cited by applicant

U.S. Appl. No. 12/044,435; Examiner-Initiated Interview Summary, dated Mar. 18, 2011; 4 pages. cited by applicant

U.S. Appl. No. 12/044,435; Final Office Action, dated Jun. 8, 2011; 22 pages. cited by applicant

U.S. Appl. No. 12/044,435; Non-Final Office Action, dated May 2, 2012; 9 pages. cited by applicant

U.S. Appl. No. 12/044,435; Non-Final Office Action, dated Nov. 24, 2010; 28 pages. cited by applicant

U.S. Appl. No. 12/044,435; Notice of Allowance, dated Sep. 12, 2012; 9 pages. cited by applicant

U.S. Appl. No. 12/044,435; Supplemental Notice of Allowability, dated Dec. 13, 2012; 2 pages. cited by applicant

U.S. Appl. No. 12/848,055; 1.132 Declaration, dated May 21, 2013; 7 pages. cited by applicant

U.S. Appl. No. 12/848,055; Advisory Action, dated May 5, 2014; 3 pages. cited by applicant

U.S. Appl. No. 12/848,055; Applicant-Initiated Interview Summary, dated Nov. 5, 2012; 6 pages. cited by applicant

U.S. Appl. No. 12/848,055; Final Office Action, dated Aug. 23, 2012; 10 pages. cited by applicant

U.S. Appl. No. 12/848,055; Final Office Action, dated Dec. 24, 2014; 18 pages. cited by applicant

U.S. Appl. No. 12/848,055; Final Office Action, dated Jul. 12, 2013; 15 pages. cited by applicant

U.S. Appl. No. 12/848,055; Non-Final Office Action, dated Apr. 4, 2012; 10 pages. cited by applicant

U.S. Appl. No. 12/848,055; Non-Final Office Action, dated Dec. 19, 2012; 13 pages. cited by applicant

U.S. Appl. No. 12/848,055; Non-Final Office Action, dated Jun. 6, 2014; 18 pages. cited by applicant

U.S. Appl. No. 12/848,055; Non-Final Office Action, dated May 13, 2016; 23 pages. cited by applicant

U.S. Appl. No. 12/848,055; Notice of Appeal, dated Jan. 7, 2014; 2 pages. cited by applicant

U.S. Appl. No. 12/848,055; Notice of Appeal, dated Jun. 22, 2015; 2 pages. cited by applicant

U.S. Appl. No. 12/848,055; Notice of Appeal, dated Nov. 11, 2016; 2 pages. cited by applicant

U.S. Appl. No. 13/249,105; 1.132 Declaration, dated Apr. 24, 2017; 26 pages. cited by applicant

U.S. Appl. No. 13/249,105; 1.132 Declaration, dated Aug. 24, 2015; 13 pages. cited by applicant

U.S. Appl. No. 13/249,105; Applicant Summary of Interview, dated Mar. 6, 2017; 2 pages. cited by applicant

U.S. Appl. No. 13/249,105; Applicant Summary of Interview, dated Oct. 9, 2015; 1 page. cited by applicant

U.S. Appl. No. 13/249,105; Applicant-Initiated Interview Summary, dated Aug. 6, 2015; 3 pages. cited by applicant

U.S. Appl. No. 13/249,105; Applicant-Initiated Interview Summary, dated Feb. 7, 2017; 3 pages.

cited by applicant
U.S. Appl. No. 13/249,105; Applicant-Initiated Interview Summary, dated Sep. 11, 2015; 3 pages.
cited by applicant
U.S. Appl. No. 13/249,105; Final Office Action, dated Apr. 11, 2018; 7 pages. cited by applicant
U.S. Appl. No. 13/249,105; Final Office Action, dated Nov. 30, 2015; 17 pages. cited by applicant
U.S. Appl. No. 13/249,105; Non-Final Office Action, dated Apr. 3, 2015; 9 pages. cited by applicant
U.S. Appl. No. 13/249,105; Non-Final Office Action, dated Sep. 8, 2017; 11 pages. cited by applicant
U.S. Appl. No. 13/249,105; Notice of Appeal, dated May 31, 2016; 2 pages. cited by applicant
U.S. Appl. No. 13/294,109; 1.132 Declaration, dated Dec. 11, 2014; 12 pages. cited by applicant
U.S. Appl. No. 13/294,109; 1.132 Declaration, dated Oct. 20, 2015; 21 pages. cited by applicant
U.S. Appl. No. 13/294,109; Examiner-Initiated Interview Summary, date of interview Mar. 24, 2016; 3 pages. cited by applicant
U.S. Appl. No. 13/294,109; Final Office Action, dated Jan. 12, 2015; 11 pages. cited by applicant
U.S. Appl. No. 13/294,109; Final Office Action, dated Nov. 13, 2013; 10 pages. cited by applicant
U.S. Appl. No. 13/294,109; Non-Final Office Action, dated Jun. 4, 2013; 9 pages. cited by applicant
U.S. Appl. No. 13/294,109; Notice of Allowance, dated Apr. 4, 2016; 9 pages. cited by applicant
U.S. Appl. No. 13/294,109; Notice of Allowance, dated Sep. 29, 2016; 5 pages. cited by applicant
U.S. Appl. No. 13/294,109; Notice of Appeal, dated Jul. 10, 2015; 2 pages. cited by applicant
U.S. Appl. No. 13/712,832; Non-Final Office Action, dated Feb. 27, 2015; 11 pages. cited by applicant
U.S. Appl. No. 13/830,521; 1.132 Declaration, dated Apr. 4, 2016; 33 pages. cited by applicant
U.S. Appl. No. 13/830,521; 1.132 Declaration, dated Dec. 23, 2014; 8 pages. cited by applicant
U.S. Appl. No. 13/830,521; Applicant-Initiated Interview Summary, dated Dec. 19, 2016; 2 pages. cited by applicant
U.S. Appl. No. 13/830,521; Applicant-Initiated Interview Summary, dated May 27, 2015; 3 pages. cited by applicant
U.S. Appl. No. 13/830,521; Final Office Action, dated Mar. 5, 2015; 13 pages. cited by applicant
U.S. Appl. No. 13/830,521; Non-Final Office Action, dated Jul. 25, 2014; 21 pages. cited by applicant
U.S. Appl. No. 13/830,521; Non-Final Office Action, dated Jun. 28, 2016; 10 pages. cited by applicant
U.S. Appl. No. 13/830,521; Notice of Appeal, dated Sep. 4, 2015; 2 pages. cited by applicant
U.S. Appl. No. 13/842,302; 1.132 Declaration, dated Feb. 22, 2016; 2 pages. cited by applicant
U.S. Appl. No. 13/842,302; 1.132 Declaration, dated Jan. 29, 2016; 46 pages. cited by applicant
U.S. Appl. No. 13/842,302; Advisory Action, dated Jul. 10, 2019; 6 pages. cited by applicant
U.S. Appl. No. 13/842,302; Applicant-Initiated Interview Summary, dated Aug. 15, 2018; 4 pages. cited by applicant
U.S. Appl. No. 13/842,302; Applicant-Initiated Interview Summary, dated Jan. 25, 2017; 4 pages. cited by applicant
U.S. Appl. No. 13/842,302; Applicant-Initiated Interview Summary, dated Jun. 23, 2015; 5 pages. cited by applicant
U.S. Appl. No. 13/842,302; Examiner-Initiated Interview Summary, date of interview Oct. 15, 2020; 1 page. cited by applicant
U.S. Appl. No. 13/842,302; Final Office Action, dated Dec. 28, 2018; 18 pages. cited by applicant
U.S. Appl. No. 13/842,302; Final Office Action, dated Feb. 18, 2015; 22 pages. cited by applicant
U.S. Appl. No. 13/842,302; Final Office Action, dated May 3, 2017; 28 pages. cited by applicant
U.S. Appl. No. 13/842,302; Non-Final Office Action, dated Apr. 29, 2020; 18 pages. cited by

applicant
U.S. Appl. No. 13/842,302; Non-Final Office Action, dated Apr. 30, 2014; 15 pages. cited by applicant
U.S. Appl. No. 13/842,302; Non-Final Office Action, dated Apr. 30, 2018; 18 pages. cited by applicant
U.S. Appl. No. 13/842,302; Non-Final Office Action, dated Jul. 6, 2016; 24 pages. cited by applicant
U.S. Appl. No. 13/842,302; Notice of Allowance, dated Oct. 22, 2020; 11 pages. cited by applicant
U.S. Appl. No. 13/842,302; Notice of Appeal, dated Aug. 17, 2015; 2 pages. cited by applicant
U.S. Appl. No. 14/531,707; Applicant-Initiated Interview Summary, dated Jan. 31, 2018; 3 pages. cited by applicant
U.S. Appl. No. 14/531,707; Corrected Notice of Allowability, dated Jul. 20, 2018; 2 pages. cited by applicant
U.S. Appl. No. 14/531,707; Examiner-Initiated Interview Summary, date of interview Jun. 5, 2018; 1 page. cited by applicant
U.S. Appl. No. 14/531,707; Non-Final Office Action, dated Oct. 3, 2017; 12 pages. cited by applicant
U.S. Appl. No. 14/531,707; Notice of Allowance, dated Jun. 18, 2018; 11 pages. cited by applicant
U.S. Appl. No. 14/684,153; Applicant-Initiated Interview Summary, dated Dec. 20, 2016; 2 pages. cited by applicant
U.S. Appl. No. 14/684,153; Non-Final Office Action, dated Jun. 30, 2016; 18 pages. cited by applicant
U.S. Appl. No. 14/684,153; Notice of Allowance, dated Jan. 26, 2017; 6 pages. cited by applicant
U.S. Appl. No. 14/723,268; Final Office Action, dated Mar. 30, 2016; 15 pages. cited by applicant
U.S. Appl. No. 14/723,268; Non-Final Office Action, dated Oct. 16, 2015; 23 pages. cited by applicant
U.S. Appl. No. 14/723,268; Notice of Appeal, dated Sep. 29, 2016; 2 pages. cited by applicant
U.S. Appl. No. 15/348,959; Applicant-Initiated Interview Summary, date of interview Apr. 4, 2019; 2 pages. cited by applicant
U.S. Appl. No. 15/348,959; Applicant-Initiated Interview Summary, date of interview Jun. 27, 2017; 2 pages. cited by applicant
U.S. Appl. No. 15/348,959; Applicant-Initiated Interview Summary, dated Mar. 29, 2018; 3 pages. cited by applicant
U.S. Appl. No. 15/348,959; Examiner-Initiated Interview Summary, date of interview Jun. 21, 2019; 2 pages. cited by applicant
U.S. Appl. No. 15/348,959; Final Office Action, dated Jan. 12, 2018; 18 pages. cited by applicant
U.S. Appl. No. 15/348,959; First-Action Interview Office Action, dated Aug. 30, 2017; 20 pages. cited by applicant
U.S. Appl. No. 15/348,959; Non-Final Office Action, dated Feb. 21, 2019; 6 pages. cited by applicant
U.S. Appl. No. 15/348,959; Notice of Allowance, dated Jul. 3, 2019; 8 pages. cited by applicant
U.S. Appl. No. 15/348,959; Pre-Interview First Office Action, dated Apr. 13, 2017; 5 pages. cited by applicant
U.S. Appl. No. 15/433,898; Non-Final Office Action, dated Sep. 28, 2017; 16 pages. cited by applicant
U.S. Appl. No. 15/433,898; Notice of Allowance, dated May 17, 2018; 6 pages. cited by applicant
U.S. Appl. No. 15/433,898; Notice of Allowance, dated May 31, 2018; 3 pages. cited by applicant
U.S. Appl. No. 15/572,137; Final Office Action, dated Feb. 8, 2022; 13 pages. cited by applicant
U.S. Appl. No. 15/572,137; Final Office Action, dated Feb. 25, 2021; 12 pages. cited by applicant
U.S. Appl. No. 15/572,137; Non-Final Office Action, dated Jul. 23, 2021; 14 pages. cited by

applicant
U.S. Appl. No. 15/572,137; Non-Final Office Action, dated Sep. 28, 2022; 22 pages. cited by applicant
U.S. Appl. No. 15/572,137; Notice of Allowance, dated Nov. 22, 2023; 8 pages. cited by applicant
U.S. Appl. No. 15/610,550; Applicant-Initiated Interview Summary, dated Mar. 3, 2020; 3 pages. cited by applicant
U.S. Appl. No. 15/610,550; Final Office Action, dated Jun. 19, 2020; 10 pages. cited by applicant
U.S. Appl. No. 15/610,550; Non-Final Office Action, dated Nov. 4, 2019; 9 pages. cited by applicant
U.S. Appl. No. 15/807,415; Applicant-Initiated Interview Summary, dated Aug. 29, 2023; 2 pages. cited by applicant
U.S. Appl. No. 15/807,415; Final Office Action, dated Aug. 30, 2022; 20 pages. cited by applicant
U.S. Appl. No. 15/807,415; Final Office Action, dated Dec. 30, 2020; 20 pages. cited by applicant
U.S. Appl. No. 15/807,415; Non-Final Office Action, dated Aug. 19, 2021; 13 pages. cited by applicant
U.S. Appl. No. 15/807,415; Non-Final Office Action, dated Jul. 29, 2020; 21 pages. cited by applicant
U.S. Appl. No. 15/807,415; Non-Final Office Action, dated May 25, 2023; 47 pages. cited by applicant
U.S. Appl. No. 15/999,192; Non-Final Office Action, dated Apr. 15, 2020; 19 pages. cited by applicant
U.S. Appl. No. 15/999,192; Notice of Allowance, dated Oct. 13, 2020; 8 pages. cited by applicant
U.S. Appl. No. 16/132,000; Examiner-Initiated Interview Summary, date of interview Jan. 7, 2022; 1 page. cited by applicant
U.S. Appl. No. 16/132,000; Notice of Allowance, dated Jan. 21, 2022; 10 pages. cited by applicant
U.S. Appl. No. 16/156,326; Final Office Action, dated Apr. 20, 2022; 34 pages. cited by applicant
U.S. Appl. No. 16/200,199; Non-Final Office Action, dated Aug. 27, 2020; 5 pages. cited by applicant
U.S. Appl. No. 16/200,199; Notice of Allowance, dated Jan. 13, 2021; 6 pages. cited by applicant
U.S. Appl. No. 16/603,180; Final Office Action, dated Mar. 1, 2023; 14 pages. cited by applicant
U.S. Appl. No. 16/603,180; Non-Final Office Action, dated Jul. 19, 2022; 13 pages. cited by applicant
U.S. Appl. No. 17/097,682; Non-Final Office Action, dated Oct. 18, 2023; 17 pages. cited by applicant
U.S. Appl. No. 17/128,776; Non-Final Office Action, dated Dec. 8, 2023; 7 pages. cited by applicant
U.S. Appl. No. 17/153,212; Non-Final Office Action, dated Nov. 14, 2024; 17 pages. cited by applicant
U.S. Appl. No. 17/228,014; Application as filed, dated Apr. 12, 2021; 63 pages. cited by applicant
Xu, J et al., "Evolution of symbiotic bacteria in the distal human intestine", PLoS Biol., 5(7)e156:1574-86, (2007). cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. provisional application 62/591,921 filed on Nov. 29, 2017; U.S. provisional application 62/636,520 filed on Feb. 28, 2018; and U.S. provisional application 62/641,607 filed on Mar. 12, 2018, all of which are incorporated by reference herein in their entirety.

INCORPORATION OF SEQUENCE LISTING

(1) The sequence listing that is contained in the file named "PTI0010-505-US," which was created Mar. 8, 2024, and which has a file size of 29,798.4 bytes as measured in Microsoft Windows operating system, is filed electronically herewith and incorporated herein by reference.

SUMMARY

(2) Provided herein are compositions comprising a plurality of antigen-major histocompatibility complexes coupled to a nanoparticle core. The compositions are useful for treating autoimmune and inflammatory disorders. Many autoimmune or inflammatory diseases are associated with an immune response directed to a particular tissue specific antigen or subset of antigens. This presents a problem for designing interventions that treat an autoimmune or inflammatory disease since each disease requires a specific medicament that targets that antigen or subset of antigens. Alternatively, nonspecific immune inhibitors can be used, but these are associated with significant systemic side-effects. Described herein are compositions comprising ubiquitous autoantigen-MHC complexes coupled to nanoparticles (uaMHC-NP), that are useful for treating autoimmune diseases or expanding populations of T regulatory cells that suppress autoreactive (i.e., autoimmune or inflammatory) T cells.

(3) Primarily, the treatments described herein are multi-purpose in that a single composition can treat multiple-autoimmune or inflammatory disorders that are not mechanistically or pathologically linked. Many treatments, for example, corticosteroids or antibodies against general inflammatory mediators, that are multi-purpose in this way result in systemic immunosuppression, thus, increasing a treated patients risk for developing secondary infections and systemic immunological complications. The compositions described herein, while being multi-purpose, also spare systemic immunity leaving intact the ability of a patient to fight off viral, bacterial, fungal infection, or tumors. Surprisingly, the antigens utilized by the methods described herein are able to treat multiple diseases that are pathologically distinct (e.g., different autoimmune diseases of the same organ can be treated by a single antigen) and distinct with regard to tissue of origin (e.g., autoimmune diseases that afflict different organ systems). These antigens are antigens that are broadly expressed across many different tissue types, but are not the primary antigens associated with the disease, which generally display expression restricted to the particular tissue.

(4) In certain aspects described herein is a composition comprising: (a) a plurality of antigen-major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, that is not a tissue specific antigen, associated with the binding groove of a MHC molecule; and (b) a nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core. In certain embodiments, the MHC molecule is a MHC class II molecule. In certain embodiments, the nanoparticle core is a metal or metal oxide. In certain embodiments, the metal is iron. In certain embodiments, the metal oxide is iron oxide. In certain embodiments, the diameter is between about 5 nanometers and about 50 nanometers. In certain embodiments, the diameter is between about 5 nanometers and about 25 nanometers. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of at least 10:1. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of no more than 150:1. In certain embodiments, the plurality of antigen-MHCs is coupled

to the nanoparticle core at a density from about 0.4 to about 13 antigen-MHCs per 100 nm² of nanoparticle surface area. In certain embodiments, the antigen-MHCs are covalently coupled to the nanoparticle. In certain embodiments, the antigen-MHCs are covalently coupled to the nanoparticle core by a polyethylene glycol (PEG) linker having a mass of less than about 5 kilodaltons. In certain embodiments, the nanoparticle core further comprises a biocompatible coating. In certain embodiments, the ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment. In certain embodiments the intracellular compartment is cytosol, mitochondria, Golgi apparatus, endoplasmic reticulum, nucleus, or plasma membrane. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the ubiquitous autoantigen is pyruvate dehydrogenase complex-E2 component (PDC-E2). In certain embodiments, the ubiquitous autoantigen is Cytochrome P450 2D6 (CYP2D6). In certain embodiments, the ubiquitous autoantigen is soluble liver antigen (SLA). In certain embodiments, the ubiquitous autoantigen is actin (ACTB). In certain embodiments, the ubiquitous autoantigen is formimidoyltransferase-cyclodeaminase (FTCD). In certain embodiments, the ubiquitous autoantigen is myeloperoxidase (MPO). In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D6.sub.5-19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301); ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386, SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD 495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285, FTCD.sub.498-512, and FTCD.sub.301-315. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-

476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the complex further comprises a second plurality of antigen-major histocompatibility complexes (MHCs) coupled to the nanoparticle core, each antigen-MHC of the second plurality comprising an antigen. In certain embodiments, the antigen of the second plurality of antigen-major histocompatibility complexes (MHCs) is a second ubiquitous autoantigen. In certain embodiments, the second ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment. In certain embodiments, the intracellular compartment is cytosol, mitochondria, Golgi apparatus, endoplasmic reticulum, nucleus, or plasma membrane. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D6.sub.65-19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301); ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386, SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD.sub.495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285, FTCD.sub.498-512, and FTCD.sub.301-315. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the composition further comprises a pharmaceutically acceptable stabilizer, excipient, diluent, or combination thereof. In certain

embodiments, the composition is formulated for intravenous administration. In certain embodiments, the composition is for use in a method of treating an autoimmune or inflammatory disease. In certain embodiments, the composition is for use in the manufacture of a medicament for treating an autoimmune or inflammatory disease. In certain embodiments, the autoimmune or inflammatory disease is selected from the group consisting of: type I diabetes, multiple sclerosis, relapsing-remitting multiple sclerosis, pemphigus, pemphigus foliaceus, pemphigus vulgaris, neuromyelitis optica spectrum disorder, rheumatoid arthritis, osteoarthritis, psoriatic arthritis, inflammatory bowel disease, Crohn's disease, celiac disease, allergic asthma, systemic lupus erythematosus, atherosclerosis, chronic obstructive pulmonary disease, emphysema, psoriasis, uveitis, Sjogren's Syndrome, scleroderma, anti-phospholipid syndrome, ANCA-associated vasculitis, primary biliary cirrhosis, autoimmune hepatitis, primary sclerosing cholangitis, and Stiff Man Syndrome. In certain embodiments, the autoimmune or inflammatory disease is multiple sclerosis. In certain embodiments, the ubiquitous autoantigen is not a polypeptide derived from myelin basic protein, myelin associated glycoprotein, myelin oligodendrocyte protein (MOG), proteolipid protein, oligodendrocyte myelin oligoprotein, myelin associated oligodendrocyte basic protein, oligodendrocyte specific protein, heat shock proteins, an oligodendrocyte specific protein, NOGO A, glycoprotein Po, peripheral myelin protein 22, or 2'3'-cyclic nucleotide 3'-phosphodiesterase. In certain embodiments, the autoimmune or inflammatory disease is type I diabetes. In certain embodiments, the ubiquitous autoantigen is a polypeptide derived from pre-proinsulin, proinsulin, islet-specific glucose-6-phosphatase (IGRP), glutamate decarboxylase (GAD), islet cell autoantigen-2 (ICA2), or insulin.

(5) In certain aspects described herein is a method of treating an autoimmune or inflammatory disease comprising administering to an individual a therapeutically effective amount of a composition comprising: (a) a plurality of antigen-major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, that is not a tissue specific antigen, associated with the binding groove of an MHC molecule; and (b) a nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core. In certain embodiments, the MHC molecule is a MHC class II molecule. In certain embodiments, the nanoparticle core is a metal or metal oxide. In certain embodiments, the metal is iron. In certain embodiments, the metal oxide is iron oxide. In certain embodiments, the diameter is between about 5 nanometers and about 50 nanometers. In certain embodiments, the diameter is between about 5 nanometers and about 25 nanometers. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of at least 10:1. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of no more than 150:1. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at a density from about 0.4 to about 13 antigen-MHCs per 100 nm² of nanoparticle core surface area. In certain embodiments, the antigen-MHCs are covalently coupled to the nanoparticle core. In certain embodiments, the antigen-MHCs are covalently coupled to the nanoparticle core by a polyethylene glycol (PEG) linker having a mass of less than about 5 kilodaltons. In certain embodiments, the nanoparticle core further comprises a biocompatible coating. In certain embodiments, the ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment. In certain embodiments, the intracellular compartment is cytosol, mitochondria, Golgi apparatus, endoplasmic reticulum, nucleus, or plasma membrane. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the ubiquitous autoantigen is pyruvate dehydrogenase complex-E2 component (PDC-E2). In certain embodiments, the ubiquitous autoantigen is Cytochrome P450 2D6 (CYP2D6). In certain embodiments, the ubiquitous autoantigen is soluble liver antigen (SLA). In certain embodiments, the ubiquitous autoantigen is actin (ACTB). In certain embodiments, the ubiquitous autoantigen is formimidoyltransferase-cyclodeaminase (FTCD). In certain embodiments, the ubiquitous

autoantigen is myeloperoxidase (MPO). In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2422-436 and PDC-E2.sub.80-94. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D65-19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301); ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386, SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD.sub.495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285, FTCD.sub.498-512, and FTCD.sub.301-315. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the complex further comprises a second plurality of antigen-major histocompatibility complexes (MHCs) coupled to the nanoparticle core, each antigen-MHC of the second plurality comprising an antigen. In certain embodiments, the antigen of the second plurality of antigen-major histocompatibility complexes (MHCs) is a second ubiquitous autoantigen. In certain embodiments, the second ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment. In certain embodiments, the intracellular compartment is cytosol, mitochondria, Golgi apparatus, endoplasmic reticulum, nucleus, or plasma membrane. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-

648, PDC-E2.sub.122-135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D6.sub.19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301); ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386, SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD.sub.495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285, FTCD.sub.498-512, and FTCD.sub.301-315. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the composition further comprises a pharmaceutically acceptable stabilizer, excipient, diluent, or combination thereof. In certain embodiments, the composition is formulated for intravenous administration. In certain embodiments, the autoimmune or inflammatory disease is selected from the group consisting of type I diabetes, multiple sclerosis, relapsing-remitting multiple sclerosis, pemphigus, pemphigus foliaceus, pemphigus vulgaris, neuromyelitis optica spectrum disorder, rheumatoid arthritis, osteoarthritis, psoriatic arthritis, inflammatory bowel disease, Crohn's disease, celiac disease, allergic asthma, systemic lupus erythematosus, atherosclerosis, chronic obstructive pulmonary disease, emphysema, psoriasis, uveitis, Sjogren's Syndrome, scleroderma, anti-phospholipid syndrome, ANCA-associated vasculitis, primary biliary cirrhosis, autoimmune hepatitis, primary sclerosing cholangitis, and Stiff Man Syndrome. In certain embodiments, the autoimmune or inflammatory disease is multiple sclerosis. In certain embodiments, the ubiquitous autoantigen is not a polypeptide derived from myelin basic protein, myelin associated glycoprotein, myelin oligodendrocyte protein (MOG), proteolipid protein, oligodendrocyte myelin oligoprotein, myelin associated oligodendrocyte basic protein, oligodendrocyte specific protein, heat shock proteins, an oligodendrocyte specific protein, NOGO A, glycoprotein Po, peripheral myelin protein 22, or 2'3'-

cyclic nucleotide 3'-phosphodiesterase. In certain embodiments, the autoimmune or inflammatory disease is type I diabetes. In certain embodiments, the ubiquitous autoantigen is a polypeptide derived from pre-proinsulin, proinsulin, islet-specific glucose-6-phosphatase (IGRP), glutamate decarboxylase (GAD), islet cell autoantigen-2 (ICA2), or insulin.

(6) In another aspect described herein is a method of treating multiple sclerosis comprising administering to an individual a therapeutically effective amount of a composition comprising: (a) a plurality of antigen-major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, which is not a multiple sclerosis specific antigen, associated with the binding groove of an MHC molecule; and (a) a nanoparticle core, the nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core. In certain embodiments, the MHC molecule is a MHC class II molecule. In certain embodiments, the nanoparticle core is a metal or metal oxide. In certain embodiments, the metal is iron. In certain embodiments, the metal oxide is iron oxide. In certain embodiments, the diameter is between about 5 nanometers and about 50 nanometers. In certain embodiments, the diameter is between about 5 nanometers and about 25 nanometers. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of at least 10:1. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of no more than 150:1. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at a density from about 0.4 to about 13 antigen-MHCs per 100 nm² of nanoparticle core surface area. In certain embodiments, the antigen-MHCs are covalently coupled to the nanoparticle core. In certain embodiments, the antigen-MHCs are covalently coupled to the nanoparticle core by a polyethylene glycol (PEG) linker having a mass of less than about 5 kilodaltons. In certain embodiments, the nanoparticle core further comprises a biocompatible coating. In certain embodiments, the ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment. In certain embodiments, the intracellular compartment is cytosol. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the ubiquitous autoantigen is pyruvate dehydrogenase complex-E2 component (PDC-E2). In certain embodiments, the ubiquitous autoantigen is Cytochrome P450 2D6 (CYP2D6). In certain embodiments, the ubiquitous autoantigen is soluble liver antigen (SLA). In certain embodiments, the ubiquitous autoantigen is actin (ACTB). In certain embodiments, the ubiquitous autoantigen is formimidoyltransferase-cyclodeaminase (FTCD). In certain embodiments, the ubiquitous autoantigen is myeloperoxidase (MPO). In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2422-436 and PDC-E2.sub.80-94. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D65-19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301);

ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386, SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD.sub.495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285, FTCD.sub.498-512, and FTCD.sub.301-315. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the complex further comprises a second plurality of antigen-major histocompatibility complexes (MHCs) coupled to the nanoparticle core, each antigen-MHC of the second plurality comprising an antigen. In certain embodiments, the antigen of the second plurality of antigen-major histocompatibility complexes (MHCs) is a second ubiquitous autoantigen. In certain embodiments, the second ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment. In certain embodiments, the intracellular compartment is cytosol. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D6.sub.65-19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301); ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386,

SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD.sub.495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285, FTCD.sub.498-512, and FTCD.sub.301-315. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the composition further comprises a pharmaceutically acceptable stabilizer, excipient, diluent, or combination thereof. In certain embodiments, the composition is formulated for intravenous administration. In certain embodiments, the ubiquitous autoantigen is not a polypeptide derived from myelin basic protein, myelin associated glycoprotein, myelin oligodendrocyte protein (MOG), proteolipid protein, oligodendrocyte myelin oligoprotein, myelin associated oligodendrocyte basic protein, oligodendrocyte specific protein, heat shock proteins, an oligodendrocyte specific protein, NOGO A, glycoprotein Po, peripheral myelin protein 22, or 2'3'-cyclic nucleotide 3'-phosphodiesterase.

(7) In another aspect described herein is a method of treating type I diabetes comprising administering to an individual a therapeutically effective amount of a composition comprising: (a) a plurality of antigen-major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, which is not a type I diabetes specific antigen, associated with the binding groove of an MHC molecule; and (b) a nanoparticle core, the nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the plurality of antigen-MHCs is coupled to the nanoparticle core. In certain embodiments, the MHC molecule is a MHC class II molecule. In certain embodiments, the nanoparticle core is a metal or metal oxide. In certain embodiments, the metal is iron. In certain embodiments, the metal oxide is iron oxide. In certain embodiments, the diameter is between about 5 nanometers and about 50 nanometers. In certain embodiments, the diameter is between about 5 nanometers and about 25 nanometers. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of at least 10:1. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of no more than 150:1. In certain embodiments, the plurality of antigen-MHCs is coupled to the nanoparticle at a density from about 0.4 to about 13 antigen-MHCs per 100 nm² of nanoparticle surface area. In certain embodiments, the antigen-MHCs are covalently coupled to the nanoparticle core. In certain embodiments, the antigen-MHCs are covalently coupled to the nanoparticle core by a polyethylene glycol (PEG) linker having a mass of less than about 5 kilodaltons. In certain embodiments, the nanoparticle core further comprises a biocompatible coating. In certain embodiments, the ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment. In certain embodiments, the intracellular compartment is cytosol. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the ubiquitous autoantigen is pyruvate dehydrogenase complex-E2 component (PDC-E2). In certain embodiments, the ubiquitous autoantigen is Cytochrome P450 2D6 (CYP2D6). In certain embodiments, the ubiquitous autoantigen is soluble liver antigen (SLA). In certain embodiments, the ubiquitous autoantigen is actin (ACTB). In certain embodiments, the ubiquitous autoantigen is formimidoyltransferase-cyclodeaminase (FTCD). In

certain embodiments, the ubiquitous autoantigen is myeloperoxidase (MPO). In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2422-436 and PDC-E2.sub.80-94. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D65-19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301); ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386, SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD.sub.495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285, FTCD.sub.498-512, and FTCD.sub.301-315. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the complex further comprises a second plurality of antigen-major histocompatibility complexes (MHCs) coupled to the nanoparticle core, each antigen-MHC of the second plurality comprising an antigen. In certain embodiments, the antigen of the second plurality of antigen-major histocompatibility complexes (MHCs) is a second ubiquitous autoantigen. In certain embodiments, the second ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment. In certain embodiments, the intracellular compartment is cytosol. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the intracellular compartment is a mitochondrion. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-

135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D6.sub.65-19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301); ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386, SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD 495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285, FTCD.sub.498-512, and FTCD.sub.301-315. In certain embodiments, the second ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the composition further comprises a pharmaceutically acceptable stabilizer, excipient, diluent, or combination thereof. In certain embodiments, the autoimmune or inflammatory disease is type I diabetes. In certain embodiments, the ubiquitous autoantigen is not a polypeptide derived from pre-proinsulin, proinsulin, islet-specific glucose-6-phosphatase (IGRP), glutamate decarboxylase (GAD), islet cell autoantigen-2 (ICA2), or insulin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features described herein are set forth with particularity in the appended claims. A better understanding of the features and advantages of the features described herein will be obtained by reference to the following detailed description that sets forth illustrative examples, in which the principles of the features described herein are utilized, and the accompanying drawings of which:

(2) FIG. 1 illustrates the expansion of Primary Biliary Cirrhosis (PBC) relevant regulatory T cells

by nanoparticles coupled to MHC class II associated with peptides derived from pyruvate dehydrogenase complex-E2 component (PDC-E2) (PBC-relevant peptide-major histocompatibility complex-nanoparticles (pMHC-NPs)). FIG. 1A shows percentage of tetramer+ CD4+ T-cells in blood of NOD vs. NOD.c3c4 mice as a function of age. FIG. 1B shows percentage of tetramer+ CD4+ T-cells in peripheral blood of pMHC-NP-treated NOD.c3c4 mice as compared to untreated NOD or NOD.c3c4 mice or NOD.c3c4 mice treated with control Cys-NPs. FIG. 1C shows percentages of tetramer+CD4+ T-cells in mice from panel B in various organs at the end of pMHC-NP therapy. FIG. 1D shows percentages of tetramer+CD4+ T-cells in NOD.c3c4 mice treated with type 1 diabetes-relevant pMHC-NPs.

(3) FIG. 1E shows percentages of tetramer+CD4+ T-cells in various lymphoid organs and liver of NOD.c3c4 mice treated with NPs coated with one of two different PBC-relevant pMHCs. FIG. 1F shows expression of TR1-like cell surface markers by the tetramer+CD4+ T-cells expanded in NOD.c3c4 mice by pMHC-NP therapy. (FIG. 1G shows cytokine secretion profile of sorted tetramer+CD4+TR1-like cells vs. tetramer-negative CD4+ T-cells ex vivo upon stimulation with peptide-pulsed DCs.

(4) FIG. 2 illustrates the expansion of PBC relevant regulatory T cells by PBC-relevant pMHC-NPs. (FIGS. 2A-B) Upregulation of TR1-like markers on tetramer+CD4+ T-cells expanded in vivo in response to PDC.sub.166/IA.sup.g7- or PDC.sub.82/IA.sup.g7-NP therapy FIG. 2A is a representative FACS profiles; FIG. 2B shows average mean fluorescence intensity values. FIG. 2C shows average mean fluorescence intensity values for TR1 cell surface markers on tetramer+CD4+ T-cells arising in 38-44 week-old NOD.c3c4 mice in response to PDC.sub.166/IA.sup.g7-NP therapy.

(5) FIG. 3 illustrates clinical, phenotypic, immunological and pathological features of liver disease in NOD.c3c4 mice. FIG. 3A shows changes in serum TB and ALT levels with age. FIG. 3B shows microscopic scoring system (left) and progression of microscopic scores of disease with age (right). FIG. 3C shows representative CBD images and progression of CBD diameter and scores with age. FIG. 3D shows representative liver images (top) and progression of liver scores and weight with age (bottom). FIG. 3E shows that NOD.c3c4 mice spontaneously develop anti-PDC-E2-specific autoantibodies (left) and ANAs (right). FIG. 3F depicts representative images of liver inflammation by CD4+ and CD8+ T-cells.

(6) FIG. 4 illustrates reversal of disease in a mouse model of PBC using PBC-relevant pMHC-NPs. FIG. 4A shows changes in total bile acid (TBA) and alanine aminotransferase (ALT) levels in serum of NOD.c3c4 mice treated with PDC.sub.166/IA.sup.g7-NPs, PDC.sub.82/IA.sub.g7-NPs or control (Cys-NP). FIG. 4B shows representative histological images of livers from PDC.sub.166/IA.sup.g7-NPs, PDC.sub.82/IA.sup.g7-NPs, or Cys-NP-treated NOD.c3c4 mice (top) and average histological scores (bottom). FIG. 4C shows representative macroscopic images of the common bile duct (top), and average common bile duct scores and diameters (bottom). FIG. 4D shows representative macroscopic images of livers (top) and average liver scores and weight (bottom). FIG. 4E shows representative whole body images of NOD.c3c4 mice treated with PDC.sub.166/IA.sup.g7-NP or control Cys-NPs. FIG. 4F shows changes in the titer of anti-mitochondrial (PDC-E2) antibodies and anti-nuclear autoantibodies (ANAs) after treatment (top left two panels and right panel, respectively), and representative images of Hep2 cells stained with sera from pMHC-NP vs Cys-NP-treated NOD.c3c4 mice (bottom). FIG. 4G shows percentages of tetramer+ cells in mice treated starting at 24 weeks of age (until week 38-44). FIGS. 4H and 4I show microscopic (FIG. 4H) and macroscopic scores (FIG. 4I) for the mice studied in FIG. 4G.

(7) FIG. 5 illustrates changes in the circulating frequency of tetramer+CD4+ T-cells in response to periodic re-treatment with PBC relevant nanoparticles. FIG. 5A shows two different mice, and FIG. 5B shows average values corresponding to cohorts of mice treated with PDC.sub.166/IA.sup.g7-NPs or left untreated. FIG. 5C shows percentage of tetramer+CD4+ T-cells, and FIG. 5D shows mean fluorescence intensity staining for TR1 markers in tetramer+CD4+ T-cells from the mice

studied in FIG. 5A. FIG. 5E shows average macroscopic CBD and liver scores for the mice studied in FIGS. 5A-D.

(8) FIG. 6 illustrates effects of treatment with PBC-relevant pMHC-NPs or the standard of care in PBC (UDCA) on macroscopic and serum ALT levels (FIG. 6A) and microscopic (FIG. 6B) disease scores when treatment is initiated early on in the disease process. FIG. 6C shows percentages of tetramer+CD4⁺ T-cells in the mice studied in FIGS. 6A-B. FIGS. 6D and E show effects of treatment with PBC-relevant pMHC-NPs or the standard of care in PBC (UDCA) on macroscopic (FIG. 6D) and microscopic (FIG. 6E) disease scores when treatment is initiated at advanced stages of disease. FIG. 6F shows percentages of tetramer+CD4⁺ T-cells in the mice studied in FIGS. 6D-E.

(9) FIG. 7 illustrates that PBC-relevant pMHC-NPs expand regulatory B-cells. FIG. 7A shows percentages of tetramer+CD4⁺ T-cells in mice treated with pMHC-NPs and rat-IgG (control) or blocking rat mAbs against mouse IL-10 or TGF-beta. FIGS. 7B and C show macroscopic (FIG. 7B) and microscopic (FIG. 7C) scores of the mice studied in FIG. 7A. FIG. 7D shows percentages of tetramer+CD4⁺ T cells in blood and lymphoid organs of NOD.c3c4.scid hosts reconstituted with whole splenocytes from untreated NOD.c3c4 donors and then transfused with splenic CD4⁺ T-cells from PDC.sub.166-181/IA.sup.g7-NP-treated NOD.c3c4 mice. The latter were either left untreated after CD4⁺ T-cell transfer or were treated with PDC.sub.166-181/IA.sup.g7-NPs. FIG. 7E shows representative FACS staining histograms (top) and average mean fluorescence intensity values for TR1 markers on tetramer+CD4⁺ vs. tetramer-CD4⁺ T-cells (bottom) of the hosts treated with PDC.sub.166-181/IA.sup.g7-NPs. FIG. 7F shows macroscopic scores and measurements of the mice studied in A. FIG. 7G shows the cytokine profile of LPS challenged CD11b⁺ cells isolated from the liver draining (PLN) or non-draining (MLN) lymph nodes of pMHC-NP vs Cys-NP-treated NOD.c3c4 mice. FIG. 7H shows the cytokine profile of liver Kupffer cells from pMHC-NP vs Cys-NP-treated NOD.c3c4 mice. FIG. 7I shows absolute numbers of B-cells in liver draining (PCLN) and non-draining (ILN) lymph nodes or liver of pMHC-NP vs Cys-NP-treated NOD.c3c4 mice. FIG. 7J shows correlation between absolute numbers of B-cells and tetramer+CD4⁺ T-cells in pMHC-NP-treated mice. FIG. 7K shows IL-10 secretion levels of LPS-challenged B-cells isolated from liver draining and non-draining lymph nodes or liver of pMHC-NP vs Cys-NP-treated NOD.c3c4 mice. FIG. 7L shows representative FACS plots showing conversion of B cells into IL-10-producing Breg cells only in the experiments arising in hosts treated with pMHC-NPs. FIG. 7M shows percentages of conventional B-cell-derived Breg cells in hosts treated with pMHC-NP or Cys-NP in liver and peripheral lymphoid organs harboring (spleen and PCLN) or lacking (MLN) pMHC-NP-induced TR1-like CD4⁺ T-cells.

(10) FIG. 8 illustrates that PBC-relevant pMHC-NPs spare systemic immunity. FIG. 8A shows average titers of rVV in ovaries of pMHC-NP-treated vs. untreated NOD.c3c4 female mice 4 and 14 days after a systemic rVV infection. FIG. 8B shows titers of influenza virus in lungs of pMHC-NP-treated vs. untreated NOD.c3c4 female mice 3 and 7 days after infection with the PR8 strain with or without prior priming with the HKx31 strain. FIG. 8C shows clinical scores and body weight changes as a function of infection and treatment. FIG. 8D shows colony-forming units of *Listeria* in the spleen or liver of pMHC-treated vs untreated NOD or NOD.c3c4 mice on days 3, 7, and 14 after a systemic *Listeria* infection. FIG. 8E shows serum anti-DNP antibody titers upon KLH-DNP immunization (n=3/group). FIGS. 8F-8K show percentages of tetramer+ cells (FIG. 8F), macroscopic PBC scores and liver weight (FIG. 8G), microscopic PBC scores (FIG. 8H), liver images (FIG. 8I), microscopic tumor scores (FIG. 8J) and survival rates (FIG. 8K) of untreated vs. PDC.sub.166-181/IA.sup.g7-NP-treated NOD.c3c4 (n=5/group) or untreated Balb/c (n=7) injected with CT26 cells. FIGS. 8L-8P show percentages of tetramer+ cells (FIG. 8L), macroscopic (FIG. 8M) and microscopic PBC scores (FIG. 8N), and liver weight and metastasis number (FIG. 8O), and liver images (FIG. 8P) of untreated vs. PDC.sub.166-181/IA.sup.g7-NP-treated NOD.c3c4 (n=5 and 4) or untreated B6 (n=6) injected with B16/F10 cells. FIG. 8Q shows the survival rates of

the mice in FIGS. 8L-8P. Data correspond to mean $\pm$ SEM. P values were compared via Mann-Whitney U except for K, Q (log-rank) or C (two-way ANOVA).

(11) FIG. 9 illustrates expansion of T regulatory cells by PBC-relevant pMHC-NPs in humanized mice. FIG. 9A shows representative tetramer stains in NSG hosts engrafted with PBMCs from DRB4*0101+ PBC patients and treated with three different PBC-relevant pMHC-NPs. FIG. 9B shows average percentages and absolute numbers of tetramer+CD4⁺ cells in responsive vs. unresponsive pMHC-NP-treated mice or untreated littermates. FIG. 9C shows mean fluorescence intensity values (top) and two-dimensional FACS plots (bottom) for the TR1 markers CD49b and LAG3 in the mice studied in FIGS. 9A-B.

(12) FIG. 10 illustrates that PBC-relevant pMHC-NPs are able to treat disease in models of liver autoimmunity distinct from PBC, in particular, primary sclerosing cholangitis (PSC) and autoimmune hepatitis (AIH). FIG. 10A shows percentage of tetramer+CD4⁺ T-cells in NOD.Abcb4^{-/-} mice in response to pMHC-NP therapy. FIG. 10B shows average Primary Sclerosing Cholangitis scores (top) and representative H&E stained or picrosirius-stained liver sections (bottom), and TBA and ALT (right column) corresponding to the mice studied in FIG. 10A. FIG. 10C shows percentage of tetramer+CD4⁺ T-cells in NOD mice infected with an hFTCD-encoding Adenovirus (developing AIH) and treated with three different pMHC-NP types. FIG. 10D shows autoimmune hepatitis histopathological scores (top) and representative H&D and picrosirius liver stained sections (bottom) for the mice studied in FIG. 10C. FIG. 10E shows serum ALT levels in the mice in FIG. 10C [n=11, 5, 9 and 10 mice (left to right); 2-3 experiments].

(13) FIG. 11 illustrates that PBC and AIH-relevant pMHC-NPs expand T regulatory cells in a mouse model of PSC. FIG. 11A shows representative FACS staining histograms, and FIG. 11B shows average mean fluorescence intensity values for TR1 markers on tetramer+CD4⁺ vs. tetramer-CD4⁺ T-cells of NOD.Abcb4^{-/-} mice (spontaneously developing PSC) treated with PBC- or AIH-relevant pMHC-NPs.

(14) FIG. 12 illustrates that PBC relevant pMHC-NPs expand T regulatory cells in a mouse model of AIH. FIG. 12A shows representative FACS staining histograms, and FIG. 12B shows average mean fluorescence intensity values for TR1 markers on tetramer+CD4⁺ vs. tetramer-CD4⁺ T-cells of NOD mice infected with an adenovirus encoding hFTCD (spontaneously developing AIH) treated with PBC- or AIH-relevant pMHC-NPs.

(15) FIG. 13. illustrates the ability of ubiquitous autoantigen based pMHC-NPs to blunt liver autoimmune disease in an organ rather than disease-specific manner. FIG. 13A shows percentage of tetramer+CD4⁺ T-cells in NOD.c3c4 mice treated with an AIH relevant pMHC-NP type. FIG. 13B shows average macroscopic liver scores in NOD.c3c4 mice treated with PBC-AIH- or T1D-relevant pMHC-NP types. FIG. 13C shows percentage of tetramer+CD4⁺ T-cells in NOD mice treated with PBC- or AIH-relevant pMHC-NP types.

(16) FIG. 14 illustrates therapeutic effects of PBC-relevant pMHCII-NPs in (NODxB6.Ifng-ARE-Del^{-/-}) F1 mice. FIG. 14A shows percentages of tetramer+CD4⁺ T-cells in mice (males and females pooled) treated with Cys-NPs or PDC.sub.166-181/IA.sup.g7-NPs. FIG. 14B shows serum TBA and ALT levels in the mice studied in A. FIG. 14C shows microscopic scores of the female mice studied in A (left) and representative H&E (top right) and picrosirius-red-stained liver sections (bottom right). Data in FIGS. 14A-C correspond to n=7 and 12 mice/treatment type, respectively, from 2-3 experiments.

(17) FIG. 15 illustrates that disease-relevant pMHCII-NPs spare general immunity despite harboring PDC-E2 autoreactive TR1-like CD4⁺ T-cells. FIG. 15A shows average percentages of PDC.sub.166-181/IA.sup.g7 tetramer+CD4⁺ T-cells in blood and lymphoid organs of PDC.sub.166-181/IA.sup.g7 in NP-treated NOD.c3c4 mice 7 days after priming with HKx31 strain. Data correspond to n=4 and 5, respectively. FIG. 15B shows average percentages of PDC.sub.166-181/IA.sup.g7 tetramer+CD4⁺ T-cells in blood and lymphoid organs of PDC.sub.166-181/A87 in NP-treated NOD.c3c4 mice 14 days after infection with *Listeria*. Data

correspond to n=3 and 5 mice, respectively. Data correspond the mean±SEM. P values were compared via Mann-Whitney U.

(18) FIG. 16 illustrates that pMHCII-based nanomedicines displaying epitopes from ubiquitous autoantigens blunt relapsing-remitting EAE in NOD mice. FIG. 16A shows percentage of tetramer+CD4⁺ cells in blood, spleen and cervical LNs of pMHCII-NP-treated NOD mice with EAE. Data correspond to (from top): (i) MOG.sub.36-50/IA.sup.g7 tetramer staining for 5 Cys-NP- and 7 MOG.sub.36-50/IA.sup.g7-NP-treated mice; (ii) PDC.sub.166-181/IA.sup.g7 tetramer staining for 11 Cys-NP- plus 14 BDC.sub.2.5mi/IA.sup.g7-NP- and 22 PDC.sub.166-181/IA.sup.g7-NP-treated mice (pooled from 2 experiments); (iii) CYPD.sub.398-412/IA.sup.g7 tetramer staining for 11 Cys-NP- and 8 CYPD.sub.398-412/IA.sup.g7-NP-treated mice (pooled from 2 experiments); and (iv) BDC.sub.2.5mi/IA.sup.g7 tetramer staining for 14 PDC.sub.166-181/IA.sup.g7-NP- and 14 BDC.sub.2.5mi/IA.sup.g7-NP-treated mice. FIG. 16B shows and MFI for TR1 markers in splenic tetramer+CD4⁺ cells. Data in FIG. 16B correspond to 3 PDC.sub.166-181/IA.sup.g7-NP- and 4 BDC.sub.2.5mi/IA.sup.g7-NP-treated mice. FIG. 16C shows EAE scores in NOD mice treated with: (i) Cys-NPs (n=9), BDC.sub.2.5mi/IA.sup.g7-NPs (n=14), or MOG.sub.36-50/IA.sup.g7-NPs (n=8) (top); and (ii) Cys-NPs (n=11, from 2 experiments) PDC.sub.166-181/IA.sup.g7-NPs (n=22, from 2 experiments) or CYPD.sub.398-412/IA.sup.g7-NPs (n=8) (bottom). FIG. 16D shows representative luxol fast blue (LFB)/H&E staining pictures of cerebellum from the mice in FIG. 16C showing infiltration of inflammatory cells and areas of demyelination (white arrows). FIG. 16E shows average blinded histopathological rank order LFB scores for the mice in FIG. 16D. Data correspond to n=5, 7, 6, 8 and 7/NP type, from left to right in the bar diagram. Data correspond to the mean±SEM. P values were calculated using two-way ANOVA (FIG. 16C) or Mann-Whitney U test (FIGS. 16A, B, and E).

(19) FIG. 17 illustrates pMHCII-based nanomedicines displaying epitopes from ubiquitous autoantigens blunt chronic progressive EAE in C57BL/6 mice. FIG. 17A shows percentages of tetramer+CD4⁺ cells in in blood, spleen and cervical LNs of pMHCII-NP-treated C57BL/6J mice with EAE. Data correspond to samples from 6 Cys-NP- and 7 pMHCII-NP-treated mice per panel, stained with cognate pMHC tetramers (top: MOG.sub.38-49/IA.sup.b; middle: PDC.sub.94-108/IA.sup.b; and bottom: CYPD.sub.353-367/IA.sup.b). FIG. 17B shows and MFI of TR1 markers in splenic tetramer+CD4⁺ cells. Data in FIG. 17B correspond to 7 pMHCII-NP-treated mice in each panel. FIG. 17C shows EAE scores and % initial body weight changes in pMHCII-NP-treated C57BL/6J mice with EAE. Data corresponds to 6 Cys-NP-, 7 MOG.sub.38-49/IA.sup.b-NP-, 7 PDC.sub.94-108/IA.sup.b-NP- and 7 CYPD.sub.353-367/IA.sup.b-NP-treated mice. FIG. 17D shows average blinded histopathological rank order LFB scores for the mice in FIG. 17D. Data correspond to n=6, 6, 6 and 7/pMHC-NP type, from left to right in the bar diagram. Data correspond to the mean±SEM. P values were calculated using two-way ANOVA (FIG. 17C) or Mann-Whitney U test (FIGS. 17A, B, and D).

(20) FIG. 18 illustrates autoantigen shedding renders peripheral autoreactive T-cells responsive to pMHCII-NP therapy. FIG. 18A shows percentage of tetramer+CD4⁺ cells in the blood of pMHCII-NP-treated vs. untreated NOD mice before (time 0) and 2.5 and 5 weeks after treatment initiation. Data correspond to (from left to right panels): (i) 5 PDC.sub.166-181/IA.sup.g7-NP treatment (2 experiments); (ii) 5 CYPD.sub.398-412/IA.sup.g7-NP treatment; and (iii) 5 untreated mice (2 experiments). Blood samples were stained with BDC.sub.2.5mi/IA.sup.g7, PDC.sub.166-181/IA.sup.g7 and CYPD.sub.398-412/IA.sup.g7 tetramers. FIG. 18B shows percentages of tetramer+CD4⁺ cells in different lymphoid organs and liver of NOD mice treated with PDC.sub.166-181/IA.sup.g7-NPs or CYPD.sub.398-412/IA.sup.g7-NPs. PCLN, portal and celiac LNs; PLN: pancreatic LNs; MLN, mesenteric LNs; BM, bone marrow. Data corresponds to 9 untreated (2 experiments), 5 PDC.sub.166-181/IA.sup.g7-NP-treated (2 experiments) and 5 CYPD.sub.398-412/IA.sup.g7-NP-treated mice. FIG. 18C shows percentages of tetramer+CD4⁺ cells in PCLN, PLN and spleen of DT-treated RIP-DTR-transgenic NOD mice following pMHCII-

NP therapy. Data correspond to 6 Cys-NP-, 5 BDC.sub.2.5mi/IA.sup.g7-NP-, 5 PDC.sub.166-181/IA.sup.g7-NP- and 5 CYPD.sub.398-412/IA.sup.g7-NP-treated mice from 2 independent experiments. Samples from mice treated with BDC.sub.2.5mi/IA.sup.g7-NPs and CYPD.sub.398-412/IA.sup.g7-NPs were stained with both pMHCII tetramers (cognate and non-cognate). Samples from mice treated with Cys-NPs, PDC.sub.166-181/IA.sup.g7-NPs and MOG.sub.36-50/IA.sup.g7-NPs were stained with PDC.sub.166-181/IA.sup.g7 or MOG.sub.36-50/IA.sup.g7 tetramers, respectively. FIG. 18 D shows mean fluorescence intensity (MFI) for TR1 markers in splenic tetramer+CD4+ cells. Data are presented as mean±SEM. P values were calculated using ANOVA (FIG. 18A) or Mann-Whitney U test (FIGS. 18B, C and D).

(21) FIG. 19 illustrates that a mouse model of psoriasis can prime ubiquitous autoantigen, and that administration of ubiquitous autoantigen pMHCII-NP therapy can reduce signs of psoriasis. FIG. 19A shows percent tetramer staining of PDC.sub.166-181/IA.sup.g7 T cells. FIG. 19B shows improvements of ear thickness in mice administered, PDC.sub.166-181/IA.sup.g7-NPs.

(22) FIG. 20 illustrates a non-limiting embodiment of a pMHC-NP of this disclosure. FIG. 20A shows an alpha chain of an MHC class II dimer fused to the CH2 and CH3 domain of an immunoglobulin molecule comprising an engineered knob (SEQ ID NO:103). FIG. 20B shows a beta chain of an MHC class II with an N-terminal ubiquitous polypeptide (E2.sub.122-135), fused to the beta chain of the MHC class II, and fused to the CH2 and CH3 domains of an immunoglobulin molecule comprising an engineered hole (SEQ ID NO:104).

(23) TABLE 1 illustrates linkers useful for coupling ubiquitous autoantigen-MHCs to nanoparticles.

(24) TABLES 2, 3, and 4 illustrate percentages and absolute numbers of tetramer+CD4+ T-cells in NSG mice engrafted with PBMCs from DRB4*0101+ PBC patients, upon treatment with three different human PBC-relevant pMHC-NP types.

DETAILED DESCRIPTION

(25) Various embodiments are described hereinafter. It should be noted that the specific embodiments are not intended as an exhaustive description or as a limitation to the broader aspects discussed herein. One aspect described in conjunction with a particular embodiment is not necessarily limited to that embodiment and can be practiced with any other embodiment(s).

(26) The use of the terms “a” and “an” and “the” and similar referents in the context of describing the elements (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context. Recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the embodiments and does not pose a limitation on the scope of the claims unless otherwise stated. No language in the specification should be construed as indicating any non-claimed element as essential.

(27) “Antigen” as used herein refers to all, part, fragment, or segment of a molecule that can induce an immune response in a subject or an expansion of an immune cell, preferably a T or B cell. Antigens can be polypeptides, lipids, carbohydrates, or nucleic acids.

(28) As used herein “individual” is synonymous with “subject” or “patient”. The individual can be diagnosed with a disease. The individual can suspected of having a particular disease based on manifesting at least one symptom of said disease, having a family history of said disease, having a genotype relevant to define risk for said disease, or having one or more phenotypic measurements or “lab tests” at or near a level that would place an individual at risk for the disease. The individual can be a mammal, such as a horse, cat, dog, pig, cow, goat, or sheep. The individual can in certain

instances be a human person.

(29) As used herein, “about” will be understood by persons of ordinary skill in the art and will vary to some extent depending upon the context in which it is used. If there are uses of the term which are not clear to persons of ordinary skill in the art, given the context in which it is used, “about” will mean up to plus or minus 10% of the particular term.

(30) As used herein, “polypeptide” refers to a plurality of amino acids joined by peptide bonds having more than about eight amino acid residues. The amino acids of the polypeptide can be naturally occurring or unnatural amino acid residues.

(31) Percent (%) sequence identity with respect to a reference polypeptide sequence is the percentage of amino acid residues in a candidate sequence that are identical with the amino acid residues in the reference polypeptide sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Alignment for purposes of determining percent amino acid sequence identity can be achieved in various ways that are known for instance, using publicly available computer software such as BLAST, BLAST-2, ALIGN or Megalign (DNASTAR) software. Appropriate parameters for aligning sequences are able to be determined, including algorithms needed to achieve maximal alignment over the full length of the sequences being compared. For purposes herein, however, % amino acid sequence identity values are generated using the sequence comparison computer program ALIGN-2. The ALIGN-2 sequence comparison computer program was authored by Genentech, Inc., and the source code has been filed with user documentation in the U.S. Copyright Office, Washington D.C., 20559, where it is registered under U.S. Copyright Registration No. TXU510087. The ALIGN-2 program is publicly available from Genentech, Inc., South San Francisco, Calif., or may be compiled from the source code. The ALIGN-2 program should be compiled for use on a UNIX operating system, including digital UNIX V4.0D. All sequence comparison parameters are set by the ALIGN-2 program and do not vary.

(32) In situations where ALIGN-2 is employed for amino acid sequence comparisons, the % amino acid sequence identity of a given amino acid sequence A to, with, or against a given amino acid sequence B (which can alternatively be phrased as a given amino acid sequence A that has or comprises a certain % amino acid sequence identity to, with, or against a given amino acid sequence B) is calculated as follows: 100 times the fraction X/Y , where X is the number of amino acid residues scored as identical matches by the sequence alignment program ALIGN-2 in that program's alignment of A and B, and where Y is the total number of amino acid residues in B. It will be appreciated that where the length of amino acid sequence A is not equal to the length of amino acid sequence B, the % amino acid sequence identity of A to B will not equal the % amino acid sequence identity of B to A. Unless specifically stated otherwise, all % amino acid sequence identity values used herein are obtained as described in the immediately preceding paragraph using the ALIGN-2 computer program.

(33) The uaMHC of the uaMHC-NP of the current disclosure, described herein, can be encoded by a nucleic acid. A nucleic acid is a type of polynucleotide comprising two or more nucleotide bases. In certain embodiments, the nucleic acid is a component of a vector that can be used to transfer the polypeptide encoding polynucleotide into a cell. As used herein, the term “vector” refers to a nucleic acid molecule capable of transporting another nucleic acid to which it has been linked. One type of vector is a genomic integrated vector, or “integrated vector,” which can become integrated into the chromosomal DNA of the host cell. Another type of vector is an “episomal” vector, e.g., a nucleic acid capable of extra-chromosomal replication. Vectors capable of directing the expression of genes to which they are operatively linked are referred to herein as “expression vectors.” Suitable vectors comprise plasmids, bacterial artificial chromosomes, yeast artificial chromosomes, viral vectors and the like. In the expression vectors regulatory elements such as promoters, enhancers, polyadenylation signals for use in controlling transcription can be derived from

mammalian, microbial, viral or insect genes. The ability to replicate in a host, usually conferred by an origin of replication, and a selection gene to facilitate recognition of transformants may additionally be incorporated. Vectors derived from viruses, such as lentiviruses, retroviruses, adenoviruses, adeno-associated viruses, and the like, may be employed. Plasmid vectors can be linearized for integration into a chromosomal location. Vectors can comprise sequences that direct site-specific integration into a defined location or restricted set of sites in the genome (e.g., AttP-AttB recombination). Additionally, vectors can comprise sequences derived from transposable elements.

(34) Any of the nucleic acids encoding the uaMHC or the vectors comprising said nucleic acids can be transferred to a suitable cell line for the production of uaMHC. In certain embodiments, the nucleic acid or vector is stably integrated into the genome of the cell line. Suitable cell lines can be as Vero (ATCC CRL 81) or CHO-K1 (ATCC CRL 61) cells, HeLa cells and L cells. Exemplary eukaryotic cells that can be used to express polypeptides include, but are not limited to, COS cells, including COS 7 cells; 293 cells, including 293-6E cells; CHO cells, including CHO—S and DG44 cells; PER.C6™ cells (Crucell); and NSO cells.

(35) Described herein in certain embodiments, is a composition comprising: a plurality of antigen-major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, that is not a tissue specific antigen, associated with the binding groove of an MHC molecule; and a nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core.

(36) Described herein in certain embodiments, is a method of treating an autoimmune or inflammatory disease comprising administering to an individual a therapeutically effective amount of a composition comprising: (a) a plurality of antigen-major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, that is not a tissue specific antigen, associated with the binding groove of an MHC molecule; (b) and a nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core.

(37) Described herein in certain embodiments, is a method of treating multiple sclerosis comprising administering to an individual a therapeutically effective amount of a composition comprising: (a) a plurality of antigen-major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, which is not a multiple sclerosis specific antigen, associated with the binding groove of an MHC molecule; (b) and a nanoparticle core, the nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core.

(38) Described herein in certain embodiments, is a method of treating type I diabetes comprising administering to an individual a therapeutically effective amount of a composition comprising: (a) a plurality of antigen-major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, which is not a type I diabetes specific antigen, associated with the binding groove of an MHC molecule; and (b) a nanoparticle core, the nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the plurality of antigen-MHCs is coupled to the nanoparticle core.

(39) Ubiquitous Autoantigens

(40) Described herein are nanoparticle compositions and methods useful for treating autoimmune diseases and inflammatory disorders. The nanoparticle compositions comprise a plurality of antigens associated with MHCs coupled to a nanoparticle. The nanoparticle compositions and methods utilize broadly expressed ubiquitous autoantigens to elicit the generation of regulatory T and B lymphocytes.

(41) In a certain aspect, the antigens that are associated with the MHC molecules are ubiquitous autoantigens. Ubiquitous autoantigens are differentiated from tissue specific antigens at least in that they are antigens commonly expressed by a plurality of different cell types that are unrelated. In

certain embodiments, a ubiquitous autoantigen is one that is commonly expressed by ontogenically distinct tissues. In certain embodiments, a ubiquitous autoantigen that is one that is expressed in at least two cell types derived from a tissue originating from the list consisting of ectoderm, mesoderm, and endoderm. In certain embodiments, a ubiquitous autoantigen is one that is commonly expressed by functionally distinct tissues. In certain embodiments, a ubiquitous autoantigen is one that is expressed in at least two tissues selected from the list consisting of neural tissue, endocrine tissue, connective tissue, hematopoietic cells, liver tissue, cardiac tissue, skin tissue, lung tissue, vascular tissue, intestinal tissue, and stomach tissue. In certain embodiments, a ubiquitous autoantigen is one that is expressed in both neural tissue and liver tissue. In certain embodiments, a ubiquitous autoantigen is one that is expressed in both neural tissue and pancreatic tissue. In a certain embodiment, the ubiquitous autoantigen is a polypeptide derived from a protein that participates in a cellular process common to at least 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more cell types. Ubiquitous autoantigens may be sequences that are common to two or more closely related proteins that are recognized as being paralogs or homologs in the same family, yet display differential expression across various tissues. In certain embodiments, the two or more closely related proteins can for example, perform the same or similar function in two different unrelated tissues. In a certain embodiment, the ubiquitous autoantigen is a polypeptide derived from a protein that participates in a cellular process, wherein the cellular process is a metabolic process selected from glycolysis, oxidative phosphorylation, glycogenesis, nucleotide biosynthesis, beta oxidation, and omega oxidation. In certain embodiments, the ubiquitous autoantigen is selected from the list consisting of pyruvate dehydrogenase complex-E2 component (PDC-E2), Cytochrome P450 2D6 (CYP2D6), formimidoyltransferase-cyclodeaminase (FTCD), soluble liver antigen (SLA), actin (ACTB), and myeloperoxidase (MPO).

(42) Ubiquitous autoantigens are often encoded by housekeeping genes that are utilized in a variety of cell types. For example, actin is a cytoskeletal protein that contributes to cell structure, motility, cell division, and vesicle motility and is also ubiquitously expressed. As such, many ubiquitous autoantigens are intracellular and reside in a particular intracellular compartment at a steady-state. An antigen exists at steady-state in the cellular location where the antigen can be found at its highest quantities, determined, for example, by microscopy or cell fractionation. For example, despite the fact that actin can be found extracellularly associated with exosomes the vast majority of actin is found in the cytosol of the cell. Likewise, many antigens may transit through different organelles but reside primarily in a single organelle. For example, many endoplasmic reticulum (ER) resident proteins will transiently transit through the cis-Golgi, but are immediately returned to the ER, where they reside at steady-state.

(43) In certain embodiments, the ubiquitous autoantigens for use with the nanoparticle compositions described herein are polypeptides derived from pyruvate dehydrogenase complex-E2 component (PDC-E2). In certain embodiments, the polypeptide derived from PDC-E2 is any one or more of: PDC-E2.sub.353-367, PDC-E2.sub.72-86 and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94 and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135 and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801. In certain embodiments, the polypeptide derived from PDC-E2 is any one or more of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94. In certain embodiments, the polypeptide derived from PDC-E2 is any one or more of: SEQ ID NOs: 1 to 12.

(44) In certain embodiments, the ubiquitous autoantigens for use with the nanoparticle compositions described herein are polypeptides derived from Cytochrome P450 2D6 (CYP2D6). In certain embodiments, the polypeptide derived from CYP2D6 is any one or more of: CYP2D6.sub.284-298, CYP2D6.sub.289-303, CYP2D6.sub.318-332, CYP2D6.sub.313-332, CYP2D6.sub.393-412, CYP2D6.sub.192-206, CYP2D65-19, CYP2D6.sub.293-307 (for DRB1*0301); CYP2D6.sub.219-233, CYP2D6.sub.237-251, CYP2D6.sub.15-29 (for DRB3*0202); CYP2D6.sub.235-249, CYP2D6.sub.317-331, CYP2D6.sub.293-307 (for

DRB4*0101); CYP2D6.sub.428-442, CYP2D6.sub.237-251, CYP2D6.sub.14-28 (for DRB5*0101); CYP2D6.sub.199-213, CYP2D6.sub.450-464, CYP2D6.sub.301-315 (for DRB1*0401); CYP2D6.sub.452-466, CYP2D6.sub.59-73, CYP2D6.sub.130-144, CYP2D6.sub.193-212, CYP2D6.sub.305-324, CYP2D6.sub.15-29 (for DRB1*0701). In certain embodiments, the polypeptide derived from CYP2D6 is any one or more of: SEQ ID NOs: 13 to 37.

(45) In certain embodiments, the ubiquitous autoantigens for use with the nanoparticle compositions described herein are polypeptides derived from soluble liver antigen (SLA). In certain embodiments, the polypeptide derived from SLA is any one or more of: SLA.sub.334-348, SLA.sub.196-210, SLA.sub.115-129, SLA.sub.373-386, SLA.sub.186-197 (for DRB1*0301); SLA.sub.342-256, SLA.sub.110-124, SLA.sub.299-313 (for DRB3*0202); SLA.sub.49-63, SLA.sub.260-274, SLA.sub.119-133 (for DRB4*0101); SLA.sub.86-100, SLA.sub.26-40, SLA.sub.331-345 (for DRB5*0101); SLA.sub.317-331, SLA.sub.171-185, SLA.sub.417-431 (for DRB1*0401); SLA.sub.359-373, SLA.sub.215-229, SLA.sub.111-125 (for DRB1*0701). In certain embodiments, the polypeptide derived from SLA is any one or more of: SEQ ID NOs: 53 to 72.

(46) In certain embodiments, the ubiquitous autoantigens for use with the nanoparticle compositions described herein are polypeptides derived from actin (ACTB). In certain embodiments, the polypeptide derived from ACTB is any one or more of: ACTB.sub.202-216, ACTB.sub.170-184, ACTB.sub.245-259, (for DRB1*0301); ACTB.sub.187-201, ACTB.sub.172-186, ACTB.sub.131-145 (for DRB3*0202); ACTB.sub.131-145, ACTB.sub.171-185, ACTB.sub.129-143 (for DRB4*0101); ACTB.sub.164-178, ACTB.sub.25-39, ACTB.sub.323-337 (for DRB5*0101). In certain embodiments, the polypeptide derived from ACTB is any one or more of: ACTB.sub.146-160, ACTB.sub.18-32, and ACTB.sub.171-185. In certain embodiments, the polypeptide derived from ACTB is any one or more of: SEQ ID NOs: 38 to 52.

(47) In certain embodiments, the ubiquitous autoantigens for use with the nanoparticle compositions described herein are polypeptides derived from formimidoyltransferase-cyclodeaminase (FTCD). In certain embodiments, the polypeptide derived from FTCD is any one or more of: FTCD.sub.439-453, FTCD.sub.381-395, FTCD.sub.297-311 (for DRB3*0202); FTCD.sub.525-539, FTCD.sub.218-232, FTCD.sub.495-509 (for DRB1*0301); FTCD.sub.262-276, FTCD.sub.300-314, FTCD.sub.259-273 (for DRB4*0101); FTCD.sub.490-504, FTCD.sub.389-403, FTCD.sub.295-309 (for DRB5*0101). In certain embodiments, the polypeptide derived from FTCD is any one or more of: FTCD.sub.271-285, FTCD 498-512, and FTCD.sub.301-315. In certain embodiments, the polypeptide derived from FTCD is any one or more of: SEQ ID NOs: 73 to 87.

(48) In certain embodiments, the ubiquitous autoantigens for use with the nanoparticle compositions described herein are polypeptides derived from myeloperoxidase (MPO). In certain embodiments, the polypeptide derived from MPO is any one or more of: MPO.sub.322-336, MPO.sub.714-728, MPO.sub.617-631 (for DRB3*0202); MPO.sub.504-518, MPO.sub.462-476, MPO.sub.617-631 (for DRB1*0301); MPO.sub.444-458, MPO.sub.689-703, MPO.sub.248-262 (for DRB4*0101); MPO.sub.511-525, MPO.sub.97-111, MPO.sub.616-630 (for DRB5*0101). In certain embodiments, the polypeptide derived from MPO is any one or more of: SEQ ID NOs: 88 to 99.

(49) Additional ubiquitous autoantigens are listed in Table 5 at the end of this disclosure. In certain embodiments, the ubiquitous autoantigens for use with the nanoparticle compositions described herein are derived from a protein or polypeptide listed in Table 5. In certain embodiments, the ubiquitous autoantigens for use with the nanoparticle compositions described herein is derived from a human homolog to protein or polypeptide listed in Table 5. In certain embodiments, a homologue or human homologue is a protein or polypeptide that displays at least about 75%, 80%, 85%, 90%, 95%, or 98% identity to a protein listed in Table 5.

(50) Tissue Specific Antigens

(51) The nanoparticle compositions and methods described herein utilize ubiquitous autoantigens that are not tissue specific antigens. Many autoimmune or inflammatory diseases are associated with an immune response directed to a tissue specific antigen. This presents a problem for the production of a medicament to treat an autoimmune or inflammatory disease, since each disease requires a specific medicament that targets that antigen. Alternatively, nonspecific immune inhibitors can be used, but these are associated with significant systemic side-effects.

(52) Tissue specific antigens are often expressed by a tissue or cell type affected by the autoimmune disease, for example a main pathological consequence of multiple sclerosis is demyelination of nervous system tissue, as a consequence tissue specific antigens for multiple sclerosis are largely restricted to the nervous system (e.g., myelin basic protein). Tissue specific antigens are those antigens that are associated with a specific cell or cell type. Tissue specific antigens can perform specialized functions or contribute to specialized tissue structures. In certain embodiments, a tissue specific antigen has expression restricted to any one of the following tissues: neural, kidney, cardiac, lung, liver, small intestine, colon, stomach, muscle, connective, and blood-vessel. In certain embodiments, a tissue specific antigen is restricted to expression of any one of the following cell types: beta cells, alpha cells, B lymphocytes, T lymphocytes, Schwann cells, adrenocortical cells.

(53) Many tissue specific antigens may be expressed at a very low level in other cell or tissue types, but the main source of expression is one specific cell or tissue type. For example, a single cell or tissue type that displays cell specific or tissue type specific expression of a certain gene will express at least 10-fold, 50-fold, 100-fold, 500-fold, 1,000 fold or more of the gene at the mRNA or protein level than any other unrelated cell-type. Additionally, some tissue specific antigens will gain ectopic expression of a cell specific antigen under a pathogenic condition or by an exogenous stimulation. It is intended that merely because a different cell-type may gain ectopic expression under pathological or exogenous conditions the tissue specific nature of the antigen is not lost. For example, insulin is a tissue specific antigen produced by beta cells, yet due to genetic instability, some tumors (known as insulinomas) will express insulin, and under these types of circumstances insulin is still considered tissue specific.

(54) Tissue specific antigens that are not ubiquitous autoantigens are primarily antigens associated with a particular tissue specific autoimmune or inflammatory disease.

(55) In certain embodiments, the autoimmune or inflammatory disease is multiple sclerosis. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from myelin basic protein, myelin associated glycoprotein, myelin oligodendrocyte protein (MOG), proteolipid protein, oligodendrocyte myelin oligoprotein, myelin associated oligodendrocyte basic protein, oligodendrocyte specific protein, heat shock proteins, an oligodendrocyte specific protein, NOGO A, glycoprotein Po, peripheral myelin protein 22, and/or 2'3'-cyclic nucleotide 3'-phosphodiesterase.

(56) In certain embodiments, the autoimmune or inflammatory disease is type I diabetes. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from pre-proinsulin, proinsulin, islet-specific glucose-6-phosphatase (IGRP), glutamate decarboxylase (GAD), islet cell autoantigen-2 (ICA2), and/or insulin.

(57) In certain embodiments, the autoimmune or inflammatory disease is Pemphigus foliaceus (PF) or Pemphigus vulgaris (PV). In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from desmoglein 3 (DG3) and/or desmoglein 1 (DG1).

(58) In certain embodiments, the autoimmune or inflammatory disease is Neuromyelitis optica spectrum disorder (NMO). In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from aquaporin 4 (AQP4).

(59) In certain embodiments, the autoimmune or inflammatory disease is Arthritis. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide

derived from a heat shock protein, immunoglobulin binding protein, heterogeneous nuclear RNPs, annexin V, calpastatin, type II collagen, glucose-6-phosphate isomerase, elongation factor human cartilage gp39, mannose binding lectin, citrullinated vimentin, type II collagen, fibrinogen, alpha enolase, anti-carbamylated protein (anti-CarP), peptidyl arginine deiminase type 4 (PAD4), BRAF, fibrinogen gamma chain, inter-alpha-trypsin inhibitor heavy chain H1, alpha-1-antitrypsin, plasma protease C1 inhibitor, gelsolin, alpha 1-B glycoprotein, ceruloplasmin, inter-alpha-trypsin inhibitor heavy chain H4, complement factor H, alpha 2 macroglobulin, serum amyloid, C-reactive protein, serum albumin, fibrogen beta chain, serotransferin, alpha 2 HS glycoprotein, vimentin, and/or Complement C3

(60) In certain embodiments, the autoimmune or inflammatory disease is allergic asthma. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from DERP1 and/or DERP2.

(61) In certain embodiments, the autoimmune or inflammatory disease is inflammatory bowel disease. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from bacteroides integrase, flagelin, flagellin 2 (Fla-2/Fla-X), or uncharacterized *E. coli* protein (YIDX).

(62) In certain embodiments, the autoimmune or inflammatory disease is systemic lupus erythematosus disease. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide H4, H2B, H1', dsDNA, RNP, Smith (Sm), Sjogren's Syndrome-related Antigen A (SS-A)/Ro, Sjogren's Syndrome-related Antigen B (SS-B)/La, and/or histones. In some embodiments, SS-A includes, but is not limited to, RO60 and RO52. In some embodiments, histones include but are not limited to H4, H2B, H1.

(63) In certain embodiments, the autoimmune or inflammatory disease is atherosclerosis. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from Apolipoprotein B (ApoB) and/or Apolipoprotein E (ApoE).

(64) In certain embodiments, the autoimmune or inflammatory disease is chronic obstructive pulmonary disease (COPD). In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from elastin.

(65) In certain embodiments, the autoimmune or inflammatory disease is psoriasis. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from human adamis-like protein 5 (ATL5), cathelicidin antimicrobial peptide (CAP18), and/or ADAMTS-like protein 5 (ADMTSL5).

(66) In certain embodiments, the autoimmune or inflammatory disease is uveitis. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from arrestin, human retinal S-antigen, and/or interphotoreceptor retinoid-binding protein (IRBP).

(67) In certain embodiments, the autoimmune or inflammatory disease is Sjogren's syndrome. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from (SS-A)/Ro, (SS-B)/La, RO60, RO52, and/or muscarinic receptor 3 (MR3).

(68) In certain embodiments, the autoimmune or inflammatory disease is scleroderma. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from centromere autoantigen centromere protein C (CENP-C), DNA topoisomerase I (TOP1), and/or RNA polymerase III.

(69) In certain embodiments, the autoimmune or inflammatory disease is anti-phospholipid syndrome. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from beta-2-glycoprotein 1 (BG2P1 or APOH).

(70) In certain embodiments, the autoimmune or inflammatory disease is stiff man syndrome. In certain embodiments, the tissue specific antigen that is not a ubiquitous autoantigen is a polypeptide derived from GAD65.

(71) Antigen-Major Histocompatibility Complexes (MHCs)

(72) The nanoparticle complexes of this disclosure comprise a nanoparticle core, with or without layers and/or coatings, coupled to a ubiquitous autoantigen-MHC. The individual MHC polypeptide(s) and the antigenic (e.g., polypeptide) components form a complex through covalent or non-covalent binding (e.g. through hydrogen bonds, ionic bonds, or hydrophobic bonds). The preparation of such complexes may require varying degrees of manipulation and such methods are well known in the literature. In some aspects, antigenic components can be associated non-covalently with the pocket portion of the MHC component by, for instance, mixing the MHC and antigenic components; this relies on the natural binding affinity between an MHC and an antigen. Alternatively, in some aspects, the MHC component may be covalently associated with the antigenic component using standard procedures, such as, but not limited to, the introduction of known coupling agents or photo affinity labelling (see e.g., Hall et al., *Biochemistry* 24:5702-5711 (1985)). In certain aspects, an antigenic component may be operatively coupled to the MHC component via peptide linkages or other methods discussed in the literature, including but not limited to, attachment via carbohydrate groups on the glycoproteins, including, e.g., the carbohydrate moieties of the alpha- and/or beta-chains. In particular embodiments, the antigenic component may be attached to the N-terminal or C-terminal end of an appropriate MHC molecule. Alternatively, in certain embodiments, the MHC complex may be recombinantly formed by incorporating the sequence of the antigenic component into a sequence encoding an MHC, such that both retain their functional properties.

(73) Multiple ubiquitous autoantigen-MHCs may be coupled to the same nanoparticle core; these complexes, MHCs, and/or antigens may be the same or different from one another.

(74) Major Histocompatibility Molecules

(75) The ubiquitous autoantigens described herein are associated with MHC molecules (to form the ubiquitous autoantigen-MHC), and coupled to nanoparticles. The antigens are bound to the binding groove of the MHC molecule. MHC molecules primarily bind antigens that are polypeptides, but polypeptides can comprise modifications such as lipidation, glycosylation, phosphorylation and the like. The MHC molecule can be an MHC class I molecule (MHCI) or an MHC class II molecule (MHCII). MHC class I molecules bind polypeptides between 8-10 amino acid residues in their binding groove, as the binding groove is closed on either side. MHC class II molecules, including those described herein, bind polypeptides at least 8 amino acids residues in length, but can bind longer peptides, with lengths of 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19 amino acid residues or longer, as the binding groove is open on either side.

(76) For use with human individuals the MHC molecules utilized herein are human (also referred to as human leukocyte antigens, abbreviated "HLA"). In certain embodiments, the MHC class I molecule is a classical or a non-classical MHC class I molecule HLA-A, HLA-B, HLA-C, HLA-E, CD1d, or a fragment or biological equivalent thereof. In certain embodiments, the MHC class II molecule is an HLA-DRB1, HLA-DRB3, HLA-DRB4, HLA-DRB5, HLA-DQB1, or HLA-DPB1, or a fragment or biological equivalent thereof. In some embodiments, the antigen-MHC (pMHC) can be a single chain construct. In some embodiments, the pMHC can be a dual-chain construct. In the case of a class II MHC, the beta chain of HLA will generally be non-covalently bound with an appropriate alpha chain to form a dual chain heterodimer with the alpha chain paired to the beta chain. Generally, the alpha chain of the MHC class II exhibits a much lower degree of polymorphism, for example, the DR alpha chain.

(77) Since MHC class II complexes are heterodimers comprising an alpha and a beta chain, the heterodimers can have problems forming under some conditions, or are inherently unstable in some circumstances. When MHC class II molecules are deployed in a method or composition herein, the MHC molecules can further comprise a knob-in-hole architecture. In general, the alpha or beta chain is fused to an antibody C.sub.H2 and C.sub.H3 domain that has been modified to comprise a protuberance, while the corresponding other alpha or beta chain of the heterodimer is fused to an

antibody C.sub.H2 and C.sub.H3 domain that has been modified to comprise a cavity.

(78) As used herein, “knob-in-hole” or “knob-into-hole” refers to a polypeptidyl architecture requiring a protuberance (or “knob”) at an interface of a first polypeptide and a corresponding cavity (or a “hole”) at an interface of a second polypeptide, such that the protuberance can be positioned in the cavity so as to promote heterodimer formation. Protuberances may be constructed by replacing small amino acid side chains from the interface of the first polypeptide with larger side chains (e.g., phenylalanine or tyrosine). Cavities of identical or similar size to the protuberances may be created in the interface of the second polypeptide by replacing large amino acid side chains with smaller ones (e.g., alanine or threonine). The protuberances and cavities can be made by synthetic means such as by altering the nucleic acid encoding the polypeptides or by peptide synthesis, using routine methods by one skilled in the art. In some embodiments, the interface of the first polypeptide is located on an Fc domain in the first polypeptide; and the interface of the second polypeptide is located on an Fc domain on the second polypeptide. Knob-in-hole heterodimers and methods of their preparation and use are disclosed in U.S. Pat. Nos. 5,731,168; 5,807,706; 5,821,333; 7,642,228; 7,695,936; 8,216,805; and 8,679,785; and in Merchant et al., *Nature Biotechnology*, 1998, 16:677-681, all of which are incorporated by reference herein in their entirety.

(79) Alternatively or in addition any of the antigens described herein can comprise a cysteine residue that interacts with a cysteine residue (engineered or natural) of an MHC class II alpha or beta chain. This is commonly known as a cysteine trap.

(80) A cysteine trap can be utilized to stabilize a heterodimer described herein. Cysteine trapping involves forming covalently joined polypeptide complexes from unbound polypeptide partners. In some embodiments, cysteine trapping comprises introducing a cysteine at a strategically selected position within the interaction interface of the polypeptide partners to form a stabilized polypeptide complex. In some embodiments, cysteine trapping may stabilize the polypeptide complex to favor a specific conformation and to prevent dissociation. Cysteine trapping is also referred to as disulfide trapping and disulfide crosslinking. Examples of methods and applications of cysteine trapping are reviewed in Kufareva, et al., *Methods Enzymol.* 570:389-420 (2016). In the context of MHC, a cysteine is engineered into a polypeptide that is known or suspected to associate in the binding groove of an MHC class II dimer. A cysteine is then engineered in or near the binding groove such that, when the polypeptide associates with the binding groove, the binding groove cysteine can come into proximity and form a disulfide linkage with a polypeptide cysteine.

(81) Provided herein, in one aspect, are isolated heterodimers comprising at least one first polypeptide and at least one second polypeptide, wherein the first polypeptide and the second polypeptide meet at an interface, wherein the interface of the first polypeptide comprises an engineered protuberance which is positionable in an engineered cavity in the interface of the second polypeptide; and (i) the first polypeptide comprises an MHC class II $\alpha 1$ domain, an MHC class II $\alpha 2$ domain, or a combination thereof; and the second polypeptide comprises an MHC class II $\beta 1$ domain, an MHC class II $\beta 2$ domain, or a combination thereof; or (ii) the first polypeptide comprises an MHC class II $\beta 1$ domain, an MHC class II $\beta 2$ domain, or a combination thereof; and the second polypeptide comprises an MHC class II $\alpha 1$ domain, an MHC class II $\alpha 2$ domain, or a combination thereof. The first polypeptide, the second polypeptide, or both can comprise an antibody C.sub.H3 domain fused to the polypeptide. Optionally, the first polypeptide, the second polypeptide, or both comprise an antibody C.sub.H2 domain located between the MHC (α or β chain) and the C.sub.H3 domain. In certain embodiments, the first polypeptide comprises an antibody C.sub.H3 domain, and the antibody C.sub.H3 domain comprises at least one mutation selected from the list consisting of S354C, T366W, and both S354C and T366W (EU numbering). In certain embodiments, the second polypeptide comprises an antibody C.sub.H3 domain, and the antibody C.sub.H3 domain comprises at least one mutation selected from the list consisting of Y349C, T366S, L368A, Y407V (EU numbering), and combinations thereof. In further

embodiments, the isolated heterodimer comprises a ubiquitous autoantigen, optionally covalently bound to either the first or the second polypeptide. Optionally, the ubiquitous autoantigen comprises a cysteine residue that interacts with a cysteine residue in either the first or second polypeptide to create a cysteine trap.

(82) In one aspect, one polypeptide of the heterodimer comprises an MHC class II $\alpha 1$ domain, an MHC class II $\alpha 2$ domain, or a combination thereof; and at least one engineered protuberance. In some embodiments, the at least one engineered protuberance is not located at the MHC class II $\alpha 1$ domain or the MHC class II $\alpha 2$ domain. In some embodiments, the engineered protuberance is located at an antibody C.sub.H3 domain fused to the polypeptide. In some embodiments, the polypeptide optionally comprises an antibody C.sub.H2 domain located between an MHC class II $\alpha 2$ domain and the C.sub.H3 domain with an engineered protuberance. In certain embodiments, the polypeptide comprises an antibody C.sub.H3 domain, and the antibody C.sub.H3 domain comprises at least one mutation selected from the list consisting of S354C, T366W, and both S354C and T366W (EU numbering). In further embodiments, the polypeptide comprises a ubiquitous autoantigen. Optionally, the ubiquitous autoantigen comprises a cysteine residue that interacts with a cysteine residue in either an MHC $\alpha 1$ or $\beta 1$ domain to create a cysteine trap.

(83) In one aspect, one polypeptide of the heterodimer comprises an MHC class II $\beta 1$ domain, an MHC class II $\beta 2$ domain, or a combination thereof; and at least one engineered protuberance. In some embodiments, the at least one engineered protuberance is not located at the MHC class II $\beta 1$ domain or the MHC class II $\beta 2$ domain. In some embodiments, the engineered protuberance is located at an antibody C.sub.H3 domain fused to the polypeptide. In some embodiments, the polypeptide optionally comprises an antibody C.sub.H2 domain located between an MHC class II $\beta 2$ domain and the C.sub.H3 domain with an engineered protuberance. In certain embodiments, the polypeptide comprises an antibody C.sub.H3 domain, and the antibody C.sub.H3 domain comprises at least one mutation selected from the list consisting of S354C, T366W, and both S354C and T366W (EU numbering). In further embodiments, the polypeptide comprises a ubiquitous autoantigen. Optionally, the ubiquitous autoantigen comprises a cysteine residue that interacts with a cysteine residue in either an MHC $\alpha 1$ or $\beta 1$ domain to create a cysteine trap.

(84) In one aspect, one polypeptide of the heterodimer comprises an MHC class II $\alpha 1$ domain, an MHC class II $\alpha 2$ domain, or a combination thereof; and at least one engineered cavity. In some embodiments, the at least one engineered cavity is not located at the MHC class II $\alpha 1$ domain or the MHC class II $\alpha 2$ domain. In some embodiments, the engineered cavity is located at an antibody C.sub.H3 domain fused to the polypeptide. In some embodiments, the polypeptide optionally comprises an antibody C.sub.H2 domain located between an MHC class II $\alpha 2$ domain and the C.sub.H3 domain with an engineered cavity. In certain embodiments, the polypeptide comprises an antibody C.sub.H3 domain, and the antibody C.sub.H3 domain comprises at least one mutation selected from the list consisting of Y349C, T366S, L368A, Y407V (EU numbering), and combinations thereof. In further embodiments, the polypeptide comprises a ubiquitous autoantigen. Optionally, the ubiquitous autoantigen comprises a cysteine residue that interacts with a cysteine residue in either an MHC $\alpha 1$ or $\beta 1$ domain to create a cysteine trap.

(85) In one aspect, one polypeptide of the heterodimer comprises an MHC class II $\beta 1$ domain, an MHC class II $\beta 2$ domain, or a combination thereof; and at least one engineered cavity. In some embodiments, the at least one engineered cavity is not located at the MHC class II $\beta 1$ domain or the MHC class II $\beta 2$ domain. In some embodiments, the engineered cavity is located at an antibody C.sub.H3 domain fused to the polypeptide. In some embodiments, the polypeptide optionally comprises an antibody C.sub.H2 domain located between an MHC class II $\beta 2$ domain and the C.sub.H3 domain with an engineered cavity. In certain embodiments, the polypeptide comprises an antibody C.sub.H3 domain, and the antibody C.sub.H3 domain comprises at least one mutation selected from the list consisting of Y349C, T366S, L368A, Y407V (EU numbering), and combinations thereof. In further embodiments, the polypeptide comprises a ubiquitous autoantigen.

Optionally, the ubiquitous autoantigen comprises a cysteine residue that interacts with a cysteine residue in either an MHC $\alpha 1$ or $\beta 1$ domain to create a cysteine trap.

(86) FIG. 20A and FIG. 20B show non-limiting embodiments of an engineered uaMHC comprising an engineered cavity and an engineered protuberance. FIG. 20A shows a human MHC class II alpha chain fused to an immunoglobulin CH2 and CH3 domain, SEQ ID NO: 103. The CH3 domain comprises an engineered knob that has been created by substituting two amino acids S354C and T366W. The alpha chain comprises an optional c-terminal cysteine to allow for conjugation to a functionalized linker, however this c-terminal cysteine can be alternatively included on the beta chain. FIG. 20B shows a human MHC class II beta chain fused to an immunoglobulin CH2 and CH3 domain, SEQ ID NO: 104. The CH3 domain comprises an engineered hole that has been created by substituting four amino acids Y349C, T366S, L368A and Y407V. The beta chain also comprises a ubiquitous autoantigen (PDC-E2.sub.122-135) that is covalently coupled to the beta chain by a peptide linker. The protuberance-cavity interaction favors assembly and allows for purification of intact uaMHC heterodimers using standard techniques to purify immunoglobulin. Once purified the uaMHC can be coupled to a suitable nanoparticle through a functionalized linker molecule (E.g., functionalized PEG molecules). In certain embodiments, the alpha chain of MHC heterodimer comprises an amino acid sequence at least about 90%, 95%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 103. In certain embodiments, the beta chain of MHC heterodimer comprises an amino acid sequence at least about 90%, 95%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 104. In certain embodiments, the MHC heterodimer comprises an alpha chain at least about 90%, 95%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 103; and a beta chain at least about 90%, 95%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 104. In certain embodiments the alpha chain and the beta chain are at least about 90%, 95%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 103 or 104, and any one or more of the CH2 and CH3 domain of SEQ ID NO: 103, the CH2 and CH3 domain of SEQ ID NO: 104, and/or the ubiquitous autoantigen is identical to that disclosed in SEQ ID NO: 104. In certain embodiments the alpha chain and the beta chain are at least about 90%, 95%, 97%, 98%, 99%, or 100% identical to SEQ ID NO: 103 or 104, and the ubiquitous autoantigen is identical to that disclosed in SEQ ID NO: 104. Those of skill in the art will recognize that the alpha and beta chains of human MHC are highly polymorphic, and thus can tolerate a relatively high degree of variability. One of skill in the art will also be able to substitute the alpha and beta chains of the MHC shown in SEQ ID NOs: 102 and 104 for a different HLA allele, and an appropriate ubiquitous autoantigen that is able to bind the specific substituted allele. Such HLA allele and ubiquitous autoantigen pairings are disclosed elsewhere in the application, for example, at the sequence listing at the end of this application.

(87) Nanoparticles

(88) The ubiquitous autoantigen-MHCs are coupled to a nanoparticle core (uaMHC-NP). The nanoparticle can be made from a variety of materials. In certain embodiments, the nanoparticle is non-liposomal and/or has a solid core. In certain embodiments, the solid core can be a metal or a metal oxide. In certain embodiments, the solid core can be iron, iron oxide, or gold. The solid core can be a high density core such that the density is greater than about 2.0 g/cm.³, about 3.0 g/cm.³, about 4.0 g/cm.³, about 5.0 g/cm.³, about 6.0 g/cm.³, or about 7.0 g/cm.³. In certain embodiments, the density of the solid core is between about 4.0 g/cm.³ and about 8.0 g/cm.³. In certain embodiments, the density of the solid core is between about 5.0 g/cm.³ and about 8.0 g/cm.³. In certain embodiments, the density of the solid core is between about 5.0 g/cm.³ and about 7.0 g/cm.³. In certain embodiments, the density of the solid core is between about 5.0 g/cm.³ and about 6.0 g/cm.³.

(89) The nanoparticle core of the uaMHC-NP comprises, or consists essentially of, or yet further consists of a core, for example a solid core, a metal core, a dendrimer core, a polymeric micelle nanoparticle core, a nanorod, a fullerene, a nanoshell, a coreshell, a protein-based nanostructure or

a lipid-based nanostructure. In some aspects, the nanoparticle core is bioabsorbable and/or biodegradable. In some aspects, the nanoparticle core is a dendrimer nanoparticle core comprising, or alternatively consisting essentially thereof, or yet further consisting of a highly branched macromolecule having a tree-like structure growing from a core. In further aspects, the dendrimer nanoparticle core may comprise, or alternatively consist essentially thereof, or yet further consist of a poly(amidoamine)-based dendrimer or a poly-L-lysine-based dendrimer. In certain aspects, the nanoparticle core is a polymeric micelle core comprising, or alternatively consisting essentially thereof, or yet further consisting of an amphiphilic block co-polymer assembled into a nano-scaled core-shell structure. In further aspects, the polymeric micelle core comprises, or alternatively consists essentially thereof, or yet further consists of a polymeric micelle produced using polyethylene glycol-diaسترoylphosphatidylethanolamine block copolymer. In a further aspect, the nanoparticle core comprises, or alternatively consists essentially of, or yet further consists of a metal. In another aspect, the nanoparticle core is not a liposome. Additional examples of core materials include but are not limited to, standard and specialty glasses, silica, polystyrene, polyester, polycarbonate, acrylic polymers, polyacrylamide, polyacrylonitrile, polyamide, fluoropolymers, silicone, celluloses, silicon, metals (e.g., iron, gold, silver), minerals (e.g., ruby), nanoparticles (e.g., gold nanoparticles, colloidal particles, metal oxides, metal sulfides, metal selenides, and magnetic materials such as iron oxide), and composites thereof. In some embodiments, an iron oxide nanoparticle core comprises iron (II, III) oxide. The core could be of homogeneous composition, or a composite of two or more classes of material depending on the properties desired. In certain aspects, metal nanoparticles will be used. These metal particles or nanoparticles can be formed from Au, Pt, Pd, Cu, Ag, Co, Fe, Ni, Mn, Sm, Nd, Pr, Gd, Ti, Zr, Si, and In, precursors, their binary alloys, their ternary alloys and their intermetallic compounds. See U.S. Pat. No. 6,712,997, which is incorporated herein by reference for such disclosure. In certain embodiments, the compositions of the core and layers (described below) may vary provided that the nanoparticles are biocompatible and bioabsorbable. The core could be of homogeneous composition, or a composite of two or more classes of material depending on the properties desired. In certain aspects, metal nanospheres will be used. These metal nanoparticles can be formed from Fe, Ca, Ga and the like. In certain embodiments, the nanoparticle comprises, or alternatively consists essentially of, or yet further consists of a core comprising metal or metal oxide such as gold or iron oxide.

(90) In another aspect, provided herein are uaMHC-NPs comprising at least one ubiquitous autoantigen-MHC described herein and a nanoparticle, wherein the nanoparticle is non-liposomal and has an iron oxide core.

(91) In another aspect, provided herein are uaMHC-NPs comprising at least one ubiquitous autoantigen-MHC described herein and a nanoparticle, wherein the nanoparticle is non-liposomal and has a gold core.

(92) In another aspect, provided herein are uaMHC-NPs comprising at least one ubiquitous autoantigen-MHC herein and a nanoparticle, wherein the nanoparticle is non-liposomal and has an iron oxide core; and the at least one ubiquitous autoantigen-MHC is covalently linked to the nanoparticle through a linker.

(93) In some aspects, the nanoparticle core has a diameter selected from the group of from about 1 nm to about 100 nm; from about 1 nm to about 75 nm; from about 1 nm to about 50 nm; from about 1 nm to about 25 nm; from about 5 nm to about 100 nm; from about 5 nm to about 50 nm; from about 5 nm to about 40 nm; from about 5 nm to about 30 nm; from about 5 nm to about 25 nm; or from about 5 nm to about 20 nm. In some embodiments, the nanoparticle core has a diameter from about 10 nm to about 100 nm; from about 10 nm to about 50 nm; from about 10 nm to about 40 nm; from about 10 nm to about 30 nm; from about 10 nm to about 25 nm; or from about 10 nm to about 20 nm. In certain embodiments, the nanoparticle core has a diameter greater than about 1 nm, 2 nm, 5 nm, 10 nm, 15 nm, 20 nm, 25 nm, 30 nm, 40 nm, or 50 nm. In certain embodiments,

the nanoparticle core has a diameter less than about 100 nm, 75 nm, 50 nm, 40 nm, 30 nm, 20 nm, or 15 nm.

(94) In some aspects, the nanoparticle core is a dendrimer nanoparticle core comprising, or alternatively consisting essentially thereof, or yet further consisting of a highly branched macromolecule having a tree-like structure growing from a core. In further aspects, the dendrimer nanoparticle may comprise, or alternatively consist essentially thereof, or yet further consist of a poly(amidoamine)-based dendrimer or a poly-L-lysine-based dendrimer. In certain aspects, the nanoparticle core is a polymeric micelle core comprising, or alternatively consisting essentially thereof, or yet further consisting of an amphiphilic block co-polymer assembled into a nano-scaled core-shell structure. In further aspects, the polymeric micelle core may comprise, or alternatively consist essentially thereof or yet further consist of, a polymeric micelle produced using polyethylene glycol-diaسترoylphosphatidylethanolamine block copolymer. The dendrimer core or polymeric micelle core may further comprise an outer coating or layer as described herein.

(95) In certain embodiments, specific means of synthesis of dendrimer nanoparticles or nanoparticles with a dendrimer nanoparticle core may require that metal ions are extracted into the interior of dendrimers and then subsequently chemically reduced to yield nearly size-monodispersed particles having dimensions of less than 3 nm, such as the method disclosed in Crooks et al., "Synthesis, Characterization, and Applications of Dendrimer-Encapsulated Nanoparticles". *The Journal of Physical Chemistry B* (109): 692-704 (2005), wherein the resulting dendrimer core component serves not only as a template for preparing the nanoparticle but also to stabilize the nanoparticle, making it possible to tune solubility, and provides a means for immobilization of the nanoparticle on solid supports.

(96) The nanoparticle cores typically consist of a substantially spherical core and optionally one or more layers or coatings. The core may vary in size and composition as described herein. In addition to the core, the particle may have one or more layers to provide functionalities appropriate for the applications of interest. The thicknesses of layers, if present, may vary depending on the needs of the specific applications. For example, layers may impart useful optical properties.

(97) Layers may also impart chemical or biological functionalities, referred to herein as chemically active or biologically active layers. These layers typically are applied on the outer surface of the particle and can impart functionalities to the pMHC-NPs. The layer or layers may typically range in thickness from about 0.001 micrometers (1 nanometer) to about 10 micrometers or more (depending on the desired particle diameter) or from about 1 nm to 5 nm, from about 1 nm to about 10 nm, from about 1 nm to about 40 nm, from about 15 nm to about 25 nm, or about 20 nm, and ranges in between.

(98) The layer or coating may comprise, or alternatively consist essentially of, or yet further consist of a biodegradable sugar or other polymer. Examples of biodegradable layers include but are not limited to dextran; poly(ethylene glycol); poly(ethylene oxide); mannitol; poly(esters) based on polylactide (PLA), polyglycolide (PGA), polycaprolactone (PCL); poly(hydroxalkanoate) of the PHB-PHV class; and other modified poly(saccharides) such as starch, cellulose and chitosan. Additionally, the nanoparticle may include a layer with suitable surfaces for attaching chemical functionalities for chemical binding or coupling sites.

(99) Surface Valency and Density of Antigen-MHCs

(100) The ubiquitous autoantigen-MHCs described herein are coupled to the nanoparticle at a certain valency. Valency is the number of pMHCs per nanoparticle core. In certain embodiments, the valency of the nanoparticle may range between about 1 pMHC per nanoparticle core to about 6,000 pMHCs per nanoparticle core. In certain embodiments, the valency of the nanoparticle may range between about 10 pMHCs per nanoparticle core to about 6,000 pMHCs per nanoparticle core. In certain embodiments, the valency of the nanoparticle may range between about 50 pMHCs per nanoparticle core to about 6,000 pMHCs per nanoparticle core. In certain embodiments, the valency of the nanoparticle may range between about 1 pMHC per nanoparticle core to about 5000,

about 4000, about 3000, about 2000, or about 1000 pMHCs per nanoparticle core. In certain embodiments, the valency of the nanoparticle may range between about 10 pMHCs per nanoparticle core to about 5000, about 4000, about 3000, about 2000, or about 1000 pMHCs per nanoparticle core. In certain embodiments, the valency of the nanoparticle may range between about 50 pMHC per nanoparticle core to about 5000, about 4000, about 3000, about 2000, or about 1000 pMHCs per nanoparticle core. In certain embodiments, the valency of the nanoparticle may range between about 1 pMHC to per nanoparticle core to about 1000 pMHCs per nanoparticle core, or between about 10:1 to about 1000:1, or between about 11:1 to about 1000:1, or between about 12:1 to about 1000:1. In certain embodiments, the valency (antigen-MHC to nanoparticle core) may range between about 10:1 to about 500:1, or between about 11:1 to about 500:1, or between about 12:1 to about 500:1. In certain embodiments, the valency (antigen-MHC to nanoparticle core) may range between about 10:1 to about 200:1, or between about 11:1 to about 200:1, or between about 12:1 to about 200:1. In certain embodiments, the valency (antigen-MHC to nanoparticle core) may range between about 10:1 to about 150:1, or between about 11:1 to about 150:1, or between about 12:1 to about 150:1. In certain embodiments, the valency (antigen-MHC to nanoparticle core) may range between about 10:1 to about 100:1, or between about 11:1 to about 100:1, or between about 12:1 to about 100:1. In certain embodiments, the valency (antigen-MHC to nanoparticle core) may range between about 10:1 to about 200:1, between about 20:1 to about 200:1, between about 30:1 to about 200:1, between about 40:1 to about 200:1, or between about 50:1 to about 200:1. In certain embodiments, the valency (antigen-MHC to nanoparticle core) may range between about 10:1 to about 150:1, between about 20:1 to about 150:1, between about 30:1 to about 200:1, between about 40:1 to about 150:1, or between about 50:1 to about 150:1. In certain embodiments, the valency (antigen-MHC to nanoparticle core) may range between about 10:1 to about 100:1, between about 20:1 to about 100:1, between about 30:1 to about 100:1, between about 40:1 to about 100:1, or between about 50:1 to about 100:1.

(101) In some aspects, the nanoparticle core has a defined valency per surface area of the core, also referred to herein as “density.” In these aspects, the pMHC density per nanoparticle is from about 0.025 pMHC/100 nm.^{sup.2} to about 100 pMHC/100 nm.^{sup.2} of the surface area of the nanoparticle core, or alternatively from about 0.406 pMHC/100 nm.^{sup.2} to about 50 pMHC/100 nm.^{sup.2}; or alternatively from about 0.05 pMHC/100 nm.^{sup.2} to about 25 pMHC/100 nm.^{sup.2}. In certain aspects, the pMHC density per nanoparticle is from about 0.4 pMHC/100 nm.^{sup.2} to about 25 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 20 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 15 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 14 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 13 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 12 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 11.6 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 11.5 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 11 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 10 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 9 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 8 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 7 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 6 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 5 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 4 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 3 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 2.5 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 2 pMHC/100 nm.^{sup.2}, or from about 0.4 pMHC/100 nm.^{sup.2} to about 1.5 pMHC/100 nm.^{sup.2}.

(102) In another aspect, the nanoparticle may have a pMHC density of from about 0.22 pMHC/100 nm.^{sup.2} to about 10 pMHC/100 nm.^{sup.2}, or from about 0.22 pMHC/100 nm.^{sup.2} to about 9 pMHC/100 nm.^{sup.2}, or from about 0.22 pMHC/100 nm.^{sup.2} to about 8 pMHC/100 nm.^{sup.2}, or from about 0.22 pMHC/100 nm.^{sup.2} to about 7 pMHC/100 nm.^{sup.2}, or from about 0.22

pMHC/100 nm.sup.2 to about 6 pMHC/100 nm.sup.2, or from about 0.22 pMHC/100 nm.sup.2 to about 5 pMHC/100 nm.sup.2, or from about 0.22 pMHC/100 nm.sup.2 to about 4 pMHC/100 nm.sup.2, or from about 0.22 pMHC/100 nm.sup.2 to about 3 pMHC/100 nm.sup.2, or from about 0.22 pMHC/100 nm.sup.2 to about 2 pMHC/100 nm.sup.2, or from about 0.22 pMHC/100 nm.sup.2 to about 1.5 pMHC/100 nm.sup.2. In some aspects, the nanoparticle has a pMHC density of from about 0.22 pMHC/100 nm.sup.2 to about 10 pMHC/100 nm.sup.2, or 0.24 pMHC/100 nm.sup.2 to about 9 pMHC/100 nm.sup.2, or from about 0.26 pMHC/100 nm.sup.2 to about 8 pMHC/100 nm.sup.2, or from about 0.28 pMHC/100 nm.sup.2 to about 7 pMHC/100 nm.sup.2, or from about 0.24 pMHC/100 nm.sup.2 to about 4 pMHC/100 nm.sup.2, or from about 0.5 pMHC/100 nm.sup.2 to about 3 pMHC/100 nm.sup.2, or from about 0.6 pMHC/100 nm.sup.2 to about 1.5 pMHC/100 nm.sup.2. In a further aspect, the nanoparticle has a pMHC density of from about 0.4 pMHC/100 nm.sup.2 to about 1.3 pMHC/100 nm.sup.2, or alternatively from about 0.5 pMHC/100 nm.sup.2 to about 0.9 pMHC/100 nm.sup.2, or alternatively from about 0.6 pMHC/100 nm.sup.2 to about 0.8 pMHC/100 nm.sup.2.

(103) Linkers

(104) In certain aspects, ubiquitous autoantigen-MHC can be coupled to the nanoparticle core by one or more of covalently, non-covalently, or cross-linked and optionally coupled through a linker. In aspects involving a linker or linkers, the linkers may be the same or different from each other on a single nanoparticle core. In some embodiments, the ubiquitous autoantigen-MHC comprises at least one ubiquitous autoantigen-MHC described herein and a nanoparticle, wherein the nanoparticle is non-liposomal and has metal or metal oxide core; and the at least one ubiquitous autoantigen-MHC is covalently linked to the nanoparticle through a linker comprising polyethylene glycol with a molecular weight of less than 5 kilodaltons (kD). In some embodiments, polyethylene glycol has a molecular weight of less than 1 kD, 2 kD, 3 kD, 4 kD, 5 kD, 6 kD, 7 kD, 8 kD, 9 kD, or 10 kD. In some embodiments, polyethylene glycol is functionalized with maleimide. In some embodiments, polyethylene glycol has a molecular weight of between about 1 kD and about 5 kD, between about 2 kD and about 5 kD, between about 3 kD and about 5 kD. In some embodiments, polyethylene glycol is functionalized with maleimide. In certain embodiments, the end of the linker that is in contact with the solid core is embedded in the solid core. In further aspects, the linker may be less than 5 kD in size, and is optionally polyethylene glycol. The linker can be any of the linkers described in Table 1.

(105) TABLE-US-00001 TABLE 1 Exemplary linker molecules

Types of M.W. Functional Nanoparticle PEG linkers (kD) group	Structure	Gold nano- particle (GNP-C)	Thiol-PEG- carboxyl
3.5 Carboxyl (—COOH)		Gold nano- particle (GNP-N)	Thiol-PEG- amine
3.5 Amine (—NH.sub.2)		Iron oxide Nanoparticle (SFP-C)	Dopamine- PEG- carboxyl
3.5 Carboxyl (—COOH)		Iron oxide Nanoparticle (SFP-N)	Dopamine- PEG- amine
3.5 Amine (—NH.sub.2)		Iron oxide Nanoparticle (SFP-Z)	Dopamine- PEG- azide
3.5 Azide (—N.sub.3)		Iron oxide Nanoparticle (SFP-M)	Dopamine- PEG- maleimide
3.5 Maleimide		Iron oxide Nanoparticle (SFP-O)	Dopamine- PEG- Orthopyridyl disulfide
3.5 Orthopyridyl disulfide		Iron oxide Nano- particle (PF-C)	carboxyl-PEG- carboxyl
3.5 Carboxyl (—COOH)		Iron oxide Nano- particle (PF-N)	Methoxy-PEG- amine
2.0 Amine (—NH.sub.2)		Iron oxide Nano- particle (PF-M)	Methoxy- PEG- maleimide
2.0 Maleimide		Iron oxide Nano- particle (PF-O)	Methoxy- PEG- Orthopyridyl disulfide
2.0 Orthopyridyl disulfide		Iron oxide Nano- particle (PF)	PEG 2.0 Hydroxyl (—OH)

(106) In order to couple the substrate or particles of the ubiquitous autoantigen-MHC to the nanoparticle, the following techniques can be applied.

(107) The binding can be generated by chemically modifying the substrate or particle which typically involves the generation of “functional groups” on the surface, said functional groups

being capable of binding to an MHC complex, and/or linking the optionally chemically modified surface of the surface or particle with covalently or non-covalently bound so-called “linking molecules,” followed by reacting the MHC or MHC complex with the particles obtained.

(108) The functional groups or the linking molecules bearing them may be selected from amino groups, carbonic acid groups, thiols, thioethers, disulfides, guanidino, hydroxyl groups, amine groups, vicinal diols, aldehydes, alpha-haloacetyl groups, mercury organyles, ester groups, acid halide, acid thioester, acid anhydride, isocyanates, isothiocyanates, sulfonic acid halides, imidoesters, diazoacetates, diazonium salts, 1,2-diketones, phosphoric acids, phosphoric acid esters, sulfonic acids, azolides, imidazoles, indoles, N-maleimides, alpha-beta-unsaturated carbonyl compounds, arylhalogenides or their derivatives.

(109) Non-limiting examples for other linking molecules with higher molecular weights are nucleic acid molecules, polymers, copolymers, polymerizable coupling agents, silica, proteins, and chain-like molecules having a surface with the opposed polarity with respect to the substrate or particle. Nucleic acids can provide a link to affinity molecules containing themselves nucleic acid molecules, though with a complementary sequence with respect to the linking molecule.

(110) In some embodiments, the linking molecule comprises polyethylene glycol. In some embodiments, the linking molecule comprises polyethylene glycol and maleimide. In some embodiments, the polyethylene glycol comprises one or more of a C_{sub}.1-C_{sub}.3 alkoxy group, —R_{sup}.10NHC(O)R—, —R_{sup}.10C(O)NHR—, —R_{sup}.10OC(O)R—, —R_{sup}.10C(O)OR—, wherein each R is independently H or C_{sub}.1-C_{sub}.6 alkyl and wherein each R_{sub}.10 is independently a bond or C_{sub}.1-C_{sub}.6 alkyl.

(111) pMHCs can be coupled to nanoparticles by a variety of methods, one non-limiting example includes conjugation to NPs produced with PEG linkers carrying distal —NH₂ or —COOH groups that can be achieved via the formation of amide bonds in the presence of 1-Ethyl-3-[3-dimethylaminopropyl]carbodiimide hydrochloride (EDC). NPs with —COOH groups are first dissolved in 20 mM MES buffer, pH 5.5. N-hydroxysulfosuccinimide sodium salt (sulpha-NHS, Thermo Scientific, Waltham, MA, final concentration 10 mM) and EDC (Thermo scientific, Waltham, MA, final concentration 1 mM) is added to the NP solution. After 20 min of stirring at room temperature, the NP solution is added drop-wise to the solution containing pMHC monomers dissolved in 20 mM borate buffer (pH 8.2). The mixture is stirred for an additional 4 hr. To conjugate MHCs to NH₂-functionalized NPs pMHCs are first dissolved in 20 mM MES buffer, pH 5.5, containing 100 mM NaCl. Sulpha-NHS (10 mM) and EDC (5 mM) are then added to the MHC solution. The activated MHC molecules are then added to the NP solution in 20 mM borate buffer (pH 8.2), and stirred for 4 hr at room temperature.

(112) To conjugate MHC to maleimide-functionalized NPs, pMHCs are first incubated with Tributylphosphine (TBP, 1 mM) for 4 hr at room temperature, pMHCs engineered to encode a free carboxyterminal Cys residue are then mixed with NPs in 40 mM phosphate buffer, pH 6.0, containing 2 mM EDTA, 150 mM NaCl, and incubated overnight at room temperature. MHCs of the pMHCs are covalently bound with NPs via the formation of a carbon-sulfide bond between maleimide groups and the Cys residue.

(113) Click chemistry can be used to conjugate pMHC or avidin to NPs functionalized with azide groups. For this reaction, MHC or avidin molecules are first incubated with dibenzocyclooctyl (DBCO, Click Chemistry Tools, Scottsdale, AZ) reagent for 2 hr at room temperature. Free DBCO molecules can be removed by dialysis overnight. MHC- or avidin-DBCO conjugates are then incubated with SFP-Z for 2 hr, resulting in formation of triazole bonds between pMHCs or avidin molecules and NPs.

(114) Unconjugated pMHCs in the different MHC-NP conjugating reactions can be removed by extensive dialysis using methods known in the art. A non-limiting example is dialysis against PBS, pH 7.4, at 4° C. though 300 kD molecular weight cut off membranes (Spectrum labs).

Alternatively, pMHC-conjugated IONPs can be purified by magnetic separation. The conjugated

NPs can be concentrated by ultrafiltration through Amicon Ultra-15 units (100 kD MWCO) and stored in PBS.

(115) The surface of the particle can be chemically modified, for instance by the binding of phosphonic acid derivatives having functional reactive groups. One example of these phosphonic acid or phosphonic acid ester derivatives is imino-bis(methylenephosphono) carbonic acid which can be synthesized according to the “Mannich-Moedritzer” reaction. This binding reaction can be performed with a substrate or a particle as directly obtained from the preparation process or after a pre-treatment (for instance with trimethylsilyl bromide). In the first case the phosphoric acid (ester) derivative may for instance displace components of the reaction medium which are still bound to the surface. This displacement can be enhanced at higher temperatures. Trimethylsilyl bromide, on the other hand, is believed to dealkylate alkyl group-containing phosphorous-based complexing agents, thereby creating new binding sites for the phosphonic acid (ester) derivative. The phosphonic acid (ester) derivative, or linking molecules bound thereto, may display the same functional groups as given above. A further example of the surface treatment of the substrate or particle involves heating in a diol such as ethylene glycol. It should be noted that this treatment may be redundant if the synthesis already proceeded in a diol. Under these circumstances the synthesis product directly obtained is likely to show the necessary functional groups. This treatment is, however, applicable to a substrate or a particle that was produced in N- or P-containing complexing agents. If such substrate or particle is subjected to an after-treatment with ethylene glycol, ingredients of the reaction medium (e.g. complexing agent) still binding to the surface can be replaced by the diol and/or can be dealkylated.

(116) It is also possible to replace N-containing complexing agents still bound to the particle surface by primary amine derivatives having a second functional group. The surface of the substrate or particle can also be coated with silica. Silica allows a relatively simple chemical conjugation of organic molecules since silica easily reacts with organic linkers, such as triethoxysilane or chlorosilane. The particle surface may also be coated by homo- or copolymers. Examples for polymerizable coupling agents are: N-(3-aminopropyl)-3-mercaptopbenzamidine, 3-(trimethoxysilyl) propylhydrazide and 3-(trimethoxysilyl) propylmaleimide. Other non-limiting examples of polymerizable coupling agents are mentioned herein. These coupling agents can be used singly or in combination depending on the type of copolymer to be generated as a coating.

(117) Another surface modification technique that can be used with substrates or particles containing oxidic transition metal compounds is conversion of the oxidic transition metal compounds by chlorine gas or organic chlorination agents to the corresponding oxychlorides. These oxychlorides are capable of reacting with nucleophiles, such as hydroxyl or amino groups as often found in biomolecules. This technique allows generating a direct conjugation with proteins, for instance, via the amino group of lysine side chains. The conjugation with proteins after surface modification with oxychlorides can also be effected by using a bi-functional linker, such as maleimidopropionic acid hydrazide.

(118) For non-covalent linking techniques, chain-type molecules having a polarity or charge opposite to that of the substrate or particle surface are particularly suitable. Examples for linking molecules which can be non-covalently linked to core/shell nanoparticles involve anionic, cationic or zwitter-ionic surfactants, acid or basic proteins, polyamines, polyamides, polysulfone or polycarboxylic acid. The hydrophobic interaction between substrate or particle and amphiphilic reagent having a functional reactive group can generate the necessary link. In particular, chain-type molecules with amphiphilic character, such as phospholipids or derivatised polysaccharides, which can be crosslinked with each other, are useful. The absorption of these molecules on the surface can be achieved by incubation. The binding between affinity molecule and substrate or particle can also be based on non-covalent, self-organizing bonds. One example thereof involves simple detection probes with biotin as linking molecule and avidin- or streptavidin-coupled molecules.

(119) Protocols for coupling reactions of functional groups to biological molecules can be found in

the literature, for instance in “Bioconjugate Techniques” (Greg T. Hermanson, Academic Press 1996). The biological molecule (e.g., MHC molecule or derivative thereof) can be coupled to the linking molecule, covalently or non-covalently, in line with standard procedures of organic chemistry such as oxidation, halogenation, alkylation, acylation, addition, substitution or amidation. These methods for coupling the covalently or non-covalently bound linking molecule can be applied prior to the coupling of the linking molecule to the substrate or particle or thereafter. Further, it is possible, by means of incubation, to effect a direct binding of molecules to correspondingly pre-treated substrate or particles (for instance by trimethylsilyl bromide), which display a modified surface due to this pre-treatment (for instance a higher charge or polar surface).

(120) Synthesis of Nanoparticles

(121) Nanoparticles may be formed by contacting an aqueous phase containing the pMHC complex and a polymer and a non-aqueous phase followed by evaporation of the non-aqueous phase to cause the coalescence of particles from the aqueous phase as taught in U.S. Pat. No. 4,589,330 or 4,818,542. Certain polymers for such preparations are natural or synthetic copolymers or polymers which include gelatin agar, starch, arabinogalactan, albumin, collagen, polyglycolic acid, polylactic acid, glycolide-L(-) lactide poly(epsilon-caprolactone), poly(epsilon-caprolactone-CO-lactic acid), poly(epsilon-caprolactone-CO-glycolic acid), poly(beta-hydroxy butyric acid), poly(ethylene oxide), polyethylene, poly(alkyl-2-cyanoacrylate), poly(hydroxyethyl methacrylate), polyamides, poly(amino acids), poly(2-hydroxyethyl DL-aspartamide), poly(ester urea), poly(L-phenylalanine/ethylene glycol/1,6-diisocyanatohexane) and poly(methyl methacrylate).

Particularly, certain polymers are polyesters, such as polyglycolic acid, polylactic acid, glycolide-L(-) lactide poly(epsilon-caprolactone), poly(epsilon-caprolactone-CO-lactic acid), and poly(epsilon-caprolactone-CO-glycolic acid). Solvents useful for dissolving the polymer include: water, hexafluoroisopropanol, methylenechloride, tetrahydrofuran, hexane, benzene, or hexafluoroacetone sesquihydrate.

(122) Gold nanoparticles (GNPs) are synthesized using chemical reduction of gold chloride with sodium citrate as described in Perrault, S. D. et al. (2009) Nano Lett 9:1909-1915. Briefly, 2 mL of 1% of HAuCl₄ (Sigma Aldrich) is added to 100 mL H₂O under vigorous stirring and the solution is heated in an oil bath. Six (for 14 nm GNPs) or two mL (for 40 nm GNPs) of 1% Na Citrate is added to the boiling HAuCl₄ solution, which is stirred for an additional 10 min and then is cooled down to room temperature. GNPs are stabilized by the addition of 1 μMol of thiol-PEG linkers (Nanocs, MA) functionalized with —COOH or —NH₂ groups as acceptors of MHC. PEGylated GNPs are washed with water to remove free thiol-PEG, concentrated and stored in water for further analysis. NP density is determined via spectrophotometry and calculated according to Beer's law.

(123) The SFP series of iron oxide NPs (SFP IONPs) can be produced by thermal decomposition of iron acetate in organic solvents in the presence of surfactants, then rendered solvent in aqueous buffers by pegylation (Xie, J. et al. (2007) Adv Mater 19:3163; Xie, J. et al. (2006) Pure Appl. Chem. 78:1003-1014; Xu, C. et al. (2007) Polymer International 56:821-826). Briefly, 2 mMol Fe(acac)₃ (Sigma Aldrich, Oakville, ON) are dissolved in a mixture of 10 mL benzyl ether and oleylamine and heated to 100° C. for 1 hr followed by 300° C. for 2 hr with reflux under the protection of a nitrogen blanket. Synthesized NPs are precipitated by addition of ethanol and resuspended in hexane. For pegylation of the IONPs, 100 mg of different 3.5 kD DPA-PEG linkers (Jenkem Tech USA) are dissolved in a mixture of CHCl₃ and HCON(CH₃)₂ (dimethylformamide (DMF)). The NP solution (20 mg Fe) is then added to the DPA-PEG solution and stirred for 4 hr at room temperature. Pegylated SFP NPs are precipitated overnight by addition of hexane and then resuspended in water. Trace amounts of aggregates are removed by high-speed centrifugation (20,000×g, 30 min), and the monodisperse SFP NPs are stored in water for further characterization and pMHC conjugation. The concentration of iron in IONP products is determined by spectrophotometry at A₄₁₀ in 2N HCL. Based on the molecular structure and diameter of SFP NPs (Fe₃O₄; 8+1 nm diameter) (Xie, J. et al. (2007) Adv Mater 19:3163; Xie, J. et al.

(2006) *Pure Appl. Chem.* 78:1003-1014), SFP solutions containing 1 mg of iron are estimated to contain 5×10^{14} NPs.

(124) The nanoparticles can also be made by thermally decomposing or heating a nanoparticle precursor. In one embodiment, the nanoparticle is a metal or a metal oxide nanoparticle. In one embodiment, the nanoparticle is an iron oxide nanoparticle. In one embodiment, the nanoparticle is a gold nanoparticle. In one embodiment, provided herein are the nanoparticles prepared in accordance with the present technology. In one embodiment, provided herein is a method of making iron oxide nanoparticles comprising a thermal decomposition reaction of iron acetyl acetonate. In one embodiment, the iron oxide nanoparticle obtained is water-soluble. In one aspect, the iron oxide nanoparticle is suitable for protein conjugation. In one embodiment, the method comprises a single-step thermal decomposition reaction.

(125) In one aspect, the thermal decomposition occurs in the presence of functionalized PEG molecules. Certain non-limiting examples of functionalized PEG linkers are shown in Table 1.

(126) In one aspect, the thermal decomposition comprises heating iron acetyl acetonate. In one embodiment, the thermal decomposition comprises heating iron acetyl acetonate in the presence of functionalized PEG molecules. In one embodiment, the thermal decomposition comprises heating iron acetyl acetonate in the presence of benzyl ether and functionalized PEG molecules] Without being bound by theory, in one embodiment, functionalized PEG molecules are used as reducing reagents and as surfactants. The method of making nanoparticles provided herein simplifies and improves conventional methods, which use surfactants that are difficult to be displaced, or are not displaced to completion, by PEG molecules to render the particles water-soluble. Conventionally, surfactants can be expensive (e.g., phospholipids) or toxic (e.g., Oleic acid or oleilamine). In another aspect, without being bound by theory, the method of making nanoparticles obviates the need to use conventional surfactants, thereby achieving a high degree of molecular purity and water solubility.

(127) In one embodiment, the thermal decomposition involves iron acetyl acetonate and benzyl ether and in the absence of conventional surfactants other than those employed herein.

(128) In one embodiment, the temperature for the thermal decomposition is about 80° C. to about 300° C., or about 80° C. to about 200° C., or about 80° C. to about 150° C., or about 100° C. to about 250° C., or about 100° C. to about 200° C., or about 150° C. to about 250° C., or about 150° C. to about 250° C. In one embodiment, the thermal decomposition occurs at about 1 to about 2 hours of time.

(129) In one embodiment, the method of making the iron oxide nanoparticles comprises a purification step, such as by using Miltenyi Biotec LS magnet column.

(130) In one embodiment, the nanoparticles are stable at about 4° C. in phosphate buffered saline (PBS) without any detectable degradation or aggregation. In one embodiment, the nanoparticles are stable for at least 6 months.

(131) In one aspect, provided herein is a method of making nanoparticle complexes comprising contacting pMHC with iron oxide nanoparticles provided herein. Without being bound by theory, pMHC encodes a cysteine at its carboxyterminal end, which can react with the maleimide group in functionalized PEG at about pH 6.2 to about pH 6.5 for about 12 to about 14 hours.

(132) In one aspect, the method of making nanoparticle complexes comprises a purification step, such as by using Miltenyi Biotec LS magnet column.

(133) Regulatory Immune Cell Types

(134) The uaMHC-NP complexes of the current disclosure reprogram or differentiate autoreactive T cells into T regulatory or TR1 cells. In certain embodiments, the TR1 cells express IL-10. In certain embodiments, the TR1 cells secrete IL-10. In certain embodiments, the TR1 cells express CD49b. In certain embodiments, the TR1 cells express LAG-3. T-cells that have these phenotypic characteristics are useful to treat inflammatory or autoimmune conditions of individuals. In certain embodiments, the uaMHC-NP complexes are useful in a method to reprogram or differentiate

autoreactive T cells into TR1 cells in an individual after administration. This method generates TR1 cells in an antigen specific way.

(135) The ubiquitous autoantigen-MHCs of the current disclosure are useful for generating B regulatory cells. In certain embodiments, the ubiquitous autoantigen-MHCs of the current disclosure are deployed in a method to generate B-cells expressing high levels of CD1d, CD5, and/or the secretion of IL-10. B-regs are also identified by expression of Tim-1. In certain embodiments, the uaMHC-NP complexes are useful in a method to induce B regulatory cells in an individual after administration. This method generates B regulatory cells in an antigen specific way.

(136) Pharmaceutical Compositions and Administration

(137) Provided herein are pharmaceutical compositions of ubiquitous autoantigen-MHC-NPs useful for the treatment and prevention of disease. The compositions comprise, or alternatively consist essentially of, or yet further consist of, a nanoparticle complex as described herein and a carrier.

(138) Compositions of the disclosure may be conventionally administered parenterally, by injection, for example, intravenously, subcutaneously, or intramuscularly. Additional formulations which are suitable for other modes of administration include oral formulations. Oral formulations include such normally employed excipients as, for example, pharmaceutical grades of mannitol, lactose, starch, magnesium stearate, sodium saccharine, cellulose, magnesium carbonate and the like. These compositions take the form of solutions, suspensions, tablets, pills, capsules, sustained release formulations or powders and contain about 10% to about 95% of active ingredient, preferably about 25% to about 70%. The preparation of an aqueous composition that contains an antigen-MHC-nanoparticle complex that modifies the subject's immune condition will be known to those of skill in the art in light of the present disclosure. In one embodiment, the ubiquitous autoantigen-MHC-nanoparticle complex is administered systemically. In specific embodiments, the ubiquitous autoantigen-MHC-NP complex or the compositions comprising a plurality of ubiquitous autoantigen-MHC-N complexes can be administered intravenously.

(139) Typically, the ubiquitous autoantigen-MHC-NPs, described herein, are administered in a manner compatible with the dosage formulation, and in such amount as will be therapeutically effective and immune modifying. The quantity to be administered depends on the subject to be treated. Precise amounts of active ingredient required to be administered depend on the judgment of the practitioner. However, suitable dosage ranges are of the order of ten to several hundred nanograms or micrograms of antigen/MHC/nanoparticle complex per administration. Suitable regimes for initial administration and boosters are also variable, but are typified by an initial administration followed by subsequent administrations.

(140) The manner of application may be varied widely. Any of the conventional methods for administration of a vaccine are applicable. These are believed to include oral application on a solid physiologically acceptable base or in a physiologically acceptable dispersion, parenterally, by injection and the like. The dosage of the antigen/MHC/nanoparticle complex will depend on the route of administration and will vary according to the size and health of the subject. The ubiquitous autoantigen-MHC-NPs can be administered by any suitable route including intravenous, subcutaneous, intradermal, intramuscular, rectally, or intraperitoneally. In certain embodiments, autoantigen-MHC-NPs are administered parenterally. In certain embodiments, autoantigen-MHC-NPs are administered intravenously. In certain embodiments, autoantigen-MHC-NPs are administered subcutaneously.

(141) In many instances, it will be desirable to have multiple administrations of a ubiquitous autoantigen-MHC-NP, about, at least about, or at most about 3, 4, 5, 6, 7, 8, 9, 10 or more administrations. The administrations will normally range from 1, 2, 3, 4, 5, 6, or 7 day to twelve week intervals, more usually from one to two week intervals. Periodic boosters at intervals of every other day, twice a week, weekly, biweekly, monthly, or 0.1, 0.2, 0.3, 0.4, 0.5, 1, 2, 3, 4 or 5 years, usually two years, will be desirable to maintain the condition of the immune system. The course of

the administrations may be followed by assays for autoreactive immune responses, cognate TR1 cells, and T cell activity.

(142) In certain aspects, a single dose of the ubiquitous autoantigen-MHC-NP without including the nanoparticle core and any bioabsorbable/biocompatible outer layer comprises about 0.001 mg/kg to about 2.0 mg/kg, or about 0.001 mg/kg to about 1.5 mg/kg, or about 0.001 mg/kg to about 1.4 mg/kg, or about 0.001 mg/kg to about 1.3 mg/kg, or about 0.001 mg/kg to about 1.2 mg/kg, or about 0.001 mg/kg to about 1.1 mg/kg, or about 0.001 mg/kg to about 1.0 mg/kg. In some embodiments, the single dose comprises from about 0.004 mg/kg to about 1.014 mg/kg, or from about 0.02 mg/kg to about 0.811 mg/kg, or from about 0.041 mg/kg to about 0.608 mg/kg, or from about 0.061 mg/kg to about 0.507 mg/kg, or from about 0.081 mg/kg to about 0.405 mg/kg, or from about 0.121 mg/kg to about 0.324 mg/kg, or from about 0.162 mg/kg to about 0.243 mg/kg. In some embodiments, the single dose comprises from about 0.004 mg/kg to about 1.015 mg/kg, or from about 0.004 mg/kg to about 1.0 mg/kg, or from about 0.004 mg/kg to about 0.9 mg/kg, or from about 0.004 mg/kg to about 0.8 mg/kg, or from about 0.004 mg/kg to about 0.7 mg/kg, or from about 0.004 mg/kg to about 0.6 mg/kg, or from about 0.004 mg/kg to about 0.5 mg/kg, or from about 0.004 mg/kg to about 0.4 mg/kg, or from about 0.004 mg/kg to about 0.3 mg/kg, or from about 0.004 mg/kg to about 0.2 mg/kg, or from about 0.004 mg/kg to about 0.1 mg/kg. Herein, mg/kg refers to milligrams of ubiquitous autoantigen-MHC or ubiquitous autoantigen without considering the MHC component, administered per kg of subject body mass.

(143) Autoimmune and Inflammatory Diseases

(144) The ubiquitous autoantigen-MHCs of the current disclosure are useful for treating an autoimmune or inflammatory disorder. Autoimmune or inflammatory disorders include diseases or disorders arising from and directed against an individual's own tissues or organs or a manifestation thereof or a condition resulting therefrom. In one embodiment, it refers to a condition that results from, or is aggravated by, the production of T cells that are reactive with normal body tissues and antigens. In one embodiment, it refers to a condition that results from, or is aggravated by, the production by antibodies that are reactive with normal body tissues and antigens. Examples of autoimmune or inflammatory disorders include, but are not limited to arthritis (rheumatoid arthritis such as acute arthritis, chronic rheumatoid arthritis, gout or gouty arthritis, acute gouty arthritis, acute immunological arthritis, chronic inflammatory arthritis, degenerative arthritis, type II collagen-induced arthritis, infectious arthritis, Lyme arthritis, proliferative arthritis, psoriatic arthritis, Still's disease, vertebral arthritis, and juvenile-onset rheumatoid arthritis, osteoarthritis, arthritis chronica progrediente, arthritis deformans, polyarthritis chronica primaria, reactive arthritis, and ankylosing spondylitis), inflammatory hyperproliferative skin diseases, psoriasis (such as plaque psoriasis, guttate psoriasis, pustular psoriasis, and psoriasis of the nails), atopy (including atopic diseases such as hay fever and Job's syndrome), dermatitis (including contact dermatitis, chronic contact dermatitis, exfoliative dermatitis, allergic dermatitis, allergic contact dermatitis, dermatitis herpetiformis, nummular dermatitis, seborrheic dermatitis, non-specific dermatitis, primary irritant contact dermatitis, and atopic dermatitis), x-linked hyper IgM syndrome, allergic intraocular inflammatory diseases, urticaria (such as chronic allergic urticaria and chronic idiopathic urticaria, including chronic autoimmune urticaria), myositis, polymyositis/dermatomyositis, juvenile dermatomyositis, toxic epidermal necrolysis, scleroderma (including systemic scleroderma), sclerosis (such as systemic sclerosis; multiple sclerosis (MS) such as spino-optical MS, primary progressive MS (PPMS), and relapsing-remitting MS (RRMS); progressive systemic sclerosis, atherosclerosis, arteriosclerosis, sclerosis disseminata, and ataxic sclerosis), neuromyelitis optica spectrum disorder (NMO, also known as Devic's Disease or Devic's Syndrome), inflammatory bowel disease (IBD) (for example, Crohn's disease; autoimmune-mediated gastrointestinal diseases; colitis such as ulcerative colitis, colitis ulcerosa, microscopic colitis, collagenous colitis, colitis polyposa, necrotizing enterocolitis, and transmural colitis; and autoimmune inflammatory bowel disease), bowel inflammation, pyoderma gangrenosum, erythema

nodosum, primary sclerosing cholangitis, respiratory distress syndrome (including adult or acute respiratory distress syndrome (ARDS)), meningitis, inflammation of all or part of the uvea, iritis, choroiditis, an autoimmune hematological disorder, rheumatoid spondylitis, rheumatoid synovitis, hereditary angioedema, cranial nerve damage as in meningitis, herpes gestationis, pemphigoid gestationis, pruritis scroti, autoimmune premature ovarian failure, sudden hearing loss due to an autoimmune condition, IgE-mediated diseases such as anaphylaxis and allergic and atopic rhinitis, encephalitis such as Rasmussen's encephalitis and limbic and/or brainstem encephalitis, uveitis (such as anterior uveitis, acute anterior uveitis, granulomatous uveitis, nongranulomatous uveitis, phacoantigenic uveitis, posterior uveitis, or autoimmune uveitis), glomerulonephritis (GN) with and without nephrotic syndrome (such as chronic or acute glomerulonephritis such as primary GN, immune-mediated GN, membranous GN (membranous nephropathy), idiopathic membranous GN or idiopathic membranous nephropathy, membrano- or membranous proliferative GN (MPGN), including Type I and Type II, and rapidly progressive GN, or proliferative nephritis), autoimmune polyglandular endocrine failure, balanitis including balanitis circumscripta plasmacellularis, balanoposthitis, erythema annulare centrifugum, erythema dyschromicum perstans, erythema multiform, granuloma annulare, lichen *nitidus*, lichen sclerosus et atrophicus, lichen simplex chronicus, lichen spinulosus, lichen planus, lamellar ichthyosis, epidermolytic hyperkeratosis, premalignant keratosis, pyoderma gangrenosum, allergic conditions and responses, allergic reaction, eczema (including allergic or atopic eczema, asteatotic eczema, dyshidrotic eczema, and vesicular palmoplantar eczema), asthma (such as asthma bronchiale, bronchial asthma, and autoimmune asthma), conditions involving infiltration of T cells and chronic inflammatory responses, immune reactions against foreign antigens such as fetal A-B-O blood groups during pregnancy, chronic pulmonary inflammatory disease, autoimmune myocarditis, leukocyte adhesion deficiency, lupus (including lupus nephritis, lupus cerebritis, pediatric lupus, non-renal lupus, extra-renal lupus, discoid lupus and discoid lupus erythematosus, alopecia lupus, systemic lupus erythematosus (SLE) such as cutaneous SLE or subacute cutaneous SLE, neonatal lupus syndrome (NLE), and lupus erythematosus disseminatus), Type I diabetes, Type II diabetes, and latent autoimmune diabetes in adults (or Type 1.5 diabetes). Also contemplated are immune responses associated with acute and delayed hypersensitivity mediated by cytokines and T-lymphocytes, sarcoidosis, granulomatosis (including lymphomatoid granulomatosis, Wegener's granulomatosis, or agranulocytosis), vasculitides (including vasculitis, large-vessel vasculitis (including polymyalgia rheumatica and giant cell (Takayasu's) arteritis), medium-vessel vasculitis (including Kawasaki's disease and polyarteritis nodosa/periarteritis nodosa), microscopic polyarteritis, immunovascularitis, CNS vasculitis, cutaneous vasculitis, hypersensitivity vasculitis, necrotizing vasculitis such as systemic necrotizing vasculitis, and ANCA-associated vasculitis (such as Churg-Strauss vasculitis or syndrome (CSS) and ANCA-associated small-vessel vasculitis)), temporal arteritis, aplastic anemia, autoimmune aplastic anemia, Coombs positive anemia, Diamond Blackfan anemia, hemolytic anemia, immune hemolytic anemia including autoimmune hemolytic anemia (AIHA), Addison's disease, autoimmune neutropenia, pancytopenia, leukopenia, diseases involving leukocyte diapedesis, CNS inflammatory disorders, Alzheimer's disease, Parkinson's disease, multiple organ injury syndrome (such as those secondary to septicemia, trauma, or hemorrhage), antigen-antibody complex-mediated diseases, anti-glomerular basement membrane disease, anti-phospholipid antibody syndrome, anti-phospholipid syndrome, allergic neuritis, Behcet's disease/syndrome, Castleman's syndrome, Goodpasture's syndrome, Reynaud's syndrome, Sjogren's syndrome, Stevens-Johnson syndrome, pemphigoid such as pemphigoid bullous and skin pemphigoid, pemphigus (including Pemphigus vulgaris, Pemphigus foliaceus, Pemphigus mucus-membrane pemphigoid, and pemphigus erythematosus), autoimmune polyendocrinopathies, Reiter's disease or syndrome, thermal injury, preeclampsia, an immune complex disorder such as immune complex nephritis, antibody-mediated nephritis, polyneuropathies, chronic neuropathy such as IgM polyneuropathies or IgM-mediated neuropathy, autoimmune or immune-mediated

thrombocytopenia such as idiopathic thrombocytopenia purpura (ITP) including chronic or acute ITP, acquired thrombocytopenia purpura, scleritis such as idiopathic cerato-scleritis, episcleritis, autoimmune disease of the testis and ovary including autoimmune orchitis and oophoritis, primary hypothyroidism, hypoparathyroidism, autoimmune endocrine diseases (including thyroiditis (such as autoimmune thyroiditis, Hashimoto's disease, chronic thyroiditis (Hashimoto's thyroiditis), or subacute thyroiditis), autoimmune thyroid disease, idiopathic hypothyroidism, or Grave's disease), polyglandular syndromes such as autoimmune polyglandular syndromes (or polyglandular endocrinopathy syndromes), paraneoplastic syndromes, including neurologic paraneoplastic syndromes such as Lambert-Eaton myasthenic syndrome or Eaton-Lambert syndrome, stiff-man or stiff-person syndrome, encephalomyelitis such as allergic encephalomyelitis or encephalomyelitis allergica and experimental allergic encephalomyelitis (EAE), myasthenia gravis such as thymoma-associated myasthenia gravis, cerebellar degeneration, neuromyotonia, opsoclonus or opsoclonus myoclonus syndrome (OMS), sensory neuropathy, multifocal motor neuropathy, Sheehan's syndrome, autoimmune hepatitis, chronic hepatitis, lupoid hepatitis, giant cell hepatitis, chronic active hepatitis or autoimmune chronic active hepatitis, lymphoid interstitial pneumonitis (LIP), bronchiolitis obliterans (non-transplant) vs NSIP, Guillain-Barre syndrome, Berger's disease (IgA nephropathy), idiopathic IgA nephropathy, linear IgA dermatosis, acute febrile neutrophilic dermatosis, subcorneal pustular dermatosis, transient acantholytic dermatosis, cirrhosis such as primary biliary cirrhosis and pneumonocirrhosis, autoimmune enteropathy syndrome, Celiac or Coeliac disease, celiac sprue (gluten enteropathy), refractory sprue, idiopathic sprue, cryoglobulinemia, amyotrophic lateral sclerosis (ALS; Lou Gehrig's disease), coronary artery disease, autoimmune ear disease such as autoimmune inner ear disease (AIED), autoimmune hearing loss, polychondritis such as refractory or relapsed or relapsing polychondritis, pulmonary alveolar proteinosis, Cogan's syndrome/nonsyphilitic interstitial keratitis, Bell's palsy, Sweet's disease/syndrome, rosacea autoimmune, zoster-associated pain, amyloidosis, a non-cancerous lymphocytosis, a primary lymphocytosis, which includes monoclonal B cell lymphocytosis (e.g., benign monoclonal gammopathy and monoclonal gammopathy of undetermined significance, MGUS), peripheral neuropathy, paraneoplastic syndrome, channelopathies such as epilepsy, migraine, arrhythmia, muscular disorders, deafness, blindness, periodic paralysis, channelopathies of the CNS, autism, inflammatory myopathy, focal or segmental or focal segmental glomerulosclerosis (FSGS), endocrine ophthalmopathy, uveoretinitis, chorioretinitis, autoimmune hepatological disorder, fibromyalgia, multiple endocrine failure, Schmidt's syndrome, adrenalitis, gastric atrophy, presenile dementia, demyelinating diseases such as autoimmune demyelinating diseases and chronic inflammatory demyelinating polyneuropathy, Dressler's syndrome, alopecia areata, alopecia totalis, CREST syndrome (calcinosis, Raynaud's phenomenon, esophageal dysmotility, sclerodactyly, and telangiectasia), male and female autoimmune infertility (e.g., due to anti-spermatozoan antibodies) mixed connective tissue disease, Chagas' disease, rheumatic fever, recurrent abortion, farmer's lung, erythema multiforme, post-cardiotomy syndrome, Cushing's syndrome, bird-fancier's lung, allergic granulomatous angiitis, benign lymphocytic angiitis, Alport's syndrome, alveolitis such as allergic alveolitis and fibrosing alveolitis, interstitial lung disease, transfusion reaction, leprosy, malaria, parasitic diseases such as leishmaniasis, kaposi's sarcoma, schistosomiasis, ascariasis, aspergillosis, Sampter's syndrome, Caplan's syndrome, dengue, endocarditis, endomyocardial fibrosis, diffuse interstitial pulmonary fibrosis, interstitial lung fibrosis, pulmonary fibrosis, idiopathic pulmonary fibrosis, cystic fibrosis, endophthalmitis, erythema elevatum et diutinum, erythroblastosis fetalis, eosinophilic fasciitis, Shulman's syndrome, Felty's syndrome, filariasis, cyclitis such as chronic cyclitis, heterochronic cyclitis, iridocyclitis (acute or chronic), or Fuchs' cyclitis, Henoch-Schönlein purpura, human immunodeficiency virus (HIV) infection, SCID, acquired immune deficiency syndrome (AIDS), echovirus infection, sepsis, endotoxemia, pancreatitis, thyrotoxicosis, parvovirus infection, rubella virus infection, post-vaccination syndromes, congenital rubella infection, Epstein-Barr virus infection, mumps, Evan's

syndrome, autoimmune gonadal failure, Sydenham's chorea, post-streptococcal nephritis, thromboangitis obliterans, thyrotoxicosis, tabes dorsalis, chorioiditis, giant cell polymyalgia, chronic hypersensitivity pneumonitis, keratoconjunctivitis sicca, epidemic keratoconjunctivitis, idiopathic nephritic syndrome, minimal change nephropathy, benign familial and ischemia-reperfusion injury, transplant organ reperfusion, retinal autoimmunity, joint inflammation, bronchitis, chronic obstructive airway/pulmonary disease, silicosis, aphthae, aphthous stomatitis, arteriosclerotic disorders, asperniogenesis, autoimmune hemolysis, Boeck's disease, cryoglobulinemia, Dupuytren's contracture, endophthalmia phacoanaphylactica, enteritis allergica, erythema nodosum leprosum, idiopathic facial paralysis, chronic fatigue syndrome, febris rheumatica, Hamman-Rich's disease, sensorineural hearing loss, haemoglobinuria paroxysmatica, hypogonadism, ileitis regionalis, leucopenia, mononucleosis infectiosa, transverse myelitis, primary idiopathic myxedema, nephrosis, ophthalmia sympathica, orchitis granulomatosa, pancreatitis, polyradiculitis acuta, pyoderma gangrenosum, Quervain's thyroiditis, acquired splenic atrophy, non-malignant thymoma, vitiligo, toxic-shock syndrome, food poisoning, conditions involving infiltration of T cells, leukocyte-adhesion deficiency, immune responses associated with acute and delayed hypersensitivity mediated by cytokines and T-lymphocytes, diseases involving leukocyte diapedesis, multiple organ injury syndrome, antigen-antibody complex-mediated diseases, antglomerular basement membrane disease, allergic neuritis, autoimmune polyendocrinopathies, oophoritis, primary myxedema, autoimmune atrophic gastritis, sympathetic ophthalmia, rheumatic diseases, mixed connective tissue disease, nephrotic syndrome, insulinitis, polyendocrine failure, autoimmune polyglandular syndrome type I, adult-onset idiopathic hypoparathyroidism (AOIH), cardiomyopathy such as dilated cardiomyopathy, epidermolysis bullosa acquisita (EBA), hemochromatosis, myocarditis, nephrotic syndrome, primary sclerosing cholangitis, purulent or nonpurulent sinusitis, acute or chronic sinusitis, ethmoid, frontal, maxillary, or sphenoid sinusitis, an eosinophil-related disorder such as eosinophilia, pulmonary infiltration eosinophilia, eosinophilia-myalgia syndrome, Löffler's syndrome, chronic eosinophilic pneumonia, tropical pulmonary eosinophilia, bronchopneumonic aspergillosis, aspergilloma, or granulomas containing eosinophils, anaphylaxis, seronegative spondyloarthritides, polyendocrine autoimmune disease, sclerosing cholangitis, sclera, episclera, chronic mucocutaneous candidiasis, Bruton's syndrome, transient hypogammaglobulinemia of infancy, Wiskott-Aldrich syndrome, ataxia telangiectasia syndrome, angiectasis, autoimmune disorders associated with collagen disease, rheumatism, neurological disease, lymphadenitis, reduction in blood pressure response, vascular dysfunction, tissue injury, cardiovascular ischemia, hyperalgesia, renal ischemia, cerebral ischemia, and disease accompanying vascularization, allergic hypersensitivity disorders, glomerulonephritis, reperfusion injury, ischemic re-perfusion disorder, reperfusion injury of myocardial or other tissues, lymphomatous tracheobronchitis, inflammatory dermatoses, dermatoses with acute inflammatory components, multiple organ failure, bullous diseases, renal cortical necrosis, acute purulent meningitis or other central nervous system inflammatory disorders, ocular and orbital inflammatory disorders, granulocyte transfusion-associated syndromes, cytokine-induced toxicity, narcolepsy, acute serious inflammation, chronic intractable inflammation, pyelitis, endarterial hyperplasia, peptic ulcer, valvulitis, emphysema, alopecia areata, adipose tissue inflammation/diabetes type II, obesity associated adipose tissue inflammation/insulin resistance, and endometriosis.

(145) In certain embodiments, the autoimmune disease or inflammatory disorder may include, but is not limited to, diabetes mellitus Type I and Type II, pre-diabetes, transplantation rejection, multiple sclerosis, a multiple-sclerosis related disorder, premature ovarian failure, scleroderma, Sjogren's disease/syndrome, lupus, vitiligo, alopecia (baldness), polyglandular failure, Grave's disease, hypothyroidism, polymyositis, pemphigus, Crohn's disease, colitis, autoimmune hepatitis, hypopituitarism, myocarditis, Addison's disease, autoimmune skin diseases, uveitis, pernicious anemia, hypoparathyroidism, and/or rheumatoid arthritis. Other indications of interest include, but are not limited to, asthma, allergic asthma, primary biliary cirrhosis, cirrhosis, Neuromyelitis

Optica Spectrum Disorder (Devic's disease, opticospinal multiple sclerosis (OSMS)), Pemphigus vulgaris, inflammatory bowel disease (IBD), arthritis, Rheumatoid arthritis, systemic lupus erythematosus (SLE), Celiac disease, psoriasis, autoimmune cardiomyopathy, idiopathic dilated cardiomyopathy (IDCM), a Myasthenia Gravis, Uveitis, Ankylosing Spondylitis, Immune Mediated Myopathies, prostate cancer, anti-phospholipid syndrome (ANCA+), atherosclerosis, dermatomyositis, chronic obstructive pulmonary disease (COPD), emphysema, spinal cord injury, traumatic injury, a tobacco-induced lung destruction, ANCA-associated vasculitis, psoriasis, sclerosing cholangitis, primary sclerosing cholangitis, and diseases of the central and peripheral nervous systems.

(146) In certain embodiments, the autoimmune disease or inflammatory disorder may include, but is not limited to, type I diabetes, multiple sclerosis, Celiac Disease, primary biliary cirrhosis, pemphigus, pemphigus foliaceus, pemphigus vulgaris, neuromyelitis optica spectrum disorder, arthritis (including rheumatoid arthritis), allergic asthma, inflammatory bowel disease (including Crohn's disease and ulcerative colitis), systemic lupus erythematosus, atherosclerosis, chronic obstructive pulmonary disease, emphysema, psoriasis, autoimmune hepatitis, uveitis, Sjogren's Syndrome, scleroderma, anti-phospholipid syndrome, ANCA-associated vasculitis, and Stiff Man Syndrome. In a further aspect, the disease-relevant antigen is a tumor- or cancer-relevant antigen.

(147) In certain embodiments, the autoimmune disease or inflammatory disorder may include a liver autoimmune disorder selected from primary biliary cirrhosis, autoimmune hepatitis, and primary sclerosing cholangitis. In certain embodiments, the uaMHC-NP complex is used in a method of treating primary biliary cirrhosis, autoimmune hepatitis, and/or primary sclerosing cholangitis. In certain embodiments, the method of treating primary biliary cirrhosis, autoimmune hepatitis, and/or primary sclerosing cholangitis comprises administering a uaMHC-NP complex, wherein the ubiquitous autoantigen is a polypeptide derived from any one or more of pyruvate dehydrogenase complex-E2 component (PDC-E2), cytochrome P450 2D6 (CYP2D6), soluble liver antigen (SLA), actin (ACTB), formimidoyltransferase-cyclodeaminase (FTCD), and myeloperoxidase (MPO). In certain embodiments, the method of treating primary biliary cirrhosis comprises administering a uaMHC-NP complex, wherein the ubiquitous autoantigen is a polypeptide derived from any one or more of cytochrome P450 2D6 (CYP2D6), soluble liver antigen (SLA), actin (ACTB), formimidoyltransferase-cyclodeaminase (FTCD), and myeloperoxidase (MPO). In certain embodiments, the method of treating autoimmune hepatitis comprises administering a uaMHC-NP complex, wherein the ubiquitous autoantigen is a polypeptide derived from any one or more of pyruvate dehydrogenase complex-E2 component (PDC-E2), soluble liver antigen (SLA), actin (ACTB), formimidoyltransferase-cyclodeaminase (FTCD), and myeloperoxidase (MPO). In certain embodiments, the method of treating primary sclerosing cholangitis comprises administering a uaMHC-NP complex, wherein the ubiquitous autoantigen is a polypeptide derived from any one or more of pyruvate dehydrogenase complex-E2 component (PDC-E2), cytochrome P450 2D6 (CYP2D6), soluble liver antigen (SLA), actin (ACTB), and myeloperoxidase (MPO).

(148) In certain embodiments, the autoimmune disease or inflammatory disorder may include multiple sclerosis. In certain embodiments, the autoimmune disease or inflammatory disorder may include relapsing-remitting multiple sclerosis. In certain embodiments, the uaMHC-NP complex is used in a method of treating multiple sclerosis. In certain embodiments, the method of treating multiple sclerosis comprises administering a uaMHC-NP complex, wherein the ubiquitous autoantigen is a polypeptide derived from any one or more of pyruvate dehydrogenase complex-E2 component (PDC-E2), cytochrome P450 2D6 (CYP2D6), soluble liver antigen (SLA), actin (ACTB), formimidoyltransferase-cyclodeaminase (FTCD), and myeloperoxidase (MPO).

(149) Pharmaceutically Acceptable Stabilizers, Excipients, and Diluents

(150) In certain embodiments, the uaMHC-NP complexes of the current disclosure are included in a pharmaceutical composition comprising one or more pharmaceutically acceptable stabilizers

excipients, carriers, and diluents. In certain embodiments, the uaMHC-NP complexes of the current disclosure are administered suspended in a sterile solution. In certain embodiments, the solution comprises 0.9% NaCl. In certain embodiments, the solution further comprises one or more of: buffers, for example, acetate, citrate, histidine, succinate, phosphate, bicarbonate and hydroxymethylaminomethane (Tris); surfactants, for example, polysorbate 80 (Tween 80), polysorbate 20 (Tween 20), and poloxamer 188; polyol/disaccharide/polysaccharides, for example, glucose, dextrose, mannose, mannitol, sorbitol, sucrose, trehalose, and dextran 40; amino acids, for example, glycine or arginine; antioxidants, for example, ascorbic acid, methionine; or chelating agents, for example, EGTA or EDTA. In certain embodiments, the uaMHC-NP complexes of the current disclosure are shipped/stored lyophilized and reconstituted before administration. In certain embodiments, the lyophilized uaMHC-NP complexes formulations comprise a bulking agent such as mannitol, sorbitol, sucrose, trehalose, or dextran 40. The lyophilized formulation can be contained in a vial comprised of glass. The uaMHC-NP complexes, when formulated, whether reconstituted or not, can be buffered at a certain pH, generally less than 7.0. In certain embodiments, the pH can be between 4.5 and 6.5, 4.5 and 6.0, 4.5 and 5.5, 4.5 and 5.0, or 5.0 and 6.0. In certain embodiments, the uaMHC-NP complexes can be formulated for intravenous injection. In certain embodiments, uaMHC-NP complexes can be formulated for oral ingestion. In certain embodiments, uaMHC-NP complexes can be formulated for parenteral, intramuscular, or intra tissue injection. In certain embodiments, uaMHC-NP complexes can be formulated and/or administered without any immunological adjuvant or other compound or polypeptide intended to increase or decrease an immune response.

EXAMPLES

(151) The following illustrative examples are representative of embodiments of the compositions and methods described herein and are not meant to be limiting in any way.

Example 1—TR1 Like CD4⁺ T-Cell Formation and Expansion by PBC-Relevant pMHC Class II-NPs

(152) NOD.c3c4 mice, which carry anti-diabetogenic B6-derived chromosome 3 and 4 regions, spontaneously develop a form of autoimmune biliary ductal disease that resembles human PBC. See Irie, J., et al. *J. Exp. Med.* 203, 1209-1219. Like >90% of patients, these mice develop pathogenic T- and B-cell responses against the E2 and E3BP components of the pyruvate dehydrogenase (PDC) complex. See Kita, H. et al. *J. Clin. Invest.* 109, 1231-1240. In NOD.c3c4 mice as well as in humans, these autoimmune responses promote the destruction of biliary epithelial cells, leading to cholestasis, small bile duct proliferation, and finally liver failure.

(153) To design PBC-relevant pMHC class II-based nanomedicines, we searched for 15mer peptides in murine PDC-E2 capable of binding to the NOD/NOD.c3c4 mouse MHC class II molecule (I-A.sup.g7) in silico. I-A.sup.g7-based pMHCs encoding two such epitopes (PDC-E2.sub.166-181 and PDC-E2.sub.82-96) were chosen for experimentation. The T1D-relevant I-A.sup.g7-binding BDC2.5 mimotope was used as a negative control. These complexes were produced in lentiviral-transduced Chinese hamster ovary (CHO) cells, purified by sequential nickel and streptag affinity chromatography, and covalently coated via a free carboxyterminal cysteine onto iron-oxide nanoparticles produced by thermal decomposition of Iron (III) acetylacetonate (Fe(acac).sub.3) in the presence of maleimide-functionalized polyethylene glycol, as described in Singha, S. et al. *Nature Nanotechnology* 12, 701-710.

(154) pMHC tetramer staining studies demonstrated that NOD.c3c4 (but not NOD) mice harbor increasing levels of both the PDC-E2.sub.166-181/IA.sup.g7 and PDC-E2.sub.82-96/IA.sup.g7-reactive T-cell subsets with age as shown in FIG. 1A (upside-down triangles, middle and right panels). In contrast, and unlike NOD mice, NOD.c3c4 mice contain negligible levels of the T1D-relevant BDC2.5mi/IA.sup.g7-reactive subset as shown in FIG. 1A (triangles, middle and right panels), an outcome that is consistent with the PBC vs T1D proclivity of these two strains. Thus, progression of liver autoimmunity in NOD.c3c4 mice is accompanied by increases in the size

and/or circulating activity of PDC-E2-specific CD4⁺ T-cells.

(155) To ascertain if PBC-relevant pMHC class II-NPs could trigger the formation and expansion of PDC-E2-specific TR1-like CD4⁺ T-cells in NOD.c3c4 mice, we first treated 15 week-old NOD.c3c4 mice (when the disease is well established) with NPs displaying the PDC-E2.sub.166-181/IA.sup.g7 pMHC or control NPs (Cys-NPs) (twice a week intravenous for up to 13.5 weeks). Treatment triggered a rapid increase (within 2.5 weeks) in the peripheral frequency of circulating PDC-E2.sub.166-181/IA.sup.g7 tetramer+CD4⁺ T-cells as compared to mice treated with bare NPs or to untreated NOD mice as shown in FIG. 1B (squares, left and middle panels). Studies of mice at the end of follow up confirmed that the PDC-E2.sub.166-181/IA.sup.g7-NP-induced expansion of cognate CD4⁺ T-cells was systemic, with increased frequencies in spleen, bone marrow, liver and liver-draining (portal and celiac lymph nodes), but not non-draining lymph nodes (vs. mesenteric lymph nodes (MLN)) as shown in FIG. 1C.

(156) In contrast, treatment of NOD.c3c4 with NPs coated with the T1D-relevant BDC2.5/I-A.sup.g7 pMHCs did not trigger TR1 cell formation or expansion (FIG. 1B right panel and FIG. 1D). This result is consistent with these nanomedicines exclusively operating on autoantigen-experienced T-cells; NOD.c3c4 mice do not develop islet inflammation and therefore are not expected to harbor antigen-activated BDC2.5mi/IA.sup.g7-autoreactive CD4⁺ T-cells.

(157) Experiments in additional cohorts of mice demonstrated that the PDC-E2.sub.166-181/IA.sup.g7-NP-induced expansion of cognate CD4⁺ T-cells was peptide-specific, without any detectable expansion of PDC-E2.sub.82-96/I-A.sup.g7-reactive CD4⁺ T-cells (FIG. 1E, top row). In fact, expansion of PDC-E2.sub.166-181/IA.sup.g7-specific CD4⁺ T-cells in these mice was accompanied by significant reductions in the frequency of PDC-E2.sub.82-96/I-A.sup.g7-reactive CD4⁺ T-cells in all organs examined, as compared to the levels detected in age-matched untreated mice (FIG. 1E, top row). This suggests that the PDC-E2.sub.166-181/IA.sup.g7-specific CD4⁺ T-cell subset somehow inhibited the proliferation of its PDC-E2.sub.82-96/I-A.sup.g7-reactive counterpart in response to endogenous autoantigen exposure.

(158) As expected, the PDC-E2.sub.166-181/IA.sup.g7 tetramer+CD4⁺ T-cells that expanded in these mice expressed the TR1 cell markers LAG-3, CD49b and LAP as shown in FIG. 1F and FIGS. 2A and 2B. Furthermore, unlike their tetramer-negative counterparts, the splenic tetramer+CD4⁺ T-cells of these mice produced the TR1 cytokine IL-10 but not IFN γ , IL-2, IL-4, IL-9 or IL-17 in response to PDC-E2.sub.166-181 (but not BDC2.5) peptide-pulsed bone marrow-derived DCs (FIG. 1G). Similar results were obtained in mice treated with NPs displaying the second PDC-E2-based pMHC (PDC-E2.sub.82-96/I-A.sup.g7), in which there was a significant expansion of cognate PDC-E2.sub.82-96/I-A.sup.g7-reactive TR1-like CD4⁺ T-cells and significant reductions in the frequency of PDC-E2.sub.166-181/IA.sup.g7-reactive CD4⁺ T-cells as shown in FIGS. 1B (center) and 1E (bottom) and FIGS. 2A and 2B, indicating that the above outcome is not a peculiarity of any particular epitope on PDC-E2.

(159) Together, the above data demonstrate that PDC-E2 peptide/IA.sup.g7-NPs efficiently trigger the formation and expansion of cognate TR1-like CD4⁺ T-cells in NOD.c3c4 mice, as described previously for T1D-, EAE- and CIA-relevant pMHC class II-NPs in the corresponding disease models.

Example 2—Reversal of Established PBC by Disease-Relevant pMHC Class II-NPs

(160) When compared to age-matched NOD mice, 6-8 week-old NOD.c3c4 mice begin to display elevated levels of serum alanine aminotransferase (ALT), microscopic biliary epithelial proliferation, biliary track leukocyte infiltration, massive bile duct involvement (near maximum number of portal triads affected) and macroscopic enlargement of the common biliary duct (CBD) (FIGS. 3A and 3C). FIG. 3B shows an exemplary scoring matrix to quantify microscopic analysis. By ~15-16 weeks these signs worsen and the mice begin to display increased total serum bilirubin (TB) levels (FIG. 3A) high titers of anti-mitochondrial/PDC-E2-specific autoantibodies (absent in NOD mice; FIG. 3E) and macroscopic signs of liver disease (bile cysts) (FIG. 3D). The severity of

all these signs of disease peaks at ~24 weeks of age (FIGS. 3A-3D), coinciding with massive infiltration of the biliary epithelium by CD4+ and CD8+ T-cells (FIG. 3F), high titers of anti-nuclear autoantibodies (ANAs) (FIG. 3E) and a nearly three-fold increase in liver weight (FIG. 3D).

(161) We tested the therapeutic properties of PDC-E2.sub.166-181/IA.sup.g7-, PDC-E2.sub.82-96/I-A.sup.g7- and BDC2.5/I-A.sup.g7-NPs in 15 week-old NOD.c3c4 mice, an age when liver autoimmunity in these mice is well established. Mice received biweekly doses of 20 ug of pMHC-NPs or an equivalent dose of control (Cys-conjugated NPs; Cys-NPs) for 9-13.5 weeks. Treatment with PDC-E2.sub.166-181/IA.sup.g7-NPs resulted in significant reductions in serum ALT and TB levels (FIG. 4A), bile duct involvement, bile duct epithelial proliferation and leukocyte infiltration (FIG. 4B), common bile duct diameter and macroscopic score (FIG. 4C), liver weight and macroscopic liver scores (FIG. 4D) and abdominal girth (FIG. 4E). Although treatment did not decrease the autoantibody titers found at the initiation of therapy, it clearly blunted the progression of autoantibody formation, as documented by significant reductions in anti-PDC-E2 and anti-nuclear autoantibody titers in mice treated with PDC-E2.sub.166-181/IA.sup.g7-NPs, PDC-E2.sub.82-96/I-A.sup.g7-NPs vs. control NPs (FIG. 4F). Additional studies with the second PBC-relevant pMHC-NP compound (PDC-E2.sub.82-96/I-A.sup.g7-NPs), and the T1D-relevant (but PBC-irrelevant) counterpart (BDC2.5/I-A.sup.g7-NPs) confirmed the disease specificity of these compounds (FIGS. 4C and 4D).

(162) Similar results were obtained when treatment was initiated at the peak of disease severity (24 weeks of age). Analyses of mice after 14-20 weeks of therapy (38-44 weeks of age) with PDC-E2.sub.166-181/IA.sup.g7-NPs revealed systemic expansions of TR1-like CD4+ T-cells (FIG. 4G and FIG. 2C) and indicated that the magnitude of the signs of disease (FIGS. 4H and 4I) were significantly lower than those seen at the age when therapy was initiated, suggesting that resolution of liver inflammation by PBC-specific nanomedicines promotes repair of pre-existing liver damage.

Example 3—Continued Versus Intermittent Treatment

(163) pMHC-NP therapy triggers the formation and expansion of cognate TR1 cells systemically, leading to accumulation of these cells in most lymphoid organs as well as at the site of autoimmune inflammation (See FIGS. 1-3). In addition, these TR1 cells circulate through the bloodstream and their presence in blood can be used as a biomarker to gauge the need for re-treatment. To investigate this, and to ascertain whether circulating levels of cognate TR1-like CD4+ T-cells can in fact be used to guide therapeutic decisions, we withdrew treatment in NOD.c3c4 mice that had been treated with PDC-E2.sub.166-181/IA.sup.g7-NPs from 15 week to 24 weeks of age. We then measured the percentage of PDC-E2.sub.166-181/IA.sup.g7 tetramer+CD4+ T-cells in peripheral blood every two weeks, and re-treated animals in which the percentage of tetramer+ cells had declined to about 50% of the value at treatment withdrawal; treatment was again withdrawn when the tetramer+ values at the next scheduled measurement had recovered, and repeated this cycle until the mice reached 50 week of age. Although there was considerable variability from mouse to mouse (FIG. 5A), in most animals the peripheral blood tetramer+ TR1 cell content progressively declined to about 50% of the original values within 4-6 weeks after treatment withdrawal but re-treatment of these animals rapidly restored these values (FIG. 5B). Intermittent treatment did not compromise the pharmacodynamic (systemic expansion of TR1-like CD4+ T-cells) (FIGS. 5C and 5D) or the therapeutic effects of treatment, including reduced common bile duct diameter/macroscopic scores and liver weight/macroscopic score (FIG. 5E), as compared to values in untreated mice or mice which had been treated continuously from 24 to 38-44 weeks of age.

Example 4—pMHC-NPs Versus the Standard of Care

(164) Ursodeoxycholic acid (UDCA, a hydrophilic bile acid) is the standard of care for PBC. See Charatcharoenwitthaya, P. et al. Long-term survival and impact of ursodeoxycholic acid treatment for recurrent primary biliary cirrhosis after liver transplantation. *Liver Transpl.* 13, 1236-1245.

UDCA possesses anti-cholestatic effects and stimulates hepatobiliary secretion, thus protecting cholangiocytes against the toxic effects of hydrophobic bile acids. Although effective in ~50% of patients when given early on in the disease process, it is ineffective at advanced stages of PBC. (165) Oral administration of UDCA to 6 weeks-old NOD.c3c4 mice for 9 consecutive weeks via UDCA-supplemented chow had a therapeutic effect on the progression of PBC, as manifested by reductions in liver scores and liver weight, albeit not ALT, CBD scores or CBD diameter (FIG. 6A) and reductions in bile duct involvement and bile duct proliferation, albeit not leukocyte infiltration (FIG. 6B) as compared to untreated mice. However, when UDCA was given at advanced stages of disease progression (24 weeks of age), it had none of these therapeutic effects, except for a very significant reduction in CBD diameter as compared to untreated animals, possibly because of its anti-cholestatic effects (FIGS. 6D and 6E).

(166) In contrast, PDC-E2.sub.166-181/IA.sup.g7-NP treatment had highly significant therapeutic effects in both 6 week-old and 24 week-old animals, as documented by significant reductions in the severity of all read-outs examined (FIGS. 6A-6E). As expected, this was associated with systemic expansion of cognate TR1-like CD4⁺ T-cells (FIG. 6F).

Example 5—Disease Suppression Requires IL-10, TGF β and CD4⁺ T Cells

(167) To ascertain if disease reversal by the TR1-like pMHC-NP-expanded PDC-E2.sub.166-181/IA.sup.g7 CD4⁺ T-cells was mediated by the TR1 cytokines IL-10 and/or TGF β , we compared the immunological and therapeutic effects of PDC-E2.sub.166-181/IA.sup.g7-NPs on 15 week-old NOD.c3c4 mice treated with blocking anti-IL10 or anti-TGF β mAbs or rat IgG for 5 weeks. Whereas cytokine blockade did not significantly inhibit the expansion of PDC-E2.sub.166-181/IA.sup.g7-specific TR1-like CD4⁺ T-cells (FIG. 7A), it suppressed their therapeutic effects, as compared to age-matched NOD.c3c4 mice treated with rat-IgG (FIGS. 7B and 7C). Purified splenic CD4⁺ T-cells from PDC-E2.sub.166-181/IA.sup.g7-NP-treated NOD.c3c4 mice could transfer disease suppression into NOD.scid.c3c4 hosts reconstituted with splenocytes from sick NOD.c3c4 mice, and treatment of the hosts with PDC-E2.sub.166-181/IA.sup.g7-NPs enhanced this effect as shown in FIGS. 7D-7F.

Example 6—Therapy-Induced Suppression of the Pro-Inflammatory Properties of Local and Proximal APCs

(168) To ascertain whether reversal of PBC by PDC-E2.sub.166-181/IA.sup.g7-NPs was associated with specific suppression of disease-fueling APCs, we compared the cytokine and chemokine profiles of portal (draining) vs. mesenteric (non-draining) lymph node CD11b⁺ cells and liver Kupffer cells isolated from PDC-E2.sub.166-181/IA.sup.g7-NP and control NP-treated animals. LPS-challenged CD11b⁺ cells from the portal lymph nodes of control NP-treated animals secreted significantly higher levels of a broad range of pro-inflammatory cytokines and chemokines than their mesenteric lymph node counterparts (FIG. 7G). Conversely, the portal lymph node CD11b⁺ cells of PDC-E2.sub.166-181/IA.sup.g7-NP-treated mice secreted significantly lower levels of such pro-inflammatory mediators than both their mesenteric lymph node counterparts and the CD11b⁺ cells isolated from control-NP-treated animals (FIG. 7G). Likewise, the Kupffer cells isolated from the livers of PDC-E2.sub.166-181/IA.sup.g7-NP-treated mice secreted significantly lower levels of some of these mediators (FIG. 7H). Thus, systemic expansion of PDC-E2-specific TR1 CD4⁺ cells in NOD.c3c4 mice by treatment with PBC-relevant pMHC class II-NPs is associated with dramatic inhibition of the pro-inflammatory properties of local and proximal APC types, likely owing to increased uptake of liver-derived PDC-E2 autoantigenic material.

Example 7—PBC-Relevant Nanomedicines Promote Local Formation of Regulatory B Cells

(169) Pancreatic beta cell-specific TR1 CD4⁺ T-cells promote the recruitment of B-cells to the pancreas and its draining lymph nodes, as well as the local formation of anti-diabetogenic IL-10-producing Breg cells. To ascertain if this was also the case for PDC-E2-specific TR1 CD4⁺ T-cells in the context of PBC, we investigated if there were statistically significant correlations between the absolute numbers of B-cells and PDC-E2.sub.166-181/IA.sup.g7-specific TR1 cells in the liver,

portal and mesenteric lymph nodes of PDC-E2.sub.166-181/IA.sup.g7-NP-treated mice. The livers and the portal, but not the mesenteric lymph nodes of PDC-E2.sub.166-181/IA.sup.g7-NP-treated mice harbored significantly higher numbers of PDC-E2.sub.166-181/IA.sup.g7-tetramer+ cells and B-cells than those from control-NP-treated animals (FIG. 7I). Furthermore, the tetramer+ and B-cell numbers in both liver and portal lymph nodes were statistically correlated; no such correlation was seen in the mesenteric lymph nodes of PDC-E2.sub.166-181/IA.sup.g7-NP-treated mice (FIG. 7J). Thus, enhanced recruitment of cognate TR1 cells to the liver and liver-draining lymph nodes of PDC-E2.sub.166-181/IA.sup.g7-NP-treated mice promotes the local accumulation of B-cells.

(170) To ascertain whether the liver and portal lymph node B-cells of PDC-E2.sub.166-181/IA.sup.g7-NP-treated mice might be enriched for Breg cells, we compared the ability of the corresponding B-cells to produce IL-10 in response to LPS stimulation. The liver and portal, but not the mesenteric lymph node B-cells of PDC-E2.sub.166-181/IA.sup.g7-NP-treated mice produced significant levels of IL-10; neither the liver nor the portal lymph node B-cells of control NP-treated animals produced IL-10 (FIG. 7K). Thus, TR1 CD4+ T-cell-enhanced recruitment of B-cells to liver and draining lymph nodes is associated with local formation of IL-10-producing B-cells.

(171) To further substantiate a direct relationship between TR1 cell recruitment and Breg cell formation, we ascertained the ability of PDC-E2.sub.166-181/IA.sup.g7-specific TR1 cells that accumulate in the spleen, liver and portal (but not mesenteric) lymph nodes of PDC-E2.sub.166-181/IA.sup.g7-treated mice to promote the differentiation of PDC-E2.sub.166-181 peptide-pulsed conventional (IL-10/eGFP-) B-cells from NOD.II10-eGFP reporter mice into CD1dhigh/CD5+/eGFP+ progeny. As shown in FIGS. 7K-7M, there was a clear formation of Breg cells in spleen, liver and portal lymph nodes (containing cognate TR1 cells) but not in the mesenteric lymph nodes (lacking cognate TR1 cells). When taken together, these results indicate that PDC-E2-specific TR1 cells promote the recruitment and differentiation of conventional B-cells into Breg-like cells.

Example 8—Systemic Immunity is Spared

(172) To ascertain if persistent expansion of PDC-E2.sub.166-181/IA.sup.g7-specific TR1 cells results in suppression of general immunity, we compared the ability of PDC-E2.sub.166-181/IA.sup.g7-NP-treated NOD.c3c4 mice to clear a systemic vaccinia virus infection, to mount effector T-cell responses against a lethal influenza infection, to mount protective immunity against an intracellular bacterial infection (*Listeria*), and to mount local immune responses against allogeneic tumor liver metastases. Cohorts of NOD.c3c4 mice received biweekly doses of PDC-E2.sub.166-181/IA.sup.g7-NPs or control NPs for 9 weeks. At the end of therapy the mice were given an intravenous injection of recombinant vaccinia virus. The viral titers in the ovaries of both cohorts of females 14 days after infection were similar in both cohorts of mice and substantially lower than those found at the peak of infection, indicating that pMHC-NP therapy did not impair cellular immunity against the virus-infected cells (FIG. 8A).

(173) To probe this further, we infected PDC-E2.sub.166-181/IA.sup.g7-NP-treated or untreated NOD.c3c4 mice with a laboratory strain of influenza HKx31 (H3N2) i.p. to induce heterologous immunity against a subsequent, potentially lethal infection with an H1N1 strain of Influenza (PR8) given via the intranasal route, that shares MHC class I-restricted epitopes. As shown in FIGS. 8B and 8C, treated and untreated mice mounted protective immunity against the PR8 infection, as documented by viral load in lung tissue and clinical signs of active infection.

(174) Similar results were obtained in mice infected with the intracellular pathogen *Listeria monocytogenes* (LM), which does not normally cause chronic infections. LM-infected PDC-E2.sub.166-181/IA.sup.g7-NP-treated and untreated NOD.c3c4 mice were equally efficient at clearing the bacteria from both the spleen and liver, consistent with unimpaired immunity against this intracellular pathogen (FIG. 8D).

(175) PDC-E2.sub.166-181/IA.sup.g7-NP-treated and untreated NOD.c3c4 mice also produced

similar titers of anti-dinitrophenyl (DNP) antibodies upon immunization with the hapten-carrier conjugate DNP-keyhole limpet haemocyanin (KLH) (FIG. 8E).

(176) Lastly, systemic expansion and liver accumulation of PBC-suppressing PDC-E2-specific TR1-like CD4⁺ T-cells (FIGS. 8F and 8L) did not impair the ability of treated NOD.c3c4 mice to mount immune responses against allogeneic colon carcinoma (CT26) and melanoma (B16) liver metastases arising upon intra-splenic injection, as compared to untreated NOD.c3c4 mice or syngeneic hosts (Balb/c and C57BL/6J, respectively) (FIGS. 8F-8K and 8L-8Q). Thus, despite targeting a ubiquitously expressed antigen, PDC-E2.sub.166-181/IA.sup.g7-specific TR1 cells do not impair immunity against foreign or tumor antigens.

Example 9—Humanized Mice with PBC

(177) DRB4*0101 and DRB1*0801 have been associated with PBC in some studies. To ascertain the HLA haplotypic diversity in PBC, we did high-resolution HLA-DRB1-typing of 154 patients with PBC from Spain. 40.3% of patients expressed DRB1*0701, 25% were DRB1*0301+ and 14% were DRB1*0801+. Since DRB1*0701+ and haplotypes carrying other DRB1 alleles carry the oligomorphic HLA-DRB4 locus, we also typed these patients for DRB4*0101. 61.7% of all PBC patients carried the DRB4*0101 allele.

(178) Several T-cell epitopes from PDC-E2 binding to two of these HLA-DRB types (DRB4*0101 and DRB1*0801) have been described, including PDC-E2.sub.249-262 (GDLLAEIETDKATI (SEQ ID NO:9); DRB4*0101-binder), PDC-E2.sub.122-135 (GDLIAEVETDKATV (SEQ ID NO:8); also DRB4*0101-binder), PDC-E2.sub.249-263 (GDLLAEIETDKATIG (SEQ ID NO:10); DRB1*0801) and PDC-E2.sub.629-643 (AQWLAEFRKYLEKPI (SEQ ID NO:7); DRB1*0801). We therefore expressed and purified PDC-E2.sub.122-135/DRB4*0101, PDC-E2.sub.249-262/DRB4*0101, and PDC-E2.sub.629-643/DRB1*0801 complexes and produced iron oxide nanoparticles displaying each of these complexes, as described.

(179) To investigate the translational significance of the above observations, we tested the ability of these three human PBC-relevant pMHC class II-NPs to expand cognate TR1-like CD4⁺ T-cells in NOD.scid/Il2rg.sup.-/- (NSG) hosts reconstituted with PBMCs from 11 DRB4*0101+ and 5 DRB1*0801+ PBC patients (PBL-NSG mice, Tables 2, 3 and 4). PBMC-transfused NSG hosts were then treated with 8-10 doses of 20 µg pMHC-NPs intravenous (twice/week for 5 weeks). One mouse did not engraft and three others died from GvHD prior to termination of treatment. As controls, we transfused a second mouse per donor and treated it with control (non-pMHC-coated NPs). Expansions of cognate CD4⁺ T-cells were analyzed in the spleens, liver, portal/ceciac and axillary lymph nodes. We saw expansion of tetramer+CD49b+LAG-3+CD4⁺ T-cells in the spleen and/or liver and LNs from all 6/6 PBL-NSG mice treated with PDC-E2.sub.122-135/DRB4*0101-NPs, 5/6 PBL-NSG mice treated with PDC-E2.sub.249-262/DRB4*0101-NPs and 4/5 PBL-NSG mice treated with PDC-E2.sub.629-643/DRB1*0801-NPs vs. the untreated controls (Tables 2, 3, and 4). Treated responsive mice had significantly higher percentages and absolute numbers of tetramer+ cells in spleen, liver and lymph nodes (FIGS. 9A and 9B) than control NP-treated or unresponsive mice, and these cells expressed the TR1 markers CD49b and LAG-3 (FIG. 9C).

(180) TABLE-US-00002 TABLE 2 Table 2 hPDC-E2(122-135)-DRB4-PFM-010616 (30 mcg/dose). Spleen Liver LN Anti- % hCD4 % hCD4 % hCD4 % hCD4 % hCD4 % hCD4 Age mitochondrial Treated Untreated Treated Untreated Treated Untreated ID Gender (yr) Abs # of cells # of cells # of cells # of cells # of cells # of cells Outcome B013 F 79 + 0.055 0.303 2.288 0.302 0.863 0.274 + (A) 63756 141561 300444 39069 23760 154 B009 F 76 + 1.253 0.291 0.199 0.307 * + (249-NP) (249-NP) 732729 308949 7047 11853 * B005 M 74 + No engraftment B001 M 59 + 0.319 0.296 1.625 0.296 1.082 0.312 + (A) 186539 291863 256354 62871 28524 1874 B021 F 68 + 0.767 0.309 0.343 0.296 0.728 0.3 + (A) 849816 509331 18691 28993 15072 5721 B011 F 71 + 0.169 0.304 0.091 0.294 0.425 0.291 + (L) 202383 209370 6658 151772 120 755 B018 F 56 + Dead B012 F 71 + 0.709 0.298 0.722 0.290 + (A) (A) 341920 124050 ND 5602 2101 DRB4*01:01 patients

(181) TABLE-US-00003 TABLE 3 hPDC-E2(249-262)-DRB4-PFM-010616 (40 mcg/dose). Spleen Liver LN Anti- % hCD4 % hCD4 % hCD4 % hCD4 % hCD4 % hCD4 Age mitochondrial Treated Untreated Treated Untreated Treated Untreated ID Gender (yr) Abs # of cells # of cells # of cells # of cells # of cells # of cells Outcome B013 F 79 + 0.078 0.31 4.925 0.295 0.621 0.245 + (L) 105174 164589 1124257 35712 33121 275 B009 F 76 + 0.028 0.307 0.515 0.297 * + (122-NP) (122-NP) 31807 192099 20215 10866 * B004 F 63 + 0.427 0.294 0.094 0.315 0.089 0.293 + (A) 268565 228575 2990 299574 526 11251 B012 F 70 + 0.264 0.297 0.078 0.3 0.059 0.313 - (L) 41467 147568 1034 24584 1513 6160 B001 M 59 + Dead B021 F 68 + 1.612 0.31 1.943 0.289 2.589 0.32 + (L) 1368420 514802 1174100 35690 9491 788 B019 F 73 + Dead B014 F 72 + 0.709 0.295 0.588 0.305 1.096 0.299 + (A) 604820 162521 76559 30198 5557 1271 416501 181855 4960 4517 3740 2232 DRB4*01:01 patients

(182) TABLE-US-00004 TABLE 4 Table 4 hPDC-E2(629-643)-DR8-PFM-010616 (30 mcg/dose). Spleen Liver LN Anti- % hCD4 % hCD4 % hCD4 % hCD4 % hCD4 % hCD4 Age mitochondrial Treated Untreated Treated Untreated Treated Untreated ID Gender (yr) Abs # of cells # of cells # of cells # of cells # of cells # of cells Outcome B022 F 72 + 0.024 0.3 0.619 0.293 No cells 0.295 + (L) 662 48190 26 29984 260 B021 F 68 + 0.725 0.317 3.185 0.3 1.103 0.301 + (A) 1095456 562089 455716 36134 22752 8080 B020 F 70 + 0.303 0.308 0.18 0.297 No cells - 38207 114160 19765 46701 B025 F 69 + 0.404 0.308 2.895 0.294 0.501 0.319 + (A) (A) 59552 49721 78551 8747 740 466 B023 F 75 + 0.745 0.292 0.283 0.292 0.371 0.297 + (A) DRB1*08:01 patients

Example 10—Disease Versus Organ Specificity

(183) Given the large autoantigenic load of an organ such as the liver as compared to smaller organs, like the endocrine pancreas, and the fact that PDC-E2 is an autoantigen expressed in virtually all cell types, our results begged the question of whether PBC-relevant nanomedicines (i.e. PDC-E2.sub.166-181/IA.sup.g7-NP) are disease (PBC)- or organ (liver)-specific, or also able to blunt liver-distal autoimmune inflammation.

(184) Primary sclerosing cholangitis (PSC) is a chronic cholestatic disease characterized by inflammation of intra- and extra-hepatic bile ducts leading to a fibro-obliterative cholangitis with periductal fibrosis around medium and large bile ducts and degenerative changes of the biliary epithelium, which progresses to portal and biliary cirrhosis and finally liver cirrhosis. Human PSC is frequently associated with inflammatory bowel disease and accompanied by a high prevalence of atypical perinuclear anti-neutrophil cytoplasmic (pANCA) but not anti-mitochondrial autoantibodies. See Fickert, P. et al. Characterization of animal models for primary sclerosing cholangitis (PSC). *J. Hepatol.* 60, 1290-1303. Abcb4 gene knockout mice spontaneously develop a form of sclerosing cholangitis that is remarkably similar to human PSC and is caused by damage of bile duct cells by impaired biliary phospholipid secretion. See Pollheimer, M. J. & Fickert, P. Animal models in primary biliary cirrhosis and primary sclerosing cholangitis. *Clin. Rev. Allergy Immunol.* 48, 207-217, doi: 10.1007/s12016-014-8442-y (2015).

(185) Autoimmune Hepatitis (AIH) is characterized by a portal mononuclear cell infiltration of the liver parenchyma that is associated with presence of anti-nuclear and/or smooth muscle (AIH type 1) or anti-liver kidney microsomal or anti-liver cytosol type 1 autoantibodies, which specifically target the microsomal cytochrome P450IID6 (CYP2D6) or formiminotransferase cyclodeaminase (FTCD), respectively (AIH Type 2). See Longhi, M. S. et al. Aetiopathogenesis of autoimmune hepatitis. *J. Autoimmun.* 34, 7-14. Recently, it has been shown that infection of NOD mice with replication-defective adenoviruses encoding the human liver autoantigen formiminotransferase cyclodeaminase (Ad-FTCD) triggers a form of chronic autoimmune hepatitis that resembles AIH type 2. See Hardtke-Wolenski, M. et al. Genetic predisposition and environmental danger signals initiate chronic autoimmune hepatitis driven by CD4+ T cells. *Hepatology* 58, 718-728.

(186) The large bile duct and parenchymal liver damage that underlie PSC and AIH may trigger the release of PDC-E2 and the priming of cognate autoreactive CD4+ T-cells capable of responding to PDC-E2.sub.166-181/IA.sup.g7-NP therapy. If so, therapy should trigger the expansion of TR1-like

PDC-E2.sub.166-181/IA.sup.g7-specific CD4⁺ T-cells and suppression of local inflammation upon recognition of local and proximal PDC-E2-loaded APCs. Alternatively, the amount of PDC-E2 shed into the inflammatory milieu in PSC and/or AIH may be insufficient to generate PDC-E2166-181/IA.sup.g7-experienced CD4⁺ T-cells, hence an immunological and therapeutic response to PDC-E2.sub.166-181/IA.sup.g7-NPs.

(187) To test these alternative possibilities, we first investigated the ability of PDC-E2.sub.166-181/IA.sup.g7-NPs to trigger the expansion of cognate TR1-like CD4⁺ T-cells and revert PSC in NOD.Abc4.sup.-/- mice. Remarkably, PDC-E2.sub.166-181/IA.sup.g7-NP triggered the systemic expansion of cognate TR1-like CD4⁺ T-cells in these animals and reverted established disease, as compared to control NP-treated controls as shown in FIGS. 10A-10B and FIGS. 11A and 11B.

(188) We next investigated whether this was also true in Ad-FTCD-induced AIH. We compared the pharmacodynamic and therapeutic activities of both mFTCD.sub.58-72/IA.sup.g7-NPs and CYPD.sub.398-412/IA.sup.g7-NPs (AIH-relevant) with those of PDC-E2.sub.166-181/IA.sup.g7-NPs (PBC-relevant) in NOD mice infected with Ad-FTCD. All three compounds triggered the formation and expansion of cognate TR1-like CD4⁺ T-cells in these animals to a similar extent (FIG. 10C and FIGS. 12A and 12B) as compared to untreated Ad-FTCD-infected animals, and this was accompanied by significant reductions in liver inflammation as shown in FIGS. 10C and 10D, and serum ALT levels as shown in FIG. 10E.

(189) This ability of ubiquitous autoantigen-based pMHC-nanomedicines to blunt liver autoimmunity in an organ rather than disease-specific manner also occurred in NOD.c3c4 mice treated with CYPD.sub.398-412/IA.sup.g7-NPs (FIGS. 13A-13C). In fact, the latter were as efficient as PDC-E2.sub.166-181/IA.sup.g7-NPs at expanding cognate TR1 cells (FIG. 13A), and blunting PBC in 15 week-old mice (FIG. 13B). In contrast, neither PDC-E2.sub.166-181/IA.sup.g7-NPs nor CYPD.sub.398-412/IA.sup.g7-NPs triggered the expansion of cognate TR1 CD4⁺ T-cells in 10 week-old pre-diabetic NOD mice (FIG. 13C), unlike the case for beta cell-specific BDC2.5/IA.sup.g7-, IGRP.sub.4-22/IA.sup.g7- or IGRP.sub.128-145/IA.sup.g7-NPs, presumably because the PDC-E2 and CYPD content in beta cells is insufficient to prime the activation of cognate CD4⁺ T-cells.

(190) Collectively, these observations suggest that abundant levels of PDC-E2 (mitochondrial), CYPD2D6 and FTCD antigens (Golgi-resident or cytoplasmic, respectively) are delivered to local and proximal APCs upon hepatocyte (AIH) or bile duct epithelial cell damage (PBC and PSC), enabling autoreactive CD4⁺ T-cell priming, cognate TR1 cell generation by pMHC-NPs and suppression of local and proximal autoantigen-loaded APCs.

Example 11—Therapeutic Effects in Another PBC Model

(191) The NOD.c3c4 model does not fully recapitulate the immunopathology of human PBC, characterized by female prevalence, progression to liver fibrosis and absence of liver cyst formation. B6 mice carrying a deletion of the IFN γ 3'-untranslated region adenylate uridylate-rich element (ARE) (ARE-Del+/-) have a dysregulated Ifng locus, and develop a form of PBC that, like the human disease, primarily affects females and is associated with up-regulation of TBA, production of anti-PDC-E2 autoantibodies, portal duct and lobular liver inflammation, bile duct damage, granuloma formation and fibrosis. FIG. 14A-C show that treatment of (NODxB6.ARE-Del-/-) F1 mice with PDC-E2.sub.166-181/IA.sup.g7-NPs suppressed the upregulation of TBA and ALT levels, liver inflammation and fibrosis, as compared to mice treated with control NPs.

Example 12—Ubiquitous Autoantigen pMHCII-NPs Spare General Immunity Despite Harboring PDC-E2 Autoreactive TR1-Like CD4⁺ T-Cells

(192) PDC-E2.sub.166-181/IA.sup.g7-NP-treated or untreated NOD.c3c4 mice were infected with a laboratory strain of influenza (HKx31-H3N2-) i.p. to induce heterologous immunity against a lethal infection with an H1N1 strain of Influenza (PR8) given via the intranasal route. As shown in FIGS. 8B-8C, pMHCII-NP treated NOD.c3c4 mice mounted protective immunity against the PR8 infection, as documented by viral load in lung tissue and clinical signs of active infection, despite

systemic presence of cognate TR1-like CD4⁺ T-cells as shown in FIG. 15A.

(193) Similar results were obtained in mice infected with the intracellular pathogen *Listeria Monocytogenes* (LM). LM-infected PDC-E2.sub.166-181/IA.sup.g7-NP-treated and untreated NOD.c3c4 mice were equally efficient at clearing the bacteria from both the spleen and liver as shown in FIG. 8D, consistent with unimpaired immunity against this intracellular pathogen as shown in FIG. 15B.

Example 13—Ubiquitous Autoantigen pMHC-NPs Treat EAE in Mice

(194) One question unanswered was whether these liver disease-relevant nanomedicines, displaying epitopes from ubiquitous antigens, were liver-specific or, alternatively, also able to blunt extra-hepatic autoimmunity. To investigate this, the ability of PDC-E2.sub.166-181/IA.sup.g7-NPs and CYPD.sub.398-412/IA.sup.g7-NPs (displaying ubiquitous epitopes) vs. BDC2.5/IA.sup.g7-NPs (displaying a beta cell-specific epitope) and MOG.sub.36-50/IA.sup.g7-NPs (displaying a central nervous system-specific epitope) were compared to see if any could blunt a MOG.sub.36-55-induced relapsing-remitting form of EAE in NOD mice. BDC2.5/IA.sup.g7-NPs expanded cognate TR1-like CD4⁺ T-cells in these animals (in blood and spleen) to a similar extent as they do in non-EAE-affected NOD mice, but these cells were absent from the CNS-draining cervical lymph nodes (CLNs) (FIG. 16B), and had no anti-encephalitogenic activity, as shown in FIG. 16C (oligodendrocytes do not express chromogranin A).

(195) Notably, both CYPD.sub.398-412/IA.sup.g7-NPs and PDC-E2.sub.166-181/IA.sup.g7-NPs also triggered cognate TR1-like cell expansion in these mice (FIG. 16B) but, unlike BDC2.5/IA.sup.g7-NPs, suppressed CNS inflammation and demyelination and the progression and severity of EAE relapses (FIGS. 16C, D and 16E). This was associated with an accumulation of cognate TR1-like cells in the CLNs as shown in FIG. 16A. Importantly, these effects were not NOD genetic background-specific, because they were also seen in C57BL/6 mice treated with CYPD and PDC epitope/IA.sup.b-based nanomedicines: PDC.sub.94-108/IA.sup.b-NPs and CYPD.sub.353-367/IA.sup.b-NPs expanded cognate TR1-like CD4⁺ T-cells in blood, spleen and CLNs (FIGS. 17A and B) and blunted the progression of a chronic form of MOG.sub.35-55-induced EAE in these mice, as compared to Cys-NPs and MOG.sub.36-49/IA.sup.b-NPs (negative and positive controls, respectively) as shown in FIGS. 17C and D.

(196) Collectively, these observations suggested that significant levels of two different ubiquitously expressed autoantigens, PDC-E2 and CYPD2D6, are delivered to local and proximal APCs upon oligodendrocyte cell damage (in EAE), enabling autoreactive CD4⁺ T-cell priming, cognate TR1 cell generation by pMHC-NPs, recruitment and accumulation of these TR1 cells in the CNS-draining lymph nodes, and suppression of local inflammation.

Example 14—Induction of Beta Cell Apoptosis Renders NOD Mice Responsive to Ubiquitous Autoantigen pMHC-NPs

(197) Treatment of pre-diabetic 10 wk-old NOD mice with PDC-E2.sub.166-181/IA.sup.g7-NPs (PBC-relevant) and CYPD.sub.398-412/IA.sup.g7-NPs (AIH-relevant) did not trigger the expansion of cognate TR1-like CD4⁺ T-cells relative to endogenous BDC2.5/IA.sup.g7-specific CD4⁺ T-cells, as measured with pMHC tetramers, in blood (FIG. 18A) or any of the organs examined, including liver (FIG. 18B).

(198) These results suggested that spontaneous beta cell killing in pre-diabetic NOD mice does not trigger the formation of PDC-E2 or CYPD autoantigen-experienced T-cells capable of responding to the corresponding nanomedicines. Three alternative possibilities could explain these results: 1) these antigens are released from cholangiocytes and hepatocytes (in liver autoimmune diseases) but not from dying beta cells (in T1D); 2) the antigens are released from NOD beta cells, but NOD mice do not export cognate autoreactive T-cells capable of responding to these ubiquitous autoantigens; or 3) the antigens are released from beta cells and the mice harbor cognate autoreactive T-cells, but the antigens are released in amounts insufficient to prime these T-cells.

(199) To distinguish among the above three possibilities, an X-chromosome-linked rat-insulin

promoter-driven human diphtheria toxin receptor (hDTR) transgene was expressed in the pancreatic beta cells of non-diabetic female NOD mice. These mice were then treated with DT to simultaneously kill 50% of beta cells (due to X-chromosome inactivation, only 50% of beta cells express the hDTR). We then treated DT-treated and DT-untreated mice with PDC-E2.sub.166-181/IA.sup.g7-NPs, CYPD.sub.398-412/IA.sup.g7-NPs (liver autoimmunity-relevant), BDC2.5/IA.sup.g7-NPs (T1D-relevant, positive control), MOG.sub.36-50/IA.sup.g7-NPs (central nervous system autoimmune disease-specific, negative control) or Cys-coated NPs (negative control) and enumerated the presence of cognate autoreactive T-cells in spleen, liver-draining lymph nodes (portal and celiac LNs, PCLNs) and pancreatic LNs (PLNs), using pMHC tetramers. As expected, MOG.sub.36-50/IA.sup.g7-NPs did not trigger the expansion of cognate tetramer+CD4+ T-cells in these mice (MOG is an oligodendrocyte-specific protein that is not expressed in pancreatic beta cells) (FIG. 18C, bottom right panel). In contrast, both PDC-E2.sub.166-181/IA.sup.g7- and CYPD.sub.398-412/IA.sup.g7-NPs triggered the expansion and accumulation of significant numbers of cognate tetramer+CD4+ T-cells expressing the TR1 markers LAG3, LAP and CD49b in the spleen and PLNs of the DT-treated mice, to an extent similar to that seen for BDC2.5/IA.sup.g7 tetramer+ T-cells in BDC2.5/IA.sup.g7-NP-treated animals (FIG. 18C, bottom left and top panels, and FIG. 18D).

(200) These findings corroborate our previous observations that pMHC-NP-induced TR1 cell formation and expansion requires the presence of autoantigen-experienced T-cells, that priming of such cells requires antigen shedding from expressing cell types and that the peripheral repertoire of NOD mice normally harbors naïve CD4+ T-cell specificities targeting ubiquitously-expressed antigens.

Example 15—Ubiquitous Autoantigen pMHC-NPs Suppress Signs of Psoriasis in a Mouse Model of Induced Psoriasis

(201) The ability of ubiquitous autoimmune disease relevant pMHC-NPs was tested in a relevant mouse model of psoriasis, the imiquimod (IMQ) induced model. This mouse model is one of the most widely used mouse models in preclinical psoriasis studies. NOD/Ltj mice were treated with 5% topical IMQ for 6 days with 8 day rest intervals. The first application was made to the back of the mice and subsequent applications on the ear. Mice were administered either PDC-E2.sub.166-181/IA.sup.g7- or CYPD.sub.398-412/IA.sup.g7-NPs starting after the second application to the ear. Mice were assayed for symptoms of psoriasis following the PASI (Psoriasis Area and Severity Index) criteria which measures erythema, scaling and skin thickness; and mice were evaluated for expansion of PDC and CYPD tetramer+CD4 T cells in spleen, cervical lymph nodes (CV LN), and hepato-celiac lymph nodes (HC LN). As shown in FIG. 19A PDC-E2.sub.166-181/IA.sup.g7 were expanded in this model, indicating that these cells were primed by the induction of inflammation brought on by IMQ treatment. FIG. 19B shows that ear thickness was improved by treatment with PDC-E2.sub.166-181/IA.sup.g7- or CYPD.sub.398-412/IA.sup.g7-NPs, while scaling and erythema were not (not shown). These experiments are preliminary, but indicate that different models of autoimmune/inflammatory disease prime T cells to ubiquitous autoantigens that are able to be modified by ubiquitous autoantigen-relevant pMHC-NPs

Example 16—Select Methods Utilized Herein

(202) Mice

(203) NOD/LtJ, BALB/c, C57BL/6, NOD.scid.II2rg^{-/-} (NSG), NOD.c3c4 and FVB/N.Abc4^{-/-} (Abcb4 or ATP-binding cassette transporter, sub-family B, member 4) mice were purchased from the Jackson Laboratory (Bar Harbor, ME). IFN γ ARE-Del^{-/-} B6 mice were obtained from H. Young (NIH, Bethesda, MD). NOD.c3c4.scid mice were generated by backcrossing (NOD.c3c4 $\times$ NOD.scid) F1 mice with NOD.c3c4 mice for five generations, followed by intercrossing of mice heterozygous for the scid mutation and homozygous for the B6 chromosome 3 and 4 intervals from NOD.c3c4 mice. NOD.Abc4^{-/-} mice were obtained by backcrossing the mutant Abcb4 allele from FVB/N-Abc4^{-/-} mice onto the NOD/Ltj background for six

generations, followed by intercrossing. (NODxB6.IFN γ ARE-Del $^{-/-}$) F1 mice were generated by intercrossing IFN γ ARE-Del $^{-/-}$ and NOD/LtJ mice. NOD.II10tm1Flv (Tiger) mice were obtained by backcrossing the II10tm1Flv allele from C57BL/6.II10tm1Flv mice (Jackson Lab) onto the NOD/Ltj background for 10 generations. RIP-DTR.NOD transgenic mice were generated by backcrossing an X-chromosome-linked rat-insulin promoter-driven human diphtheria toxin receptor (RIP-DTR) transgene from transgenic B6 mice into the NOD background for more than 10 generations.

(204) Cell Lines, Pathogens, and Tumors

(205) CHO—S, BSC-1, MDCK, 293T, B16/F10 and CT26 cell lines were purchased from the ATCC (Manassas, VA). *Listeria monocytogenes* was obtained from DMX Corporation (Philadelphia, PA).

(206) Antibodies and Flow Cytometry

(207) FITC, PE, APC, PerCP or biotin-conjugated mAbs against mouse CD4 (RM4-5), CD5 (53-7.3), CD19 (1D3), B220 (RA36B2) and CD49b (HMa2) and streptavidin-PerCP were purchased from BD Biosciences (San Diego, CA). Anti-murine LAG-3 mAb (C9B7W) was purchased from eBioscience (San Diego, CA). Anti-latent-associated-TGF- β (LAP) antibody (TW7-16B4) was from BioLegend (San Diego, CA). PE-conjugated pMHC class II tetramers were produced using biotinylated pMHC monomers. pMHC class II tetramer staining and phenotypic marker analysis were done essentially as described with minor modifications. Briefly, after avidin incubation (15 min at RT), blood leukocytes, and single cell suspensions from spleen, lymph node, liver mononuclear cells, and bone marrow cells were stained first with pMHC tetramer (5 μ g ml.sup.-1) in FACS buffer (0.05% sodium azide and 1% FBS in PBS) for 60 min at 37° C., and later with FITC-conjugated anti-mouse CD4 (5 μ g ml.sup.-1) and PerCP-conjugated anti-mouse B220 (2 μ g ml.sup.-1; as a 'dump' channel) for 30 min at 4° C. After washing, cells were fixed (1% paraformaldehyde in PBS) and analyzed with FACScan, FACSaria, or BD LSRII flow cytometers. For phenotypic analyses, the cells were incubated with anti-FcR Abs, and then stained with cell surface marker antibodies diluted 1:100 in FACS buffer (at 4° C. for anti-CD49b and anti-LAP Abs, and at 37° C. for anti-LAG-3 Abs) followed by pMHC tetramer, FITC-conjugated anti-mouse CD4 (5 μ g ml.sup.-1) and PerCP-conjugated anti-mouse B220. Upon staining, cells were washed, fixed, and analyzed by flow cytometry. FlowJo software was used for all analyses.

(208) NSG-engrafted human T cells were analyzed using the following mAbs: FITC-conjugated anti-CD4 (OKT4, BioLegend), APC-conjugated anti-CD19 (HIB19, BD Biosciences, San Jose, CA), PerCP-conjugated polyclonal goat anti-LAG-3 IgG (R&D Systems, Minneapolis, MN), biotin-conjugated anti-CD49b (AK7, Pierce Antibodies, Thermo Fisher Scientific, Waltham, MA), and eFluor 450-conjugated streptavidin (eBioscience). Briefly, splenocytes and pancreatic lymph node cells were incubated with avidin (0.25 mg ml.sup.-1 in FACS buffer) for 30 min at room temperature, washed and stained with tetramer (5 μ g ml.sup.-1) for 1 h at 37° C., washed, and incubated with FITC-conjugated anti-CD4 (2/100), APC-conjugated anti-CD19 (5/100; used as a 'dump' channel), PerCP-conjugated anti-LAG-3 (8/100) and biotin-conjugated anti-CD49b (4/100) at 4° C. for 45 min. After washing, the cells were incubated with eFluor 450-conjugated streptavidin for 30 min at 4° C., washed, fixed in 1% PFA in PBS and cells within the hCD4.sup./hCD19.sup.- gate analysed with a FACSCanto II (BD Bioscience).

(209) pMHC Monomers and Peptides

(210) Recombinant pMHC class II monomers were purified from supernatants of CHO—S cells transduced with lentiviruses encoding a monocistronic message in which the peptide-MHCb and MHCa chains of the complex were separated by the ribosome skipping P2A sequence. The peptide was tethered to the amino terminal end of the MHCb chain via a flexible GS linker and the MHCa chains were engineered encode a BirA site, a 6 $\times$ His tag, a twin strep-tag and a free Cys at their carboxyterminal end. The secreted, self-assembled pMHC class II complexes were purified by sequential nickel and Strep-Tactin® chromatography and used for coating onto NPs or processed

for biotinylation and tetramer formation as described above. The epitopes encoded in the murine monomeric constructs were selected based on predicted MHCII-binding capacity using RANKPEP (http://imed.med.ucm.es/cgi-bin/rankpep_mif.cgi) using 7.54 as the threshold score. PDC-E2.sub.166-181 had a score that fell below the threshold but was selected for experimentation because it is contained within one of the lipoyl-binding domains of PDC-E2, an antigenic target for AMAs. For CYPD and FTCD epitope prediction, we used a second online algorithm (GPS-MBA) (mba.biocuckoo.org/) and peptides predicted by both RANKPEP and GPS-MBA were selected for experimentation. hPDC-E2.sub.122-135, hPDC-E2.sub.249-262 (both contained within the lipoyl-binding domain of PDC-E2), and hPDC-E2.sub.629-643 have been described previously. The sequences of the different epitopes are: PDC-E2.sub.166-181/IA.sup.g7 (LAEIETDKATIGFEVQ, (SEQ ID NO: 105), PDC-E2.sub.82-96/IA.sup.g7 (EKPQDIEAFKKNYTLT, (SEQ ID NO: 106), FTCD.sub.58-72/IA.sup.g7 (VVEGALHAARTASQL, (SEQ ID NO:107), CYPD.sub.398-412/IA.sup.g7 (LITNLSSALKDETVW, (SEQ ID NO:108), 2.5 mi/IA.sup.g7 (AHHPIWARMDA, (SEQ ID NO: 109), hPDC-E2.sub.122-135/DRB4*0101 (GDLIAEVETDKATV, (SEQ ID NO:8), hPDC-E2.sub.249-262/DRB4*0101 (GDLLAEIETDKATI, (SEQ ID NO:9), and hPDC-E2.sub.629-643/DRB1*0801 (AQWLAEFRKYLEKPI, (SEQ ID NO:7). Synthetic PDC-E2.sub.166-181, 2.5 mi, and mMOG.sub.36-55 (EVGWYRSPFSRVVHLYRNGK, (SEQ ID NO:110) peptides were purchased from Genscript (Piscataway, NJ). The amino acid residue numbers for each peptide correspond to those found in the mature form of the corresponding antigens.

(211) Nanoparticles, pMHCII-NP Synthesis, and Purification

(212) pMHCs were coated onto pegylated iron oxide NPs (PFM-NPs), produced as described (2). Briefly, PFM-NPs were produced by thermal decomposition of Fe(acac).sub.3 in the presence of 2 kD methoxy-PEG-maleimide. The NPs were purified using magnetic (MACS) columns (Miltenyi Biotec, Auburn, CA). Free Cysteines (controls) or pMHCs, carrying a free carboxyterminal Cys, were conjugated to the maleimide-functionalized PFMs in 40 mM phosphate buffer, pH 6.0, containing 2 mM EDTA, 150 mM NaCl overnight at room temperature. The pMHC-conjugated NPs were separated from free pMHC using magnetic columns, sterilized by filtration through 0.2 μ m filters and stored in water or PBS at 4° C. Quality control was done using transmission electron microscopy, dynamic light scattering, and native and denaturing gel electrophoresis. pMHC content was measured using Bradford assay (Thermo Fisher Scientific) and SDS-PAGE.

(213) Generation of FTCD-Expressing Adenovirus

(214) A replication-deficient adenovirus expressing human formiminotransferase cyclodeaminase (Ad-hFTCD) (a target autoantigen in AIH Type 2) was generated by cloning the hFTCD DNA sequence directly into Adeno-X Adenoviral System 3 (CMV) using In-Fusion® HD cloning technology and Stellar Competent cells (Clontech, Mountain View, CA). Cloned Ad-FTCD was amplified in Ad-293 T cells and purified using Adeno-X Maxi Purification Kit (Clontech). The viral titer was measured using the End-point Dilution Assay or Adeno-X Rapid Titer Kit (Clontech).

(215) Ursodeoxycholic Acid Treatment

(216) Cohorts of 5-6 or 24 wk-old male and/or female NOD.c3c4 mice were left untreated, fed a diet supplemented with 0.5% UDCA (BOC Sciences, Upton, NY; TestDiet, Richmond, IN), or treated with pMHCII-NPs for 14 or 9 wk, respectively, and sacrificed for pMHCII tetramer staining, PBC scoring and biochemical testing.

(217) pMHCII-NP Therapy in NOD.c3c4, (NOD×B6.IFNg ARE-Del-/-) F1 and NOD/Ltj Mice

(218) Cohorts of 15 wk-old male and/or female NOD.c3c4 mice with established PBC were left untreated or treated with 20 mg of pMHCII-NPs or Cys-NPs (i.v.) twice weekly for 9 wk unless indicated otherwise. Liver disease scoring involved macroscopic evaluation of cyst content (0-5), liver weight and CBD diameter (0-4), as well as microscopic evaluation of bile duct involvement (0-4), bile duct proliferation (0-4) and mononuclear cell infiltration (0-4), essentially as described

(223). In other experiments, treatment was initiated at the peak of disease (24 wk of age) and given twice a week for 14-20 wk. Intermittent treatment involved treating mice twice a wk from 15 to 24 wk of age, then withdrawing treatment until the percentages of tetramer+ cells dropped to ~50% of the levels seen at treatment withdrawal (measurements in peripheral blood were done once every two wk), re-treating mice twice a wk until the percentages of tetramer+ cells reached original values, and repeating this cycle until 50 wk of age.

(219) In in vivo cytokine blocking experiments, mAbs against HRPN (rIgG1), IL-10 (JES5-2A5) or TGF- β (1D11) (BioXcell, West Lebanon, NH) were given i.p. twice a week at 500 mg per dose for 2 wk, followed by 200 mg per dose for 7 additional wk. Mice were randomized into cytokine-neutralizing mAb-treatment (anti-IL-10 or -TGF β) or HRPN rat-IgG1 groups.

(220) In experiments involving (NOD $\times$ B6.IFNg ARE-Del.sup.-/-) F1 mice, 10-wk-old male and female mice were treated for 5-6 wk. Histopathologic scoring in the liver was performed as described. Briefly, severity scores were obtained by scoring portal inflammation, lobular inflammation and granuloma formation from 0-4, and bile duct damage from 0-2. The extent of portal inflammation and bile duct damage were scored from 0-4 based on the ratio between affected vs unaffected area. The extent of lobular inflammation and granuloma formation were scored from 0-4 based on number of lesions per specimen. The severity of fibrosis was scored on a 0-6 scale.

(221) Studies using NOD mice involved treating cohorts of 10-wk-old pre-diabetic female NOD/Ltj mice with 20 mg of pMHCII-NPs or Cys-NPs i.v. twice weekly for 5 wk.

(222) pMHCII-NP Therapy for EAE in NOD.c3c4 Mice

(223) Male and female 12-14 wk-old NOD.c3c4 mice were immunized with 250 mg of pMOG.sub.36-55 in CFA s.c. on either side of the flank region above the base of the tail under isofluorane anesthesia. The mice received 350 ng of Pertussis toxin i.v. on days 0 and 2 relative to peptide immunization. Mice were weighed and scored on day 0 and then daily starting on day 14 after immunization, and scores were plotted on a 5-point scale. On day 32 of immunization, all the mice having scores <3 were randomized into treatment with Cys-NPs or pMHCII-NPs. All mice were treated twice a week for 9 wk. Scoring of mice for signs of PBC was done as described above.

(224) pMHCII-NP Therapy for PSC in NOD.Abc4^{-/-} Mice

(225) Cohorts of 5-6 wk-old male and/or female NOD.Abc4^{sup.-/-} mice with established PSC were treated with 20 mg of pMHCII-NPs or Cys-NPs i.v. twice weekly for 5-6 wk. Histopathologic lesions were graded using the Ishak scoring system, which evaluates both fibrosis (0-6) as well as necroinflammatory sequelae of biliary cholangitis, including interface hepatitis (0-4), confluent necrosis (0-6), lobular inflammation (0-4) and portal inflammation (0-4).

(226) pMHCII-NP Therapy for AIH in NOD Mice

(227) AIH was induced by infecting 5-6 wk-old female NOD/Ltj mice with an adenovirus encoding human FTCD (Ad-hFTCD, 1010 plaque forming units (PFU) i.v.). Four wks later, cohorts of mice with established AIH were treated with 20 mg of pMHCII-NPs or Cys-NPs (i.v.) twice weekly for 5-6 wk. Histopathological scoring was done using the Ishak scale as above.

(228) pMHCII-NP Therapy in Human PBMC-Reconstituted NSG Hosts

(229) PBMCs from HLA-DRB4*0101+ PBC patients (recruited under informed consent approved by the Institutional Review Board at Hospital Clinic) were depleted of CD8+ T-cells using anti-CD8 mAb-coated magnetic beads (Miltenyi Biotech, Auburn, CA) and injected i.v. (2×10^6) into 8-10 wk-old NSG hosts. Mice were treated with 30-40 mg pMHC-NPs starting on day 5 after PBMC transfusion, twice a wk for 5 consecutive wks, or left untreated. Therapy-induced expansion of cognate CD4+ T-cells was measured in liver, peripheral LNs, spleen and bone marrow (not shown). A mouse was considered a responder if the percentage of tetramer+ T-cells in at least two different organs were higher than the mean $\pm$ 10 standard deviation values seen in untreated hosts.

(230) Evaluation of General Adaptive Immunity

(231) pMHCII-NP-treated and untreated female mice were injected i.v. with 2×10^6 PFU of

recombinant Vaccinia Virus (rVV) and sacrificed on days 4 and 14 after infection. Samples were processed for pMHCII tetramer staining and rVV titer measurements. Briefly, both ovaries were collected in DMEM containing 2% FBS, homogenized, freeze-thawed 3 times followed by sonication (3 rounds, 20 sec each). Serial dilutions of the lysates were added to confluent BSC-1 cell cultures at 37° C. for 1 h, washed twice with serum free DMEM and then overlaid with DMEM containing 2% FCS and 0.4% carboxymethyl cellulose (CMC; Sigma, Saint Louis, MO). On day 3, the overlay was discarded, and the cell layers were stained with crystal violet to count the number of plaques.

(232) To evaluate cellular responses to Influenza infection, pMHCII-NP-treated and untreated mice were first primed i.p. with the HKx31 (H3N2) strain at $10^{5.6}$ EID₅₀ per mouse. One cohort of mice was sacrificed 7d after priming and processed for tetramer staining to confirm presence of pMHC-NP-specific TR1-like cells during priming. Other cohorts of primed mice were re-infected 30d later with an intranasal dose of PR8 virus, a lethal H1N1 strain of Influenza ($8 \times 10^{4.4}$ EID₅₀ per mouse), under anesthesia. PR8-challenged mice were weighed daily and scored clinically from 0-4 based on the extent of ruffled fur, reduced motility, huddled appearance, and rapid and/or labored breathing. Mice were sacrificed 7d later and processed for tetramer staining and influenza titer measurement. Briefly, lungs were collected in serum free DMEM, homogenized and freeze-thawed 3 times. Serial dilutions of the lysates were added to confluent MDCK cell cultures at RT for 1 h and washed. Cultures were then overlaid with DMEM containing 0.4% CMC for 2-3 days, washed, fixed and stained with crystal violet to count the number of plaques.

(233) Cellular immunity to intracellular bacteria was determined by infecting pMHCII-NP-treated and untreated mice i.v. with 10³ colony forming units (cfu) of *Listeria monocytogenes*. Mice were sacrificed 7d or 14d after infection and samples processed for tetramer staining and bacterial load measurements. Briefly, spleen and liver were cut into several pieces, weighed and homogenized in PBS containing 0.35% Triton X-100. Serial dilutions of the lysates were then plated onto Bovine Heart Infusion agar containing 5 mg/ml erythromycin, incubated for 24-48 h at 37° C. and the number of colonies counted.

(234) Cellular immunity to liver metastatic tumors was ascertained upon intra-splenic injections of B16/F10 melanoma and CT26 colon carcinoma tumors into syngeneic (C57BL/6J or Balb/c, respectively) or allogeneic hosts (pMHC-NP-treated or untreated NOD.c3c4 mice). A small incision was made in the abdomen, under isoflurane inhalational anesthesia, to partially expose the spleen. Tumor cells ($0.2 \times 10^{5.6}$ and $0.1 \times 10^{5.6}$ for B16/F10 and CT26, respectively, in 100 µL of PBS) were injected slowly for Imin into the exposed spleen. Ten minutes later, the spleen was removed and the peritoneal and skin layers sutured. pMHCII-NP therapy was resumed within 5-7d after surgery and continued until the end of follow-up. Mice were monitored for up to 19-21d and euthanized for tetramer staining, PBC scoring, and tumor burden measurements. In B16/F10-injected mice, tumor burden was assessed by measuring liver weight and counting the number of metastases, easily distinguishable from liver parenchyma. In CT26-injected animals, tumor burden was scored histologically by measuring the hepatic area (HPA) occupied by metastatic tumors.

(235) To evaluate humoral immunity, pMHCII-NP-treated and untreated mice were immunized i.p. with 100 mg of DNP-KLH (Alpha Diagnostic International, San Antonio, TX) in CFA and boosted again 3 wk later. Mice were sacrificed 10d later, to measure serum anti-DNP antibody titers using an anti-DNP Ig ELISA Kit (Alpha Diagnostic International).

(236) Cytokine Secretion Assays

(237) Splenic and portal/cealic lymph node (PCLN) cell suspensions from pMHCII-NP-treated mice were enriched for CD4⁺ T-cells depleting CD19⁺ B-cells (EasySep™ Mouse CD19 Positive Selection Kit, Stem Cell Technologies, Vancouver, BC) and CD8⁺ T-cells (CD8 Magnetic Particles, BD Biosciences). Cells were stained with pMHCII tetramers and sorted by flow cytometry. The sorted cells ($2-3 \times 10^{4.4}$) were challenged with bone marrow-derived DCs ($2 \times 10^{4.4}$) pulsed with 2 µg ml⁻¹ peptide. Forty-eight hours later, supernatants were harvested for measurement

of cytokine content via Luminex®.

(238) To ascertain whether pMHCII-NP therapy promoted the recruitment/formation of IL-10-secreting B-cells, mesenteric LNs, PCLNs and liver cell suspensions were enriched for B-cells using a CD19 enrichment kit (Stem Cell Technologies). The cells ($2-3 \times 10^5$ in 200 mL/well) were stimulated in duplicate with LPS ($1 \mu\text{g mL}^{-1}$, Sigma) for 24 h in RPMI-1640 media containing 10% FCS. The levels of IL-10 in the supernatants were measured via Luminex®.

(239) Isolation and In Vitro Stimulation of Lymph Node CD11b⁺ Cells and Liver Kupffer Cells

(240) CD11b⁺ cells from LNs were obtained by digestion in collagenase D ($1.25 \mu\text{g mL}^{-1}$) and DNase I ($0.1 \mu\text{g mL}^{-1}$) for 15 min at 37° C., washed, incubated with anti-FcR Abs, and purified using anti-CD11b mAb-coated magnetic beads (BD Biosciences). The purified cells ($2-3 \times 10^5$ in 200 mL/well) were stimulated with LPS ($2 \mu\text{g mL}^{-1}$) for 3 days, and the supernatants analyzed for cytokine content using a Luminex® multiplex cytokine assay.

(241) To isolate Kupffer cells (KCs), livers from treated and untreated mice were minced and digested in 15 ml of 0.05% collagenase solution in HBSS for 20-30 min at 37° C. The resulting cell suspension was filtered through a nylon mesh ($0.7 \mu\text{m}$) and centrifuged at $50 \times g$ for 3 min at 4° C., to remove tissue debris and hepatocytes. Cells in the supernatant were pelleted by centrifugation at $300 \times g$ for 5 min at 4° C. The cell pellet, mainly composed of non-parenchymal liver immune cells, KCs, sinusoidal endothelial cells and stellate cells, was re-suspended in 33% Percoll® solution and centrifuged at $350 \times g$ for 30 min to isolate mononuclear cells. The pellets were re-suspended in DMEM containing 10% FCS (5×10^6 cells mL^{-1}) and plated in 6-well plate at $1-3 \times 10^7$ cells/well and incubated for 2-3 h in a 5% CO₂ atmosphere at 37° C. Non-adherent cells were removed by gentle washing with PBS. The adherent fraction (enriched for KCs) was harvested by trypsin digestion (5 min, 0.25% trypsin). The resulting cell suspension was plated in 96 well plates at $1-2 \times 10^5$ /200 mL/well and stimulated with LPS (2 mg mL^{-1}) for 3 d. The supernatants were analyzed for cytokine content using a Luminex® multiplex cytokine assay.

(242) Adoptive Transfer of Suppression

(243) Splenic CD4⁺ T-cells (10^7) from untreated mice or mice treated with 12 doses of PDC-E2.sub.166-181/IA.sup.g7-NPs were adoptively transferred (i.v.) into 10-14 wk-old, sex-matched NOD.c3c4.scid hosts. One day later, the recipients were adoptively transferred with 4×10^7 whole splenocytes from sex-matched NOD.c3c4 donor mice with established PBC (>35 wk-old). One of the cohorts of mice transfused with CD4⁺ T-cells from pMHCII-NP-treated donors was further treated with 12 doses of PDC-E2.sub.166-181/IA.sup.g7-NPs. The recipients were sacrificed 6 wk later for tetramer staining and PBC scoring.

(244) In Vivo Breg Induction Assay

(245) Splenic B-cells from NOD.II10.sup.tm1Flv (Tiger) mice were enriched using an EasySep Mouse B-cell Isolation Kit (Stem Cell Technologies) and pulsed with BDC2.5 mi or PDC.sub.166-181 peptides ($10 \mu\text{g mL}^{-1}$) for 2 h at 37° C. The peptide-pulsed B-cells were washed twice with PBS, labeled with PKH26 (Sigma) and transfused (3×10^6) into pMHC-NP-treated or untreated mice. The hosts were killed 7d later and their spleens, MLNs, PCLNs and liver mononuclear cells were labeled with anti-B220-APC and biotinylated anti-CD1d or anti-CD5 mAbs followed by Streptavidin-PerCP. PKH26⁺ B-cells were analyzed for presence of eGFP⁺/CD1d.sup.high and eGFP⁺/CD5⁺ cells by flow cytometry.

(246) Histology and Immunohistochemistry

(247) Livers were fixed in 10% formalin for 2d, embedded in paraffin, cut into $5 \mu\text{m}$ sections and stained with H&E or Picrosirius Red. For immunohistochemistry, liver tissues were embedded in Tissue-Tek OCT, sectioned into $30 \mu\text{m}$ cryosections and stored on slides at -80° C. Slides were fixed in chilled acetone, washed with PBS, treated with a 1:10 dilution of 30% H.sub.2O.sub.2 in PBS, washed with PBS, blocked with 10% normal goat serum in PBS, washed again, and stained with anti-mouse CD4 (GK1.5) or CD8 (Lyt-2) antibodies (1.5 h, 4° C.). After washing, the slides were stained with a biotinylated goat anti-rat secondary antibody (1:200 dilution), incubated with

Horseradish Peroxidase (HRP)-conjugated streptavidin, followed by 3,3-diaminobenzidine (DAB) substrate. Slides were counterstained with hematoxylin before mounting.

(248) ALT and TBA Assays

(249) Alanine aminotransferase (ALT) levels in serum were determined using a kit from Thermo Fisher Scientific following the manufacturer's protocol. Briefly, serum samples were mixed with pre-warmed (37° C.) Infinity™ ALT (GPT) Liquid Stable Reagent at 1:10 ratio and OD readings were taken for 3 min at 1 min intervals in a nanodrop at a 340 nm wavelength, 37° C. The slope was calculated by plotting absorbance vs. time using linear regression and multiplied with a factor to obtain ALT levels in serum (U/L) as described in the kit. Serum total bile acid (TBA) levels were analyzed using a TBA Enzymatic Cycling Assay Kit (Diazyme, Poway, CA) following a modified manufacturer's protocol as described.

(250) Anti-Nuclear and Anti-Mitochondrial Autoantibody Measurements

(251) Presence of anti-nuclear autoantibodies (ANAs) in serum was ascertained using NOVA Lite®HEp-2 Slides kit (Inova Diagnostics, San Diego, CA). A semi-quantitative approach was followed to measure ANA titers. Briefly, serum samples were serially diluted in PBS (at 1:160, 1:320, 1:640, 1:1280 and 1:2560) and then added to pre-fixed Hep-2 substrate slides, washed, stained with FITC-conjugated goat anti-mouse IgG in PBS containing 5% normal donkey serum (1:200 dilution), washed, mounted and read under a fluorescent microscope.

(252) Serum levels of anti-mitochondrial PDC-E2 antibodies were determined via ELISA. Briefly, ELISA plates were coated with PDC-E2 protein (5 µg ml.sup.-1, 100 mL) (SurModics Inc., Eden Prairie, MN) overnight at RT. Plates were washed, blocked using 3% dry skim milk in PBS (pH 7.4, 150 ml), and incubated with serially-diluted serum samples (100 ml, at 1:250 dilutions prepared using reagent diluent) for 2 h at RT. Wells were washed and incubated with 100 mL of HRP-conjugated anti-mouse IgG (1:2000 in reagent diluent) for 2 h at RT, and washed. Finally, wells were incubated in the dark with 100 mL of DAB substrate for 20 min at RT. Upon stopping the enzymatic reaction with 50 mL 2N H.sub.2SO.sub.4, the absorption was measured at a 450 nm wavelength using an ELISA plate reader. The Positive Antibody Activity (PAA) levels were calculated by calculating the mean OD±2 SD of the control NOD serum samples (positive index) and by dividing the OD values corresponding to NOD.c3c4 serum samples by the positive index, whereby values >1.0 correspond to PAA.

(253) Statistical Analyses

(254) Unless specified, sample size values mentioned in the figure legends correspond to the total number of mice examined, pooled from different experiments. Data were compared in GraphPad Prism 6 using Mann-Whitney U-test, Chi-Square, Log-Rank (Mantel-Cox), Pearson correlation, two-way ANOVA or multiple t test analyses using the Holm-Sidak correction. P values <0.05 were considered statistically significant. Only statistically significant P values are displayed on Figures.

(255) While preferred embodiments of the present invention have been shown and described herein, it will be obvious to those skilled in the art that such embodiments are provided by way of example only. Numerous variations, changes, and substitutions will now occur to those skilled in the art without departing from the invention. It should be understood that various alternatives to the embodiments of the invention described herein may be employed in practicing the invention.

(256) All publications, patent applications, issued patents, and other documents referred to in this specification are herein incorporated by reference as if each individual publication, patent application, issued patent, or other document was specifically and individually indicated to be incorporated by reference in its entirety. Definitions that are contained in text incorporated by reference are excluded to the extent that they contradict definitions in this disclosure.

Embodiments of the Current Description

(257) The following embodiments provide a non-exhaustive list of specific compositions and uses of the current invention.

(258) 1. In a certain aspect described herein is a composition comprising: a) a plurality of antigen-

major histocompatibility complexes (MHCs), each antigen-MHC of the plurality comprising a ubiquitous autoantigen, that is not a tissue specific antigen, associated with a binding groove of an MHC molecule; and b) a nanoparticle core possessing a diameter of between 1 and 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core.

(259) 2. An embodiment of the composition of embodiment 1, wherein the MHC molecule is an MHC class II molecule.

(260) 3. An embodiment of the composition of embodiment 1, wherein the nanoparticle core is a metal or metal oxide.

(261) 4. An embodiment of the composition of embodiment 3, wherein the metal is iron.

(262) 5. An embodiment of the composition of embodiment 3, wherein the metal oxide is iron oxide.

(263) 6. An embodiment of the composition of any one of embodiments 1 to 5, wherein the diameter is between about 5 nanometers and about 50 nanometers.

(264) 7. An embodiment of the composition of embodiment 6, wherein the diameter is between about 5 nanometers and about 25 nanometers.

(265) 8. An embodiment of the composition of any one of embodiments 1 to 7, wherein the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of at least 10:1.

(266) 9. An embodiment of the composition of any one of embodiments 1 to 8, wherein the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of no more than 150:1.

(267) 10. An embodiment of the composition of any one of embodiments 1 to 9, wherein the plurality of antigen-MHCs is coupled to the nanoparticle core at a density from about 0.4 to about 13 antigen-MHCs per 100 nm.² of nanoparticle core surface area.

(268) 11. An embodiment of the composition of any one of embodiments 1 to 10, wherein the antigen-MHCs are covalently coupled to the nanoparticle core.

(269) 12. An embodiment of the composition of any one of embodiments 1 to 11, wherein the antigen-MHCs are covalently coupled to the nanoparticle core by a polyethylene glycol (PEG) linker having a mass of less than about 5 kilodaltons.

(270) 13. An embodiment of the composition of any one of embodiments 1 to 12, wherein the nanoparticle core further comprises a biocompatible coating.

(271) 14. An embodiment of the composition of any one of embodiments 1 to 13, wherein the ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an intracellular compartment.

(272) 15. An embodiment of the composition of embodiment 14, wherein the intracellular compartment is cytosol, mitochondria, Golgi apparatus, endoplasmic reticulum, nucleus, or plasma membrane.

(273) 16. An embodiment of the composition of embodiment 14, wherein the intracellular compartment is a mitochondrion.

(274) 17. An embodiment of the composition of any one of embodiments 1 to 13, wherein the ubiquitous autoantigen is pyruvate dehydrogenase complex-E2 component (PDC-E2).

(275) 18. The composition of any one of embodiments 1 to 13, wherein the ubiquitous autoantigen is Cytochrome P450 2D6 (CYP2D6).

(276) 19. An embodiment of the composition of any one of embodiments 1 to 13, wherein the ubiquitous autoantigen is actin (ACTB).

(277) 20. An embodiment of the composition of any one of embodiments 1 to 13, wherein the ubiquitous autoantigen is soluble liver antigen (SLA).

(278) 21. An embodiment of the composition of any one of embodiments 1 to 13, wherein the ubiquitous autoantigen is formimidoyltransferase-cyclodeaminase (FTCD).

(279) 22. An embodiment of the composition of any one of embodiments 1 to 13, wherein the

ubiquitous autoantigen is myeloperoxidase (MPO).

(280) 23. An embodiment of the composition of embodiment 17, wherein the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367; PDC-E2.sub.72-86; PDC-E2.sub.422-436; PDC-E2.sub.353-367; PDC-E2.sub.80-94; PDC-E2.sub.535-549; PDC-E2.sub.629-648; PDC-E2.sub.122-135 PDC-E2.sub.249-263; and PDC-E2.sub.249-263.

(281) 24. An embodiment of the composition of embodiment 17, wherein the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94.

(282) 25. An embodiment of the composition of embodiment 18, wherein the ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298; CYP2D6.sub.289-303; CYP2D6.sub.318-332; CYP2D6.sub.313-332; CYP2D6.sub.393-412; CYP2D6.sub.192-206; CYP2D6.sub.5-19; CYP2D6.sub.293-307; CYP2D6.sub.219-233; CYP2D6.sub.237-251; CYP2D6.sub.15-29; CYP2D6.sub.235-249; CYP2D6.sub.317-331; CYP2D6.sub.293-307; CYP2D6.sub.428-442; CYP2D6.sub.237-251; CYP2D6.sub.14-28; CYP2D6.sub.199-213; CYP2D6.sub.450-464; CYP2D6.sub.301-315; CYP2D6.sub.452-466; CYP2D6.sub.59-73; CYP2D6.sub.130-144; CYP2D6.sub.193-212; CYP2D6.sub.305-324; and CYP2D6.sub.15-29.

(283) 26. An embodiment of the composition of embodiment 19, wherein the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216; ACTB.sub.170-184; ACTB.sub.245-259; ACTB.sub.187-201; ACTB.sub.172-186; ACTB.sub.131-145; ACTB.sub.131-145; ACTB.sub.171-185; ACTB.sub.129-143; ACTB.sub.164-178; ACTB.sub.25-39; and ACTB.sub.323-337.

(284) 27. An embodiment of the composition of embodiment 19, wherein the ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160; ACTB.sub.18-32; and ACTB.sub.171-185.

(285) 28. An embodiment of the composition of embodiment 20, wherein the ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348; SLA.sub.196-210; SLA.sub.115-129; SLA.sub.373-386; SLA.sub.186-197; SLA.sub.342-256; SLA.sub.110-124; SLA.sub.299-313; SLA.sub.49-63; SLA.sub.260-274; SLA.sub.119-133; SLA.sub.86-100; SLA.sub.26-40; SLA.sub.331-345; SLA.sub.317-331; SLA.sub.171-185; SLA.sub.417-431; SLA.sub.359-373; SLA.sub.215-229; and SLA.sub.111-125.

(286) 29. An embodiment of the composition of embodiment 21, wherein the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453; FTCD.sub.381-395; FTCD.sub.297-311; FTCD.sub.525-539; FTCD.sub.218-232; FTCD.sub.495-509; FTCD.sub.262-276; FTCD.sub.300-314; FTCD.sub.259-273; FTCD.sub.490-504; FTCD.sub.389-403; and FTCD 295-309.

(287) 30. An embodiment of the composition of embodiment 21, wherein the ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285; FTCD.sub.498-512; and FTCD.sub.301-315.

(288) 31. An embodiment of the composition of embodiment 22, wherein the ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336; MPO.sub.714-728; MPO.sub.617-631; MPO.sub.504-518; MPO.sub.462-476; MPO.sub.617-631; MPO.sub.444-458; MPO.sub.689-703; MPO.sub.248-262; MPO.sub.511-525; MPO.sub.97-111; and MPO.sub.616-630.

(289) 32. An embodiment of the composition of any one of embodiments 1 to 31, further comprising a second plurality of antigen-major histocompatibility complexes (MHCs) coupled to the nanoparticle core, each antigen-MHC of the second plurality comprising an antigen.

(290) 33. An embodiment of the composition of embodiment 32, wherein the antigen of the second plurality of antigen-major histocompatibility complexes (MHCs) is a second ubiquitous autoantigen.

(291) 34. An embodiment of the composition of embodiment 32, wherein the second ubiquitous autoantigen comprises a polypeptide derived from a protein that at steady-state exists in or on an

intracellular compartment.

(292) 35. An embodiment of the composition of embodiment 34, wherein the intracellular compartment is cytosol, mitochondria, Golgi apparatus, endoplasmic reticulum, nucleus, or plasma membrane.

(293) 36. An embodiment of the composition of embodiment 34, wherein the intracellular compartment is a mitochondrion.

(294) 37. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.353-367; PDC-E2.sub.72-86; PDC-E2.sub.422-436; PDC-E2.sub.353-367; PDC-E2.sub.80-94; PDC-E2.sub.535-549; PDC-E2.sub.629-648; PDC-E2.sub.122-135 PDC-E2.sub.249-263; and PDC-E2.sub.249-263.

(295) 38. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94.

(296) 39. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: CYP2D6.sub.284-298; CYP2D6.sub.289-303; CYP2D6.sub.318-332; CYP2D6.sub.313-332; CYP2D6.sub.393-412; CYP2D6.sub.192-206; CYP2D6.sub.65-19; CYP2D6.sub.293-307; CYP2D6.sub.219-233; CYP2D6.sub.237-251; CYP2D6.sub.15-29; CYP2D6.sub.235-249; CYP2D6.sub.317-331; CYP2D6.sub.293-307; CYP2D6.sub.428-442; CYP2D6.sub.237-251; CYP2D6.sub.14-28; CYP2D6.sub.199-213; CYP2D6.sub.450-464; CYP2D6.sub.301-315; CYP2D6.sub.452-466; CYP2D6.sub.59-73; CYP2D6.sub.130-144; CYP2D6.sub.193-212; CYP2D6.sub.305-324; and CYP2D6.sub.15-29.

(297) 40. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.202-216; ACTB.sub.170-184; ACTB.sub.245-259; ACTB.sub.187-201; ACTB.sub.172-186; ACTB.sub.131-145; ACTB.sub.131-145; ACTB.sub.171-185; ACTB.sub.129-143; ACTB.sub.164-178; ACTB.sub.25-39; and ACTB.sub.323-337.

(298) 41. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: ACTB.sub.146-160; ACTB.sub.18-32; and ACTB.sub.171-185.

(299) 42. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: SLA.sub.334-348; SLA.sub.196-210; SLA.sub.115-129; SLA.sub.373-386; SLA.sub.186-197; SLA.sub.342-256; SLA.sub.110-124; SLA.sub.299-313; SLA.sub.49-63; SLA.sub.260-274; SLA.sub.119-133; SLA.sub.86-100; SLA.sub.26-40; SLA.sub.331-345; SLA.sub.317-331; SLA.sub.171-185; SLA.sub.417-431; SLA.sub.359-373; SLA.sub.215-229; and SLA.sub.111-125.

(300) 43. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.439-453; FTCD.sub.381-395; FTCD.sub.297-311; FTCD.sub.525-539; FTCD.sub.218-232; FTCD.sub.495-509; FTCD.sub.262-276; FTCD.sub.300-314; FTCD.sub.259-273; FTCD.sub.490-504; FTCD.sub.389-403; and FTCD.sub.295-309.

(301) 44. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: FTCD.sub.271-285; FTCD.sub.498-512; and FTCD.sub.301-315.

(302) 45. An embodiment of the composition of embodiment 34, wherein the second ubiquitous autoantigen is selected from the group consisting of: MPO.sub.322-336; MPO.sub.714-728; MPO.sub.617-631; MPO.sub.504-518; MPO.sub.462-476; MPO.sub.617-631; MPO.sub.444-458; MPO.sub.689-703; MPO.sub.248-262; MPO.sub.511-525; MPO.sub.97-111; and MPO.sub.616-630.

(303) 46. An embodiment of the composition of any one of embodiments 1 to 45, further comprising a pharmaceutically acceptable stabilizer, excipient, diluent, or combination thereof.

(304) 47. An embodiment of the composition of any one of embodiments 1 to 46, formulated for

intravenous administration.

(305) 48. An embodiment of the composition of any one of embodiments 1 to 47, for use in a method of treating an autoimmune or inflammatory disease.

(306) 49. Use of the composition of any one of embodiments 1 to 47, in the manufacture of a medicament for treating an autoimmune or inflammatory disease.

(307) 50. An embodiment of the composition or use of embodiment 49, wherein the ubiquitous autoantigen or the second ubiquitous autoantigen is not a polypeptide derived from myelin basic protein, myelin associated glycoprotein, myelin oligodendrocyte protein (MOG), proteolipid protein, oligodendrocyte myelin oligoprotein, myelin associated oligodendrocyte basic protein, oligodendrocyte specific protein, heat shock proteins, an oligodendrocyte specific protein, NOGO A, glycoprotein Po, peripheral myelin protein 22, or 2'3'-cyclic nucleotide 3'-phosphodiesterase.

(308) 51. An embodiment of the composition or use of embodiment 49, wherein the ubiquitous autoantigen or the second ubiquitous autoantigen is not a polypeptide derived from pre-proinsulin, proinsulin, islet-specific glucose-6-phosphatase (IGRP), glutamate decarboxylase (GAD), islet cell autoantigen-2 (ICA2), or insulin.

(309) TABLE-US-00005 TABLE 5 additional ubiquitous autoantigens Cellular location Uniprot Code Description
Cytoplasm P14152 Mdh1 Malate dehydrogenase, cytoplasmic (EC 1.1.1.37) (Cytosolic malate dehydrogenase)
Cytoplasm P63260 Actg1 Actin, cytoplasmic 2 (Gamma-actin) [Cleaved into: Actin, cytoplasmic 2, N-terminally processed]
Cytoplasm P20152 Vim Vimentin
Cytoplasm P06151 Ldha L-lactate dehydrogenase A chain (LDH-A) (EC 1.1.1.27) (LDH muscle subunit) (LDH-M)
Cytoplasm P16858 Gapdh Glyceraldehyde-3-phosphate dehydrogenase (GAPDH) (EC 1.2.1.12) (Peptidyl- cysteine S-nitrosylase GAPDH) (EC 2.6.99.—)
Cytoplasm P63101 Ywhaz 14-3-3 protein zeta/delta (Protein kinase C inhibitor protein 1) (KCIP-1) (SEZ-2)
Cytoplasm P11404 Fabp3 Fatty acid-binding protein, heart (Fatty acid-binding protein 3) (Heart-type fatty acid-binding protein) (H-FABP) (Mammary-derived growth inhibitor) (MDGI)
Cytoplasm O08997 Atox1 Copper transport protein ATOX1 (Metal transport protein ATX1)
Cytoplasm P35700 Prdx1 Peroxiredoxin-1 (EC 1.11.1.15) (Macrophage 23 kDa stress protein) (Osteoblast-specific factor 3) (OSF-3) (Thioredoxin peroxidase 2) (Thioredoxin- dependent peroxide reductase 2)
Cytoplasm Q9CQM5 Txndc17 Thioredoxin domain-containing protein 17 (14 kDa thioredoxin-related protein) (TRP14) (Protein 42-9-9) (Thioredoxin-like protein 5)
Nucleus P09405 Ncl Nucleolin (Protein C23)
Nucleus Q9Z2X1 Hnrnpf Heterogeneous nuclear ribonucleoprotein F (hnRNP F) [Cleaved into: Heterogeneous nuclear ribonucleoprotein F, N-terminally processed]
Nucleus Q3U898 Cops9 COP9 signalosome complex subunit 9
Nucleus P62322 Lsm5 U6 snRNA-associated Sm-like protein LSm5
Nucleus P17918 PcnA Proliferating cell nuclear antigen (PCNA) (Cyclin)
Nucleus O88569 Hnrnpa2b1 Heterogeneous nuclear ribonucleoproteins A2/B1 (hnRNP A2/B1)
Nucleus P40142 Tkt Transketolase (TK) (EC 2.2.1.1) (P68)
Nucleus Q60972 Rbbp4 Histone-binding protein RBBP4 (Chromatin assembly factor 1 subunit C) (CAF-1 subunit C) (Chromatin assembly factor I p48 subunit) (CAF-I 48 kDa subunit) (CAF-I p48) (Nucleosome- remodeling factor subunit RBAP48) (Retinoblastoma-binding protein 4) (RBBP-4) (Retinoblastoma-binding protein p48)
Nucleus Q60973 Rbbp7 Histone-binding protein RBBP7 (Histone acetyltransferase type B subunit 2) (Nucleosome-remodeling factor subunit RBAP46) (Retinoblastoma- binding protein 7) (RBBP-7) (Retinoblastoma-binding protein p46)
Nucleus P15532 Nme1 Nucleoside diphosphate kinase A (NDK A) (NDP kinase A) (EC 2.7.4.6) (Metastasis inhibition factor NM23) (NDPK-A) (Tumor metastatic process-associated protein) (nm23-M1)
Plasma More Abundant Membrane Plasma P68040 Rack1 Receptor of activated protein C kinase Membrane 1 (12-3) (Guanine nucleotide-binding protein subunit beta-2-like 1) (Receptor for activated C kinase) (Receptor of activated protein kinase C 1) (p205) [Cleaved into: Receptor of activated protein C kinase 1, N-terminally processed (Guanine nucleotide-binding protein subunit beta-2-like 1, N-terminally processed)]
Plasma Q62351 Tfrc Transferrin receptor protein 1 Membrane (TR) (TfR) (TfR1) (Trfr) (CD antigen CD71)
Plasma Q9QYY0 Gab1 GRB2-

associated-binding protein 1 Membrane (GRB2-associated bound 1) (Growth factor receptor bound protein 2- associated protein 1) Plasma P42703 Lifr Leukemia inhibitory factor receptor Membrane (LIF receptor) (LIF-R) (D-factor/LIF receptor) (CD antigen CD118) Plasma Q01279 Egfr Epidermal growth factor receptor Membrane (EC 2.7.10.1) Plasma Q62351 Tfrc Transferrin receptor protein 1 Membrane (TR) (TfR) (TfR1) (Trfr) (CD antigen CD71) Plasma P14069 S100a6 Protein S100-A6 (5B10) (Calcyclin) Membrane (Prolactin receptor-associated protein) (S100 calcium-binding protein A6) Plasma Q61160 Fadd FAS-associated death domain protein Membrane (FAS-associating death domain- containing protein) (Mediator of receptor induced toxicity) (Protein FADD) Plasma Q5M8N0 Cnrip1 CB1 cannabinoid receptor-interacting Membrane protein 1 (CRIP-1) Plasma Q60902 Eps1511 Epidermal growth factor receptor Membrane substrate 15-like 1 (Epidermal growth factor receptor pathway substrate 15- related sequence) (Eps15-rs) (Eps15-related protein) (Eps15R) Plasma P97300 Nptp Nuroplastin Membrane Mitochondria Q64433 Hspe1 10 kDa heat shock protein, mitochondrial (Hsp10) (10 kDa chaperonin) (Chaperonin 10) (CPN10) Mitochondria Q07813 Bax Apoptosis regulator BAX Mitochondria P38647 Hspa9 Stress-70 protein, mitochondrial (75 kDa glucose-regulated protein) (GRP-75) (Heat shock 70 kDa protein 9) (Mortalin) (Peptide-binding protein 74) (PBP74) (p66 MOT) Mitochondria P19157 Gstp1 Glutathione S-transferase P 1 (Gst P1) (EC 2.5.1.18) (GST YF-YF) (GST class-pi) (GST-piB) (Preadipocyte growth factor) Mitochondria Q9CR21 Ndubf1 Acyl carrier protein, mitochondrial (ACP) (CI-SDAP) (NADH-ubiquinone oxidoreductase 9.6 kDa subunit) Mitochondria P08249 Mdh2 Malate dehydrogenase, mitochondrial (EC 1.1.1.37) Mitochondria P63038 Hspd1 60 kDa heat shock protein, mitochondrial (EC 3.6.4.9) (60 kDa chaperonin) (Chaperonin 60) (CPN60) (HSP-65) (Heat shock protein 60) (HSP-60) (Hsp60) (Mitochondrial matrix protein P1) Mitochondria Q03265 Atp5f1a ATP synthase subunit alpha, mitochondrial (ATP synthase F1 subunit alpha) Mitochondria P63038 Hspd1 60 kDa heat shock protein, mitochondrial (EC 3.6.4.9) (60 kDa chaperonin) (Chaperonin 60) (CPN60) (HSP-65) (Heat shock protein 60) (HSP-60) (Hsp60) (Mitochondrial matrix protein P1) Mitochondria P56382 Atp5f1e ATP synthase subunit epsilon, mitochondrial (ATPase subunit epsilon) (ATP synthase F1 subunit epsilon) Golgi More Abundant Golgi P61205 Arf3 ADP-ribosylation factor 3 Golgi P61750 Arf4 ADP-ribosylation factor 4 Golgi P84084 Arf5 ADP-ribosylation factor 5 Golgi Q99LT0 Dpy30 Protein dpy-30 homolog (Dpy-30-like protein) (Dpy-30L) Golgi P53811 Pitpnb Phosphatidylinositol transfer protein beta isoform (PI-TP-beta) (PtdIns transfer protein beta) (PtdInsTP beta) Golgi O35643 Ap1b1 AP-1 complex subunit beta-1 (Adaptor protein complex AP-1 subunit beta-1) (Adaptor-related protein complex 1 subunit beta-1) (Beta-1-adaptin) (Beta-adaptin 1) (Clathrin assembly protein complex 1 beta large chain) (Golgi adaptor HA1/AP1 adaptin beta subunit) Golgi P61211 Arl1 ADP-ribosylation factor-like protein 1 Golgi Q3UPH1 Prrc1 Protein PRRC1 (Proline-rich and coiled- coil-containing protein 1) Golgi P61924 Copz1 Coatomer subunit zeta-1 (Zeta-1-coat protein) (Zeta-1 COP) Golgi Q9CQC9 Sar1b GTP-binding protein SAR1b ER O55022 Pgrmc1 Membrane-associated progesterone receptor component 1 (mPR) ER P33267 Cyp2f2 Cytochrome P450 2F2 (EC 1.14.14.—) (CYPIIF2) (Cytochrome P450-NAH-2) (Naphthalene dehydrogenase) (Naphthalene hydroxylase) ER O55143 Atp2a2 Sarcoplasmic/endoplasmic reticulum calcium ATPase 2 (SERCA2) (SR Ca(2+)-ATPase 2) (EC 3.6.3.8) (Calcium pump 2) (Calcium-transporting ATPase sarcoplasmic reticulum type, slow twitch skeletal muscle isoform) (Endoplasmic reticulum class 1/2 Ca(2+) ATPase) ER P45878 Fkbp2 Peptidyl-prolyl cis-trans isomerase FKBP2 (PPIase FKBP2) (EC 5.2.1.8) (13 kDa FK506- binding protein) (13 kDa FKBP) (FKBP-13) (FK506-binding protein 2) (FKBP-2) (Immunophilin FKBP13) (Rotamase) ER P56395 Cyb5a Cytochrome b5 ER Q9D1Q6 Erp44 Endoplasmic reticulum resident protein 44 (ER protein 44) (ERp44) (Thioredoxin domain- containing protein 4) ER P35564 Canx Calnexin ER P08113 Hsp90b1 Endoplasmic (94 kDa glucose-regulated protein) (GRP-94) (Endoplasmic reticulum resident protein 99) (ERp99) (Heat shock protein 90 kDa beta member 1) (Polymorphic tumor rejection antigen 1) (Tumor rejection antigen gp96) ER Q01853

Vcp Transitional endoplasmic reticulum ATPase (TER ATPase) (EC 3.6.4.6) (15S Mg(2+)-ATPase p97 subunit) (Valosin-containing protein) (VCP) Q9D0F3 Lman1 Protein ERGIC-53 (ER-Golgi intermediate compartment 53 kDa protein) (Lectin mannose-binding 1) (p58) Also contemplated are the human proteins or human homologues disclosed in this table

(310) TABLE-US-00006 SEQUENCES DISCLOSED HEREIN HLA Position in protein Amino Acid Sequence SEQ. ID Allele bound start end PDC-E2 derived peptides NO:

DRB3*0202 353 367 GRVFSVPLAKKLAVE 1 DRB3*0202 72 86
 RLLQLLGSPGRRYY 2 DRB3*0202 422 436 DIPISNIRRVIAQRL 3 DRB5*0101 353 367
 GRVFSVPLAKKLAVE 4 DRB5*0101 80 94 SPGRRYYSLPPHQKV 5 DRB5*0101
 535 549 ETIANDVVSLATKAR 6 DRB4*0101 629 643 AQWLAEFRKYLEKPI 7
 DRB4*0101 122 135 GDLIAEVETDKATV 8 DRB4*0101 249 262 GDLLAEIETDKATI 9
 DRB1*0801 249 263 GDLLAEIETDKATIG 10 DRB3*0202 422 436 DIPISNIRRVIAQRL 11
 DRB5*0101 80 94 SPGRRYYSLPPHQKV 12 13 CYP2D6 derived peptides DRB3*01:01
 284 298 GNPESSFNDENLRIV 14 DRB3*01:01 289 303 SFNDENLRIVVADLF 15 DRB3*01:01
 318 332 LLLMILHPDVQRRVQ 16 DRB1*03:01 313 332 TLAWGLLLMILHPDVQRRVQ 17
 DRB1*03:01 393 412 TTLITNLSVLKDEAVWEKP 18 DRB1*03:01 192 206
 GRRFEYDDPRFLRLL 19 DRB1*03:01 5 19 ALVPLAVIVAIFLLL 20 DRB1*03:01 293
 307 ENLRIVVADLFSAGM 21 DRB3*0202 219 233 FLREVLNAVVPVLLHI 22 DRB3*0202 237
 251 AGKVLRFQKAFLTQL 23 DRB3*0202 15 29 IFLLLVDLMHRRQRW 24 DRB4*0101
 235 249 ALAGKVLRFQKAFLT 25 DRB4*0101 317 331 GLLLMILHPDVQRRV 26
 DRB4*0101 293 307 ENLRIVVADLFSAGM 27 DRB5*0101 428 442 VKPEAFLPFSAGRRA 28
 DRB5*0101 237 251 AGKVLRFQKAFLTQL 29 DRB5*0101 14 28 AIFLLVDLMHRRQR
 30 DRB1*04:01 199 213 DPRFLRLDLAQEGL 31 DRB1*04:01 450 464
 RMELFLFFTSLLQHF 32 DRB1*04:01 301 315 DLFSAGMVTSTTLA 33 DRB1*07:01 452
 466 ELFLFFTSLLQHFSF 34 DRB1*07:01 59 73 DQLRRRFGDVFSLQL 35 DRB1*07:01
 130 144 EQRRFSVSTLRNLGL 36 DRB1*07:01 193 212 RFEYDDPRFLRLDLAQEG 37
 DRB1*07:01 305 324 AGMVTSTTLAWGLLLMILH 38 ACTB derived peptides
 DRB1*0301 202 216 TTAEREIVRDIKEKL 39 DRB1*0301 170 184 ALPHAILRLDLAGRDL 40
 DRB1*0301 245 259 SGRTTGIVMDSGDGV 41 DRB3*02:02 187 201 DYLMKILTERGYST
 42 DRB3*02:02 172 186 PHAILRLDLAGRDLT 43 DRB3*02:02 131 145
 AMYVAIQAVLSLYAS 44 DRB4*0101 131 145 AMYVAIQAVLSLYAS 45 DRB4*0101 171 185
 LPHAILRLDLAGRDL 46 DRB4*0101 129 143 TPAMYVAIQAVLSLY 47 DRB5*0101 164 178
 PIYEGYALPHAILRL 48 DRB5*0101 25 39 DAPRAVFPSIVGRPR 49 DRB5*0101 323 337
 STMKIKIIPPERKY 50 DRB3*0101 146 160 GRTTGIVMDSGDGV 51 DRB3*0101 18
 32 KAGFAGDDAPRAVFP 52 DRB4*0101 171 185 LPHAILRLDLAGRDL 53 SLA derived
 peptides DRB1*0301 334 348 YKKLLKERKEMFSYL 54 DRB1*0301 196 210
 DELRTDLKAVEAKVQ 55 DRB1*0301 115 129 NKITNSLVLDIIKLA 56 DRB1*0301 373 386
 NRLDRCLKAVRKER 57 DRB1*0301 186 197 LIQQGARVGRID 58 DRB3*0202 342 356
 KEMFSYLSNQIKKLS 59 DRB3*0202 110 124 GSSLLNKITNSLVLD 60 DRB3*0202 299 313
 NDSFIQEISKMYPGR 61 DRB4*0101 49 63 STLELFLHELAIMDS 62 DRB4*0101 260
 274 SKCMHLIQQGARVGR 63 DRB4*0101 119 133 NSLVLDIIKLAGVHT 64 DRB5*0101
 86 100 RRHYRFIHGIGRSGD 65 DRB5*0101 26 40 RSHEHLIRLLLEKKGK 66 DRB5*0101
 331 345 SNGYKKLLKERKEMF 67 DRB1*0401 317 331 SPSLDVLITLLSLGS 68 DRB1*0401
 171 185 DQKSCFKSMITAGFE 69 DRB1*0401 417 431 YTFRGFMSHTNNYPC 70 DRB1*0701
 359 373 YNERLLHTPHNPISL 71 DRB1*0701 215 229 DCILCIHSTTSCFAP 72 DRB1*0701
 111 125 SSSLNKITNSLVLDI 73 FTCD derived peptides DRB3*0202 439 453
 ALQEGLRRAVSPLT 74 DRB3*0202 381 395 RRQFQSLDTTMRRLI 75 DRB3*0202 297 311
 EQRIRLVVSRLGLDS 76 DRB1*0301 525 539 AKTQAALVLDCLCTR 77 DRB1*0301 218 232
 KVQGIGWYLDEKNLA 78 DRB1*0301 495 509 YFNVLINLRDITDEA 79 DRB4*0101 262 276
 LPVVGSQLVGLVPLK 80 DRB4*0101 300 314 IRLVVSRLGLDSLCP 81 DRB4*0101 259 273

ELSLPVVGSQVLGLV 82 DRB5*0101 490 504 GVFGAYFNVLINLRD 83 DRB5*0101 389 403
TTMRRLIPPFREASA 84 DRB5*0101 295 309 EEEQRIRLVVSRLGL 85 DRB3*0101 271 285
GLVPLKALLDAAAFY 86 DRB3*0101 498 512 VLINLRDITDEAFKD 87 DRB3*0101 301 315
RLVVSRLGLDSLCPF 88 MPO derived peptides DRB3*0202 322 336 SNITIRNQINALTSF
89 DRB3*0202 714 728 KNNIFMSNSYPRDFV 90 DRB3*0202 617 631
LGTVLRNLKLARKLM 91 DRB1*0301 504 518 LIQPFMFRLDNRYQP 92 DRB1*0301 462
476 YLPLVLGPTAMRKY 93 DRB1*0301 617 631 LGTVLRNLKLARKLM 94 DRB4*0101
444 458 QEARKIVGAMVQIIT 95 DRB4*0101 689 703 QQRQALAQISLPRII 96 DRB4*0101
248 262 RSLMFMQWGQLLDHD 97 DRB5*0101 511 525 RLDNRYQPMENPRV 98
DRB5*0101 97 111 ELLSYFKQPVAATRT 99 DRB5*0101 616 630 QLGTVLRNLKLARKL
100 Any of the peptides described herein may encompass a naturally occurring variant that does
not affect binding to an MHC class II molecule or recognition by a T cell receptor. Such variants
are contemplated herein and included with reference to the sequence.

(311) TABLE-US-00007 SEQ ID NO: 103

MAIYLILLFTAVRGIKEEHVIIQAEFYLNPDQSGEFMFDFD
GDEIFHVDMAKKETVWRLEEFGRFASFEAQGALANIAVDKAN
LEIMTKRSNYTPITNVPPEVTVLTNSPVELREPNVLICFIDK
FTPPVVNVTWLRNGKPVTTGVSETVFLPREDHLFRKFHYLPF
LPSTEDVYDCRVEHWGLDEPLLKHWEFDAPSPLPETTESGGG
GGDKTHTCPPCPAPEAAGGPSVFLEPPKPKDTLMISRTPEVT
CVVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRV
VSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYITLPPCRDELTKNQVSLWCLVKGFYPSDIAVEWESNGQ
PENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGNVFSCSVM
HEALHNHYTQKSLSLSPGKSGSGSGSGSC 104
MGS LQPLATLYLLGMLVASSLGGDLIAEVETDKATVGGGGGS
GGSGSGSGDTQPRFLEQAKCECHFLNGTERVWNLIRYIYNQE
EYARYNSDLGEYQAVTELGRPDAEYWNSQKDLLERRRAEVD
YCRYNYGVVESFTVQRRVQPKVTVYPSKTQPLQHNLVCSV
NGFYPGSIEVRWFRNSQEEKAGVVSTGLIQNGDWTFQTLVML
ETVPRSGEVYTCQVEHPSMMSPLTVQWSARSESAQSKSGGGG
GDKTHTCPPCPAPEAAGGPSVFLEPPKPKDTLMISRTPEVTC
VVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV
SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPRE
PQVCTLPPSRDELTKNQVSLSCAVKGFYPSDIAVEWESNGQP
ENNYKTTTPVLDSGDSFFLVSKLVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK

Claims

1. A method for treating an autoimmune or inflammatory disease in a patient in need thereof, comprising administering to the patient a composition comprising: a) a plurality of antigen-major histocompatibility class II complexes (antigen-MHCII), each antigen-MHCII of the plurality comprising a ubiquitous autoantigen associated with a binding groove of an MHC class II molecule, wherein the ubiquitous autoantigen is chosen from PDC-E2.sub.353-367, PDC-E2.sub.72-86, and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94, and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2.sub.122-135, and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801; and b) a nanoparticle core possessing a diameter of between 1 and about 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core or a biocompatible layer surrounding the nanoparticle core; and wherein the autoimmune or inflammatory disease is chosen from multiple

sclerosis and psoriasis.

2. The method of claim 1, wherein the nanoparticle core is a metal or metal oxide.

3. The method of claim 2, wherein the metal is iron.

4. The method of claim 2, wherein the metal oxide is iron oxide.

5. The method of claim 1, wherein the diameter is between about 5 nanometers and about 50 nanometers.

6. The method of claim 5, wherein the diameter is between about 5 nanometers and about 25 nanometers.

7. The method of claim 1, wherein the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of at least 10:1.

8. The method of claim 1, wherein the plurality of antigen-MHCs is coupled to the nanoparticle core at an antigen-MHC to nanoparticle core ratio of no more than about 150:1.

9. The method of claim 1, wherein the plurality of antigen-MHCs is coupled to the nanoparticle core at a density from about 0.4 to about 13 antigen-MHCs per 100 nm.² of nanoparticle core surface area.

10. The method of claim 1, wherein the antigen-MHCs are covalently coupled to the nanoparticle core.

11. The method of claim 1, wherein the antigen-MHCs are coupled to the nanoparticle core by a polyethylene glycol (PEG) linker having a mass of less than about 5 kilodaltons.

12. The method of claim 1, wherein the nanoparticle core further comprises a biocompatible coating.

13. The method of claim 1, wherein the ubiquitous autoantigen is selected from the group consisting of: PDC-E2.sub.422-436 and PDC-E2.sub.80-94.

14. The method of claim 1 further comprising a pharmaceutically acceptable stabilizer, excipient, diluent, or combination thereof.

15. The method of claim 1, formulated for intravenous administration.

16. A method of treating an autoimmune or inflammatory disease in an individual comprising administering to an individual a composition comprising: a) a plurality of antigen-major histocompatibility class II complexes (antigen-MHCII), each antigen-MHCII of the plurality comprising a ubiquitous autoantigen associated with a binding groove of an MHC class II molecule, wherein the ubiquitous autoantigen is chosen from PDC-E2353-367, PDC-E2.sub.72-86, and PDC-E2.sub.422-436 for DRB3*0202; PDC-E2.sub.353-367, PDC-E2.sub.80-94, and PDC-E2.sub.535-549 for DRB5*0101; PDC-E2.sub.629-648, PDC-E2122-135, and PDC-E2.sub.249-263 for DRB4*0101; and PDC-E2.sub.249-263 for DRB1*0801; and b) a nanoparticle core possessing a diameter of between 1 and about 100 nanometers; wherein the antigen-MHCs are coupled to the nanoparticle core or a biocompatible layer surrounding the nanoparticle core; and wherein the autoimmune or inflammatory disease is chosen from multiple sclerosis and psoriasis.
